

Convex relaxation gaps and spatial certificates in nonlinear optimization

Abstract

Separate convexification of nonlinear terms can lose information about their shared variables. We develop a probability formulation of this loss and explain its relation to spatial optimality certificates. For bilinear functions with arbitrary coefficient signs, an elementary cut argument bounds the ratio of the McCormick gap to the scalar graph-hull gap by four times the square root of the largest induced edge density. A random-sign construction gives the matching order on each individual interaction graph. Sharper maximum-degree and bipartite bounds, exactness criteria, and the distinction between exact termwise convexification and recursive product relaxations identify the scope of these comparisons. The density order is also a consequence of earlier Schur-multiplier theory; the explicit transfer and proof are included to make that attribution precise.

1 Introduction

Convex relaxations support global optimization by providing bounds on nonlinear functions over a region. Their strength depends on how the model is divided into pieces. Even when each piece is replaced by its exact convex hull, the resulting bounds can be weaker than the hull of their sum: separate pieces may use incompatible representations of their common variables. Spatial subdivision introduces a second question, namely how many regions are needed before a specified bounding rule certifies a desired tolerance. These questions concern different mathematical objects. A pointwise envelope gap comparison is not an approximation ratio for a constrained optimization problem, and it does not by itself bound a branch-and-bound tree.

McCormick’s construction provides the basic recursive mechanism for factorable functions [18]. For multilinear functions, the convex hull of the scalar graph has a finite representation using box vertices. Luedtke, Namazifar, and Linderoth [17] compare this hull with termwise and recursive relaxations, and establish several exactness and positive-coefficient bounds. Boland et al. [4] show that signed bilinear gaps can grow with the square root of dimension and relate the comparison to cuts in induced interaction graphs.

We use the vertex representation to state the comparisons consistently. For positive multilinear functions on nonnegative boxes, a common probability distribution attains all upper term envelopes, so the entire gap question reduces to a single simultaneous rounding problem. For signed bilinear functions, an exact cut-range characterization instead permits graph density and degree bounds. The elementary density proof makes the constants explicit, while the accompanying Schur-multiplier transfer explains which asymptotic content was already implicit in established theory [7]. The framework also specifies the bounding oracle and the certified-region count needed for spatial lower bounds.

2 Envelope gaps as probability problems

2.1 The scalar graph and the specified termwise relaxation

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box. Fixed coordinates are removed when necessary. A function is *multiaffine* if it is affine in each coordinate separately; equivalently, it is a multilinear polynomial with no repeated variable in a monomial. For a continuous scalar function f on B , let $\text{vex}_B f$ be its greatest convex underestimator and $\text{cav}_B f$ its least concave overestimator. The vertical section of the convex hull of its graph is

$$\{y : (x, y) \in \text{conv}\{(z, f(z)) : z \in B\}\} = [\text{vex}_B f(x), \text{cav}_B f(x)].$$

Define the *scalar graph-hull gap* and, for a specified decomposition $f = \sum_{e \in E} f_e$, the *termwise gap* by

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2.1)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B f_e(x) - \text{vex}_B f_e(x)]. \quad (2.2)$$

The notation T_B always refers to the displayed decomposition. It is the width obtained by giving each scalar factor its exact graph hull and imposing $y = \sum_e y_e$. It can change when a factor is split into smaller pieces, whereas H_B does not. Always,

$$0 \leq H_B(f; x) \leq T_B(f; x), \quad (2.3)$$

because the sum of convex underestimators is a convex underestimator of the sum, and similarly for concave overestimators. Ratios are taken only where $H_B(f; x) > 0$; gap inequalities apply at every point.

For a positive multilinear polynomial, the factors are its displayed monomials $a_e \prod_{i \in e} x_i$, with $a_e \geq 0$. For a signed bilinear function, they are the signed products $a_{ij} x_i x_j$. A bilinear factor has the familiar four-inequality McCormick hull. For a higher-degree factor, its exact hull and a recursive McCormick relaxation are different objects. A recursive product relaxation is exact for one monomial on a unit box, but need not be exact on a general positive box [17, Sections 2–3]. A bound for (2.2) on that box therefore does not automatically bound the gap of a recursive bilinear factorization.

Adding an affine function to a factor, or adding a separate affine factor, changes neither gap. An affine bijection between boxes preserves both envelopes after composition and hence preserves the gaps for the transformed factors. These facts follow from the graph-hull representation.

2.2 Vertex distributions and compatible marginals

Let $\mathcal{V}(B)$ denote the vertices of B , and let $\mathcal{P}_B(x)$ be the distributions on $\mathcal{V}(B)$ with mean x . On the unit cube these are joint Bernoulli distributions with the prescribed singleton means; coordinates need not be independent.

Lemma 2.1 (Vertex representation). *For a multiaffine f on B ,*

$$\text{vex}_B f(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X), \quad (2.4)$$

$$\text{cav}_B f(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X). \quad (2.5)$$

Both extrema are attained. The envelopes are continuous piecewise-affine functions on B .

Proof. Write $x_i = \ell_i + (u_i - \ell_i)p_i$. Independently choose $X_i = u_i$ with probability p_i and $X_i = \ell_i$ otherwise. Multiaffinity gives $\mathbb{E}X = x$ and $\mathbb{E}f(X) = f(x)$. Every graph point therefore belongs to the convex hull of vertex graph points, and the reverse inclusion in the full graph hull is immediate. The vertical sections give the formulas. The feasible distributions form a nonempty compact polytope, so both extrema are attained. The lower and upper surfaces of the vertex graph polytope are respectively maxima and minima of finitely many affine functions on B , which also proves continuity at its boundary. $\square$

These representations are classical [17, Section 3]. They expose the compatibility issue: different factors may use different minimizing distributions in (2.2), whereas (2.4) requires one distribution for their sum.

For $m_e(x) = \prod_{i \in e} x_i$ with $e \neq \emptyset$, the unit-cube formulas are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max\{0, \sum_{i \in e} x_i - |e| + 1\}. \quad (2.6)$$

To prove them, interpret $\mathbb{E}m_e(X)$ as the probability of joint success. The upper bound is attained by $X_i = \mathbf{1}\{U \leq x_i\}$ for one uniform $U \in [0, 1]$. For the lower bound, place failure intervals of lengths $1 - x_i$ consecutively on the unit circle. If their total length is below one they are disjoint, and if it is at least one they cover the circle. Their union therefore has measure $\min\{1, \sum_{i \in e} (1 - x_i)\}$. Each interval has its prescribed measure, so the complement gives the claimed minimum joint-success probability. Unused coordinates can be sampled independently to complete the mean vector.

Proposition 2.2 (Common upper attainment and deficiencies). *Let $f(x) = \sum_e a_e m_e(x)$ on $[0, 1]^n$, with $a_e \geq 0$. Discard supports of size at most one. For each remaining support choose an anchor i_e of smallest mean and put*

$$u_e = x_{i_e}, \quad S_e = \sum_{j \in e \setminus \{i_e\}} (1 - x_j), \quad t_e = \min\{u_e, S_e\}, \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i.$$

Then D_e is the nonnegative indicator that the anchor succeeds and some other support coordinate fails. Moreover,

$$\text{cav } f(x) = \sum_e a_e u_e, \quad (2.7)$$

$$T(f; x) = \sum_e a_e t_e, \quad (2.8)$$

$$H(f; x) = \max_{P \in \mathcal{P}_{[0,1]^n}(x)} \sum_e a_e \mathbb{E}_P D_e(X). \quad (2.9)$$

Proof. The common-threshold law attains every single-product upper envelope simultaneously. Positivity gives (2.7), and subtracting the two single-product envelopes gives (2.8). Finally, every admissible law has $\mathbb{E}X_{i_e} = u_e$, so minimizing the expected polynomial value is equivalent to maximizing the weighted sum of deficiencies. $\square$

The common-upper statement is classical [17, Theorems 4–5 in the authors’ manuscript]. A distribution used to prove a gap bound must work for all terms together; independently optimizing the distribution for each term merely recovers the termwise relaxation. Mixtures of admissible laws preserve the means, and nonnegative deficiencies allow the guarantees of different laws to be added.

Lemma 2.3 (Two classes handled by independence). *Put $c_0 = 1 - e^{-1}$. Independent Bernoulli rounding gives $\mathbb{E}D_e \geq c_0 t_e/2$ if either every support mean exceeds $1/2$, or at least two support means are at most $1/2$.*

Proof. Independence gives $\mathbb{E}D_e = u_e(1 - \prod_{j \neq i_e} x_j)$. In the second case a nonanchor mean is at most $1/2$, giving at least $u_e/2 \geq t_e/2$. In the first case the expression is at least $u_e(1 - e^{-S_e}) \geq c_0 u_e \min(1, S_e)$. If $S_e \leq 1$, this is at least $c_0 S_e/2 \geq c_0 t_e/2$; otherwise it is at least $c_0 u_e = c_0 t_e$. $\square$

Proposition 2.4 (Nonnegative-box transfer). *For a positive multilinear polynomial on B with $0 \leq \ell_i < u_i$, substitute $x_i = \ell_i + (u_i - \ell_i)p_i$ and expand the original monomials. The unit-cube polynomial f has nonnegative coefficients and no larger degree. With its expanded monomials as factors,*

$$H_B(f; x) = H(\tilde{f}; p), \quad T_B(f; x) \leq T(\tilde{f}; p). \quad (2.10)$$

One common-threshold endpoint law attains the upper envelope of every original monomial and their sum.

Proof. Expansion coefficients are products of nonnegative endpoints and interval widths. The full-envelope equality follows from the affine bijection. For an original factor expanded as $\sum_q g_q$, the sum of the lower convex envelopes is a convex underestimator and the sum of the upper concave envelopes is a concave overestimator. Subtract and sum over the original factors to obtain (2.10). Common-threshold rounding attains all expanded upper envelopes together and therefore also those of each original factor and their sum. $\square$

Expansion is a proof device and need not be an efficient representation. It changes the factor incidence structure. Degree bounds transfer through (2.10); bounds based on the original incidence graph require separate arguments. For bilinear functions, arbitrary box endpoints cause no such difficulty: an affine change introduces only scaled bilinear terms and affine summands, regardless of signs.

2.3 What a spatial certificate counts

Let $\mathcal{F}$ be a nonempty compact feasible set, let f be continuous, and put $f^* = \min_{x \in \mathcal{F}} f(x)$. A region oracle assigns a valid lower bound $\text{LB}(S)$ to feasible points in each region S . A *certificate at target* T_* is a finite cover $S_1, \dots, S_N$ of $\mathcal{F}$ with $\text{LB}(S_j) \geq T_*$ for all j . Together with a feasible value U and an absolute tolerance $\epsilon \geq 0$, the target $U - \epsilon$ certifies absolute gap at most ϵ . For $U > 0$ and a relative tolerance $0 \leq \theta < 1$, the target $(1 - \theta)U$ certifies $(U - f^*)/U \leq \theta$. A size lower bound may grant the best incumbent $U = f^*$, requiring $f^* > 0$ in this relative convention. Under these tolerance ranges, either target is nondecreasing in U , so worse incumbents require weakly higher targets.

A complete spatial split tree supplies a cover by its final regions. A lower bound for arbitrary covers is thus independent of branching order and node selection within the specified region and oracle model. If a procedure discards feasible regions by propagation or additional cuts, those deductions must also be included in the certificate model. A lower bound for one oracle does not automatically survive stronger coupled cuts, different branch coordinates, or analytic solution of separate components.

3 Signed bilinear gaps and interaction density

Let $G = (V, E)$ be a finite simple graph and $b_a(x) = \sum_{ij \in E} a_{ij} x_i x_j$ on $[0, 1]^V$, with arbitrary real coefficients. Zero-coefficient edges may be removed. Let $c^*(a)$ be the least $c \geq 0$ with $T(b_a; x) \leq cH(b_a; x)$ everywhere, and set $c^*(0) = 0$. Define

$$\rho_G(G) = \max_{\emptyset \neq W \subseteq V} \frac{|E(G[W])|}{|W|}, \quad (3.1)$$

with value zero for an empty graph. This is half the maximum subgraph average degree, not fractional arboricity, whose denominator is $|W| - 1$. Let d_G be degeneracy and Δ_G maximum degree. Then

$$d_G/2 \leq \rho_G(G) \leq d_G, \quad \rho_G(G) \leq \Delta_G/2. \quad (3.2)$$

The first lower bound uses a subgraph of minimum degree d_G ; the upper bounds use a degeneracy orientation and the degree sum identity.

Theorem 3.1 (Bilinear upper bounds). *For every real coefficient vector on G ,*

$$c^*(a) \leq \min\{4\sqrt{\rho_G(G)}, 4\sqrt{d_G}, 2\sqrt{\Delta_G}\}. \quad (3.3)$$

If G is bipartite with parts P, Q and has an edge, let Δ_P and Δ_Q be the maximum degrees in the respective parts. Then

$$c^*(a) \leq \sqrt{2 \min\{\Delta_P, \Delta_Q\}} \leq \sqrt{2 \min\{|P|, |Q|\}}. \quad (3.4)$$

The factor $\sqrt{2}$ in the first bipartite bound is sharp. The conclusions transfer to arbitrary finite boxes after removing fixed coordinates.

The proof refines the cut argument of Boland et al. [4]. The density order also follows from older operator theory, as shown in Section 3.6; no new general graph-norm principle is claimed.

3.1 An exact characterization by induced cuts

For $W \subseteq V$, set

$$Q_{a,W}(s) = \sum_{ij \in E(G[W])} a_{ij} s_i s_j, \quad L_W(a) = \sum_{ij \in E(G[W])} |a_{ij}|,$$

and define

$$R_W(a) = \frac{1}{2} \left(\max_{s \in \{-1,1\}^W} Q_{a,W}(s) - \min_{s \in \{-1,1\}^W} Q_{a,W}(s) \right). \quad (3.5)$$

A cut encoded by s has weight $\frac{1}{2} \sum_{ij} a_{ij} - \frac{1}{2} Q_{a,W}(s)$, so R_W is exactly the maximum cut weight minus the minimum cut weight. If $L_W > 0$ then $R_W > 0$: distinct degree-two sign characters are orthogonal under uniform signs, and their nonzero combination cannot be constant. Set $L_W/R_W = 0$ when both vanish.

Lemma 3.2 (Induced-cut characterization). *For every coefficient vector,*

$$c^*(a) = \max_{W \subseteq V} \frac{L_W(a)}{R_W(a)}. \quad (3.6)$$

Each nonzero induced ratio is attained at $x_i = 1/2$ on W and $x_i = 0$ outside W .

Proof. At that point any admissible vertex law fixes outside coordinates to zero. Write $X_i = (1+s_i)/2$ on W . Since $\mathbb{E}s_i = 0$,

$$\mathbb{E}b_a(X) = \frac{1}{4} \left(\sum_{ij \in E(G[W])} a_{ij} + \mathbb{E}Q_{a,W}(s) \right).$$

An extremizing sign vector and its negative, each of probability $1/2$, attain either extreme of $Q_{a,W}$ with the prescribed means. Thus $H = R_W/2$, whereas each active product has width $1/2$ and

$T = L_W/2$. These identities hold at every point of $\{0, 1/2, 1\}^V$, with W its set of half-valued coordinates: fixing coordinates to one instead of zero adds only affine terms on W .

To extend the comparison, the width of the product $x_i x_j$ is

$$\min(x_i, x_j) - \max(0, x_i + x_j - 1) = \min(x_i, x_j, 1 - x_i, 1 - x_j).$$

Partition the cube by the hyperplanes $x_i = x_j$, $x_i + x_j = 1$ ($i \neq j$), and $x_i = 1/2$. The termwise gap is affine on every closed cell. Every cell vertex is in $\{0, 1/2, 1\}^V$. Indeed, active equations fix a coordinate to 0, 1/2, or 1, or relate two coordinates by equality or complementation. In a connected component of these relations, one fixed coordinate determines all values. If none is fixed, a cycle with an odd number of complementations forces the value 1/2. Otherwise a free parameter remains and a small perturbation preserves all active equations, contradicting extremality. This accounts for every cell vertex.

The graph-hull gap is concave, being $\text{cav } b_a - \text{vex } b_a$. Hence $T - cH$ is convex on each cell for $c \geq 0$. Its value at any convex combination of cell vertices is at most the combination of its values there. The grid-point bound therefore proves the upper direction of (3.6), and the attained ratios prove the other. $\square$

The grid identities and upper implication are those of Boland et al. [4, Lemma 1 and Corollary 1], following Luedtke et al. [17]. The cell argument above supplies the boundary details.

3.2 Cut ranges, row norms, and fractional orientations

Let A be the symmetric weighted adjacency matrix on W , with zero diagonal and zero entries at nonedges. Write $a_{i,T} = (A_{ij})_{j \in T}$ for $T \subseteq W$ and $a_i = a_{i,W}$. For a rectangular matrix define $\|M\|_{\infty \rightarrow 1} = \max_u \text{sign} \|Mu\|_1 = \max_{u,v} \text{sign } v^\top Mu$; empty matrices have norm zero.

Lemma 3.3 (Polarization and row estimates). *Writing $R = R_W(a)$,*

$$R = \max_{S \subseteq W} \|A_{S,W \setminus S}\|_{\infty \rightarrow 1}, \quad (3.7)$$

$$R \geq \frac{1}{\sqrt{2}} \max_{S \subseteq W} \sum_{i \in S} \|a_{i,W \setminus S}\|_2, \quad (3.8)$$

$$R \geq \frac{1}{4} \sum_{i \in W} \|a_i\|_2. \quad (3.9)$$

Proof. For signs s, t , put $u = (s + t)/2$, $v = (s - t)/2$. Their supports are complementary, and symmetry gives $Q(s) - Q(t) = 2v^\top Au$. Conversely, arbitrary signs on complementary supports arise from $s = u + v$, $t = u - v$. Maximizing half the difference proves (3.7).

The sharp real Khinchin inequality of Szarek [22] states that independent uniform signs satisfy

$$\mathbb{E} \left| \sum_j c_j \varepsilon_j \right| \geq \frac{1}{\sqrt{2}} \left(\sum_j c_j^2 \right)^{1/2}. \quad (3.10)$$

Apply it to every row of $A_{S,T}$ for a fixed partition S, T , and sum. The best sign vector performs at least as well as its average, proving (3.8). Transposition gives the same estimate for side T .

Choose a partition locally maximizing the cut weight with nonnegative weights a_{ij}^2 . At every vertex at least half its squared incident weight crosses, since otherwise moving the vertex improves the cut. Thus

$$X = \sum_{i \in S} \|a_{i,T}\|_2, \quad Y = \sum_{i \in T} \|a_{i,S}\|_2 \quad \text{satisfy} \quad X + Y \geq \frac{1}{\sqrt{2}} \sum_i \|a_i\|_2.$$

The two row estimates yield $R \geq \max(X, Y)/\sqrt{2} \geq (X + Y)/(2\sqrt{2})$, which proves (3.9). $\square$

Lemma 3.4 (Fractional orientation). *Every finite graph H admits nonnegative fractions $\theta_{ij} + \theta_{ji} = 1$ on each edge such that $\sum_j \theta_{ij} \leq \rho_G(H)$ for every vertex. This is the least possible uniform bound on the vertex loads.*

Proof. For $m = |E(H)| > 0$, form a flow network with a source, edge nodes, vertex nodes, and a sink. Source-to-edge capacities are one; arcs from an edge node to its endpoints have capacity $m + 1$; vertex-to-sink capacities are t . A cut of capacity at most m cannot cut an edge-to-endpoint arc. For a fixed set U of vertex nodes on the source side, placing all edge nodes internal to U on that side minimizes capacity, giving $m - |E(H[U])| + t|U|$. At $t = \rho_G(H)$ every cut has capacity at least m . Max-flow/min-cut supplies a flow saturating the source arcs; outgoing flows at each edge define its fractions. Conversely, the loads over U sum to at least $|E(H[U])|$, proving minimality. For $m = 0$ use zero loads. $\square$

Proof of Theorem 3.1. Fix an induced graph H with nonempty weighted support and abbreviate $L = \sum_{ij} |a_{ij}|$, $R = R_{V(H)}(a)$. Weighted Cauchy–Schwarz gives

$$\sum_j \theta_{ij} |a_{ij}| \leq \left(\sum_j \theta_{ij} \right)^{1/2} \left(\sum_j \theta_{ij} a_{ij}^2 \right)^{1/2} \leq \sqrt{\rho_G(H)} \|a_i\|_2.$$

Summing counts each absolute edge weight once. By (3.9), $L \leq 4\sqrt{\rho_G(H)}R$. Also $\|a_i\|_2 \geq \|a_i\|_1/\sqrt{\Delta_H}$ at every nonisolated vertex, and $\sum_i \|a_i\|_1 = 2L$. The same row bound therefore gives $R \geq L/(2\sqrt{\Delta_H})$. The graph parameters do not increase in induced subgraphs, so Lemma 3.2 proves (3.3).

For bipartite H use its inherited parts P', Q' in (3.8). Every edge crosses, giving

$$R \geq \frac{1}{\sqrt{2}} \sum_{i \in P'} \|a_i\|_2 \geq \frac{L}{\sqrt{2\Delta_P}}.$$

Transpose to obtain the bound with Δ_Q and apply the same induced characterization. On a unit-weight four-cycle with one negative edge, $L = 4$, $R = 2$, and $\Delta_P = \Delta_Q = 2$. Its center ratio attains two, so the stated bipartite constant is sharp. An affine box transformation scales each bilinear coefficient by the two interval widths and introduces only affine summands, proving the final assertion. $\square$

The degeneracy bound has a distinct direct proof. Orient along a degeneracy order with outdegree at most d_G and color vertices in reverse order. Place a vertex opposite the side receiving at least half the squared weight of its outgoing edges. Its crossing row norm is at least its outgoing row norm divided by $\sqrt{2}$. Summing and applying Cauchy–Schwarz to at most d_G outgoing entries gives $X + Y \geq L/\sqrt{2d_G}$. The final step of Lemma 3.3 yields $R \geq L/(4\sqrt{d_G})$. This proof is weaker than the density theorem but uses an ordinary acyclic orientation.

3.3 Sharp order on every interaction graph

Define

$$\Gamma(G) = \sup_{a \in \mathbb{R}^E \setminus \{0\}} c^*(a), \quad \Gamma_{\pm}(G) = \max_{a \in \{-1, 1\}^E} c^*(a),$$

with both values zero for edgeless graphs.

Theorem 3.5 (Worst coefficients on a fixed graph). *For every graph with an edge,*

$$\max \left\{ 1, \sqrt{\frac{\rho_G(G)}{2 \log 2}} \right\} \leq \Gamma_{\pm}(G) \leq \Gamma(G) \leq 4\sqrt{\rho_G(G)}. \quad (3.11)$$

Requiring every coefficient to be nonzero does not change $\Gamma(G)$. It also equals

$$\sup_{a \neq 0} L_V(a)/R_V(a). \quad (3.12)$$

The lower bound has a full-support signing and a witness in $\{0, 1/2\}^V$.

Proof. On an h -vertex, m -edge graph with $m > 0$, choose independent edge signs σ_e . For fixed vertex signs s and $\lambda > 0$, $\mathbb{E}e^{\lambda Q_{\sigma}(s)} = \cosh(\lambda)^m \leq e^{m\lambda^2/2}$, and similarly for $-\lambda$. Put $Z = \max_s |Q_{\sigma}(s)|$. Global sign reversal preserves Q_{σ} , so

$$e^{\lambda Z} \leq \sum_{s: s_1=1} (e^{\lambda Q_{\sigma}(s)} + e^{-\lambda Q_{\sigma}(s)}).$$

Taking expectations and using Jensen gives $\mathbb{E}Z \leq h \log 2/\lambda + m\lambda/2$. At $\lambda = \sqrt{2h \log 2/m}$, at least one signing satisfies

$$\|Q_{\sigma}\|_{\infty} \leq \sqrt{2mh \log 2}. \quad (3.13)$$

The values at different sign vectors need not be independent.

Apply this to a densest induced graph $H = G[W]$. Since $L_W = m$ and $R_W \leq \|Q_{\sigma}\|_{\infty}$, $L_W/R_W \geq \sqrt{\rho_G(G)/(2 \log 2)}$. Give all remaining edges arbitrary signs; they do not affect the witness face of Lemma 3.2. The lower bound one follows from $R_W \leq L_W$, and the upper bound is Theorem 3.1.

Call the supremum in (3.12) C . The full center shows $\Gamma \geq C$. Conversely, every induced ratio is a whole-graph ratio after zeroing all coefficients outside the induced subgraph. Thus $\Gamma \leq C$. On the compact set $L_V(a) = 1$, R_V is continuous and strictly positive, so the ratio attains a maximum. Perturb zero entries of a maximizer to nonzero values. Continuity of the full-graph ratio shows that the full-support supremum is also C . This does not assume continuity of c^* as edges disappear or attainment under the full-support restriction. $\square$

Uniform boundedness of all signed factors in a graph family is therefore equivalent to uniform boundedness of its induced density. No closure assumption on the family is needed: fixing outside coordinates exposes a densest subgraph. Global average density is insufficient. Attaching q^2 leaves to one vertex of $K_{q,q}$ produces a connected graph with total edge-to-vertex ratio less than two but induced density at least $q/2$.

The density is unweighted. Scaling all coefficients by a nonzero scalar preserves the gap ratio, whereas a sum of coefficient magnitudes scales. A single edge of arbitrarily small magnitude therefore rules out replacing ρ_G by $\max_W \sum_{e \in E(W)} |a_e|/|W|$. The row estimates remain coefficient-sensitive certificates; a uniform gap bound must use them on every induced subgraph, not only the full graph.

The order may equivalently be expressed using degeneracy or arboricity. If α_G is the minimum number of forests partitioning the edges, then $\rho_G(G) \leq \alpha_G \leq d_G \leq 2\rho_G(G)$. For the middle inequality, label outgoing edges in a degeneracy orientation with distinct labels from 1 to d_G . Each label class is a forest: an undirected cycle would have an earliest vertex with two outgoing edges of that label. The other inequalities follow by counting forest edges and (3.2).

3.4 Positive coefficients, exactness, and finite constants

For positive coefficients a proper coloring with $\chi \geq 2$ colors gives the classical stronger bound

$$c^*(a) \leq \begin{cases} 2 - 2/\chi, & \chi \text{ even,} \\ 2 - 2/(\chi + 1), & \chi \text{ odd.} \end{cases} \quad (3.14)$$

Choose uniformly $\lfloor \chi/2 \rfloor$ colors for one side of a cut. Every edge crosses with probability $p_\chi = 2\lfloor \chi/2 \rfloor \lceil \chi/2 \rceil / [\chi(\chi - 1)]$. Thus the maximum cut is at least $p_\chi L$, and the minimum cut is zero. The same coloring works on every induced subgraph, so Lemma 3.2 gives (3.14). This is the bound of Luedtke et al. [17, Theorem 8 in the authors' manuscript]. In particular, the positive bilinear factor is at most two and is one when its nonzero support is nonempty and bipartite. For the all-positive K_n with $n \geq 2$, the center ratio is $\binom{n}{2} / \lfloor n^2/4 \rfloor$, tending to two. Hence the worst-over-signs quantifier in Theorem 3.5 is essential.

Assume the effective support graph, obtained by deleting zero-coefficient edges, has at least one edge. Then $c^*(a) = 1$ if and only if every cycle of this graph has an even number of positive and an even number of negative edges [4, Theorem 4]. Boland et al. identify this as a consequence of the convex-envelope criterion in Theorem 3.10 of Misener, Smadbeck, and Floudas, *Dynamically generated cutting planes for mixed-integer quadratically constrained quadratic programs and their incorporation into GloMIQO 2*; our attribution to that predecessor is through Boland et al., who give the alternative cut proof used here. Indeed, $R_V = L_V$ exactly when one cut contains all positive edges and no negative edges, and another contains all negative edges and no positive edges. An edge set is a cut exactly when it intersects every cycle evenly. Necessity follows by traversing a cycle. For sufficiency, label vertices by the parity of specified edges on a path from a root in each component; cycle parity makes the label path-independent and supplies the cut. The criterion passes to induced subgraphs, proving sufficiency for $c^* = 1$. Conversely, $c^* = 1$ implies $R_V = L_V$ by Lemma 3.2. In particular, every forest with at least one edge has factor one for every signing. For empty effective support the gaps coincide at zero and $c^* = 0$ by convention.

The best universal constant in $\Gamma(G) \leq K_\rho \sqrt{\rho_G(G)}$ lies in $[2, 4]$: the lower bound is the four-cycle with one negative edge, with density one and factor two. For degeneracy the corresponding interval is $[\sqrt{2}, 4]$. These are finite-parameter questions, distinct from the best constants along sequences with density or degeneracy tending to infinity; the four-cycle does not give a growing-density lower bound.

The bounds imply $c^* \leq 4\sqrt{2}$ for series-parallel graphs by degeneracy, $c^* \leq 4\sqrt{3}$ for planar graphs by density, and $c^* \leq \sqrt{2 \min(p, q)}$ for $K_{p, q}$. For K_n , the maximum-degree bound is $2\sqrt{n - 1}$, compared with $600\sqrt{n}$ in Boland et al. [4, Theorem 2]. These are scalar vertical-gap comparisons, not approximation guarantees for a constrained objective.

3.5 Exact finite full-signing constants

For $n \geq 2$, distinguish the largest full-center ratio among complete-graph signings from the largest ratio over all points:

$$M_n = \max_{a \in \{-1, 1\}^{E(K_n)}} \frac{\binom{n}{2}}{R_V(a)}, \quad F_n = \Gamma_\pm(K_n).$$

All edges here have magnitude one. These maxima differ from the arbitrary-real-coefficient supremum in (3.12).

Proposition 3.6 (Finite exhaustive computation). *The exact values are*

n	3	4	5	6	7
$\min_a R_V(a)$	2	4	4	5	8
M_n	3/2	3/2	5/2	3	21/8
F_n	3/2	3/2	5/2	3	3

Proof. The first row is established by the exhaustive integer computation in Appendix A, which proves completeness of the enumeration, gives executable code, and supplies attaining signings. Dividing $\binom{n}{2}$ by that row gives M_n . Lemma 3.2 gives

$$F_n = \max_{2 \leq k \leq n} M_k.$$

Indeed, each induced support of a full signing is a full signing of some K_k , giving the upper bound. Conversely, any signing of K_k extends to a full signing of K_n by assigning signs to all remaining edges, and its ratio persists on the face fixing the new coordinates to zero. Since $M_2 = 1$, this proves the last row. $\square$

In particular $M_7 < M_6$ while $F_7 = F_6 = 3$. An explicit witness is the K_6 signing in Table 3, extended by a new vertex with all incident coefficients positive. At the point whose six old coordinates are $1/2$ and whose new coordinate is zero, its ratio is three; at the full center its ratio is only $21/10$. The finite computation optimizes precisely these listed full-signing problems; it makes no claim about larger n or the optimal constants for arbitrary real coefficients.

3.6 The relation to Schur-multiplier theory

We record the older implication to separate the density order from the explicit elementary constant. For a real matrix M , its Schur multiplier acts by entrywise multiplication, with norm $\|M\|_{\text{Schur}}$ measured in the Euclidean operator norm. Define the real projective norm

$$\pi(M) = \inf \left\{ \sum_k \|u_k\|_\infty \|v_k\|_\infty : M = \sum_k u_k v_k^T \right\}.$$

The established results of Davidson and Donsig [7, Theorems 1.2 and 2.4] give

$$\|M\|_{\text{Schur}} \leq 2\sqrt{\beta(P)}, \quad \pi(M) \leq K_G \|M\|_{\text{Schur}}, \quad (3.15)$$

for $|M_{ij}| \leq 1$ supported on a pattern P , where K_G is the real Grothendieck constant and

$$\beta(P) = \max_{R, C: |R|+|C|>0} \frac{|P \cap (R \times C)|}{|R| + |C|}.$$

The first bound uses the weighted matrix theorem, which permits continuous row and column bounds. Using the integer pattern theorem instead would introduce rounding. These are established norm inequalities, not statements about sizes of convex extended formulations.

For the symmetric adjacency pattern,

$$|P \cap (R \times C)| \leq |E(G[R \cup C])| + |E(G[R \cap C])| \leq \rho_G(G)(|R| + |C|).$$

An edge is counted at most once unless both endpoints lie in $R \cap C$, when both matrix orientations are counted. Taking $R = C$ equal to a densest set proves $\beta(P) = \rho_G(G)$. For weighted adjacency A , take $M = \text{sign}(A)$, with zeros where A vanishes. The definition of π gives

$$2L_V = \langle A, M \rangle \leq \pi(M) \|A\|_{\infty \rightarrow 1}.$$

For sign vectors u, v , put $p = (u+v)/2$, $q = (u-v)/2$. Symmetry yields $u^\top Av = 2[Q_{a,V}(p) - Q_{a,V}(q)]$. Multiaffinity puts these two values between the vertex extrema on $[-1, 1]^V$, so $\|A\|_{\infty \rightarrow 1} \leq 4R_V$. Consequently

$$L_V \leq 4K_G \sqrt{\rho_G(G)} R_V. \quad (3.16)$$

Apply this on every induced graph to recover the density upper order. The direct random-sign argument supplies the matching bilinear lower order.

Equivalently, let $S(G) = \sup_{a \neq 0} L_V(a) / \|Q_{a,V}\|_\infty$ be the real Sidon constant of the edge-supported degree-two sign characters, with $S(G) = 0$ for edgeless graphs. Because Q has mean zero, $\|Q\|_\infty / 2 \leq R_V \leq \|Q\|_\infty$ and hence $S(G) \leq \Gamma(G) \leq 2S(G)$. For bipartite graphs, flipping all signs in one part negates Q , making its range symmetric and giving $S(G) = \Gamma(G)$. The constant four improves the particular transfer (3.16); it is not asserted to be the best constant available in all prior literature.

4 Universal bounds for positive multilinear functions

Throughout this section the factors are the positive monomials, with equal supports collected. Let R_d be the supremum of T/H over all dimensions, positive multilinear polynomials of maximum degree at most d , finite nonnegative boxes, and points with $H > 0$. Let C_n be the corresponding supremum in n variables, with no additional degree restriction. Fixed coordinates and affine summands may be removed. Proposition 2.4 shows that both suprema can equivalently be taken on the unit cube: expansion does not increase dimension or degree and can only increase the termwise gap. All logarithms without a subscript are natural.

Theorem 4.1 (Sharp leading growth). *As the indicated integer parameter tends to infinity,*

$$R_d \sim \frac{\log d}{\log \log d}, \quad C_n \sim \frac{\log n}{\log \log n}. \quad (4.1)$$

For every $d \geq 2$, put $\Lambda = \max\{16, 1 + \log d\}$ and $c_0 = 1 - e^{-1}$. The explicit finite bound

$$T \leq \frac{24\Lambda}{c_0 \log \Lambda} H \quad (4.2)$$

holds on every finite nonnegative box. On the unit cube the upper bounds are witnessed by distributions defined from the complete marginal vector and degree allowance, independently of the supports and coefficients.

We first exhibit the obstruction to a constant bound, then construct a common distribution that matches its order. The lower family disproves Conjecture 1 in the author manuscript of Luedtke et al. [17, p. 22], which asks for a uniform constant for positive multilinear functions on nonnegative boxes. The signed bilinear examples of Section 3.6 concern a different coefficient class.

4.1 A sparse multiscale obstruction

Theorem 4.2 (Exact dyadic hull gap). *For every integer $L \geq 2$, there is a unit-coefficient polynomial in $n_L = 2^L + L$ variables, with $2^{L+1} - 2$ monomials, maximum degree $d_L = 2^{L-1} + 1$, and an interior point such that*

$$T = L, \quad H = s + (L - s)2^{-s}, \quad (4.3)$$

where $s \in \{1, \dots, L - 1\}$ satisfies

$$(L - s + 1)2^{-(s+1)} \leq 1 \leq (L - s + 2)2^{-s}. \quad (4.4)$$

Either adjacent choice at a cutoff equality gives the same value. The total number of variable occurrences is $L2^L + 2^{L+1} - 2$.

Proof. Put $m = 2^L$. Use leaf variables $z_1, \dots, z_m$ and anchors $a_1, \dots, a_L$. For each j , partition the leaves into 2^j blocks of size $k_j = 2^{L-j}$, and define

$$f_L(a, z) = \sum_{j=1}^L \sum_{B \in \mathcal{P}_j} a_j \prod_{i \in B} z_i, \quad a_j = 2^{-j}, \quad z_i = 1 - 1/m. \quad (4.5)$$

The upper and lower envelopes of each monomial at this point are 2^{-j} and $\max(0, 2^{-j} - k_j/m) = 0$. Thus $T = \text{cav } f_L = L$. The support and occurrence counts follow by summing 2^j and $2^j(k_j + 1)$, respectively.

For an arbitrary binary law with these means, let R count failed leaves and N_j count blocks hit by a failure. Then $\mathbb{E}R = 1$ and $N_j \leq \min(2^j, R)$. The deficiency representation gives

$$H = \max \mathbb{E} \sum_j A_j N_j \leq \sup \mathbb{E} \sum_j A_j \min(2^j, R), \quad \mathbb{E}A_j = 2^{-j}. \quad (4.6)$$

For a fixed law of R , an anchor of mass u maximizes its expectation against any increasing function of R by selecting the upper u -tail, splitting an atom if necessary. Indeed, moving selected mass from a smaller value to a larger unselected value cannot reduce the objective. A common nonincreasing quantile $r(t)$, $0 < t < 1$, therefore makes every anchor $A_j = \mathbf{1}\{t \leq 2^{-j}\}$ optimal simultaneously. Consequently the right side of (4.6) equals the supremum of $\int_0^1 S_{l(t)}(r(t)) dt$, where

$$S_l(r) = \sum_{j=1}^l \min(2^j, r), \quad w_l = \mathbb{P}(l(t) = l) = \begin{cases} 2^{-l-1}, & 0 \leq l < L, \\ 2^{-L}, & l = L. \end{cases}$$

Here r is integer valued, bounded by m , and has integral one.

Define $B_q = (L - q + 2)2^{-q}$. The sequence decreases strictly, with $B_1 \geq 1 \geq B_L$, so an s as in (4.4) exists. For every $r \geq 0$,

$$S_l(r) \leq sr + F_l(s), \quad F_l(s) = \begin{cases} 0, & l \leq s, \\ 2^{l-s+1} - 2, & l > s. \end{cases} \quad (4.7)$$

For $l \leq s$, bound each term by r . For $l > s$, cap the first $l - s$ terms at 2^j and the remaining s terms at r . Integrating yields

$$H \leq s + \sum_l w_l F_l(s) = s + (L - s)2^{-s}.$$

To attain this bound, choose l with probabilities w_l , let $A_j = \mathbf{1}\{j \leq l\}$, and for $q \in \{s, s+1\}$ put

$$R_l^{(q)} = \begin{cases} 0, & l < q, \\ 2^{l-q+1}, & l \geq q. \end{cases}$$

Each anchor has its required mean and $\mathbb{E}R^{(q)} = B_q$. Mix $q = s$ and $q = s+1$ with respective probabilities $(1 - B_{s+1})/(B_s - B_{s+1})$ and $(B_s - 1)/(B_s - B_{s+1})$. This gives $\mathbb{E}R = 1$. For $l > s$ the two failure counts are the endpoints of $[2^{l-s}, 2^{l-s+1}]$, where (4.7) is an equality. For $l = s$ they are 0, 2, also equality points, and for $l < s$ both are zero.

It remains to realize the hit counts, a step requiring nested partitions. Label leaves by L -bit strings, with level- j blocks defined by the first j bits. The first R strings in bit-reversal order have exactly $\min(2^j, R)$ different prefixes of length j : their prefixes are the reversed residues $0, \dots, R-1$ modulo 2^j . Declare these leaves failed and apply a uniform bitwise XOR shift to every label. The shift preserves all hit counts and gives each leaf failure probability R/m , conditional on R . Averaging gives failure mean $1/m$ for every leaf. The constructed law therefore attains the bound in the original polynomial. $\square$

The cutoff conditions imply $s = \log_2 L + O(1)$ and $(L-s)2^{-s} = O(1)$. Thus the ratio in (4.3) is asymptotic to $L/\log_2 L$. The exact formula uses nested dyadic partitions. For arbitrary equal-size partitions the same surrogate proves the weaker but still divergent lower bound

$$\frac{T}{H} \geq \frac{L}{5 + \log_2(1 + (L-1)\log 2)}; \quad (4.8)$$

Appendix B gives its distinct integral proof and the dense symmetric predecessor.

The construction may be made homogeneous of degree d_L : introduce $d_L - 2$ padding variables and multiply a degree- k term by the first $d_L - k$ of them. At padding means one both gaps agree exactly with those above, since all representing laws fix the padding coordinates. Moving them sufficiently close to one makes all means interior with ratio arbitrarily close to (4.3). This adds $O(n_L)$ variables and keeps $O(n_L)$ monomials, but the occurrence count after padding is $O(n_L^2)$, not the original $O(n_L \log n_L)$.

4.2 Harmonic failure distributions

Call a coordinate low when $x_i \leq 1/2$ and high otherwise. Independence handles the terms with zero or at least two low coordinates by Lemma 2.3. For the remaining terms we require failures at several probability scales simultaneously.

Fix a real $M \geq d-1$, and put

$$\Lambda = 1 + \log M, \quad \eta = (d-1)/M \in (0, 1]. \quad (4.9)$$

We use η for this cutoff parameter, to distinguish it from graph density and physical box aspect ratio. Both laws below draw one uniform $U \in (0, 1)$ and set every low coordinate to $\mathbf{1}\{U \leq x_i\}$. In the *threshold law*, a high coordinate with failure mean $p = 1 - x_i$ fails exactly when $U \leq p$. In the *harmonic law*, a coordinate with $p = 0$ never fails; otherwise set

$$h_p = \min(Mp, 1), \quad A_p = 1 + \log(h_p/p), \quad q_p(t) = \frac{\min(1, p/t)}{A_p} \mathbf{1}\{t \leq h_p\}. \quad (4.10)$$

Conditional on $U = t$, high coordinates fail independently with probabilities $q_p(t)$. Since $M \geq 1$, $1 \leq A_p \leq \Lambda$ and $h_p \geq p$. In particular these probabilities are valid, and

$$\int_0^1 q_p(t) dt = \frac{p + p \log(h_p/p)}{A_p} = p.$$

Thus the two laws preserve the full marginal vector, including boundary means, and require no information about a particular term.

Lemma 4.3 (Complete harmonic guarantee). *Consider a term with one low anchor of mean u , and high failure means $p_1, \dots, p_r$, where $r \leq d-1$. Put $t_e = \min(u, \sum_j p_j) > 0$ and $z =$*

$\min(\max_j p_j/t_e, 1)$. The threshold and harmonic deficiencies are at least zt_e and $J_{\Lambda,\eta}(z)t_e$, respectively, where

$$F_{\Lambda,\eta}(v) = 1 - \exp\left(-\frac{1-\eta v}{\Lambda v}\right), \quad J_{\Lambda,\eta}(z) = \int_z^1 F_{\Lambda,\eta}(v) dv, \quad F_{\Lambda,\eta}(0) = 1. \quad (4.11)$$

Proof. Threshold deficiency is $\min(u, \max_j p_j) \geq zt_e$. If $\max_j p_j \geq t_e$, the harmonic assertion is nonnegativity. Otherwise integrate over $\max_j p_j \leq t \leq t_e$. The anchor succeeds there and all $p_j \leq t$. An inactive coordinate has $p_j < t/M$, so the inactive failure mass is at most $rt/M \leq \eta t$. Active mass is consequently at least $\sum_j p_j - \eta t \geq t_e - \eta t$. On active coordinates, $q_{p_j}(t) \geq p_j/(\Lambda t)$. Conditional independence gives union failure probability at least $1 - \exp(-(t_e - \eta t)/(\Lambda t))$. Substitution $v = t/t_e$ proves the claim. Zero term gaps require no division and satisfy every asserted nonnegative bound automatically. $\square$

Theorem 4.4 (Optimized finite harmonic bound). *For the parameters in (4.9), let ζ be the unique solution of $J_{\Lambda,\eta}(\zeta) = \zeta$ in $(0, 1)$. Then*

$$R_d \leq \frac{1}{\zeta} + \frac{2}{c_0}. \quad (4.12)$$

If $w > 0$ solves $we^w = \Lambda e^{-\eta}$ and $w \geq 1$, then

$$\zeta \geq \frac{w - 1/(2w)}{\Lambda}, \quad R_d \leq \frac{\Lambda}{w - 1/(2w)} + \frac{2}{c_0}. \quad (4.13)$$

The parameter M can be optimized over $[d-1, \infty)$ in either bound. For $N = 1 + \log(d-1)$, choosing $M = (d-1)N$ gives, as $d \rightarrow \infty$,

$$R_d \leq \frac{N}{\log N - \log \log N + o(1)} + \frac{2}{c_0}. \quad (4.14)$$

This is an upper certificate, with no matching second-order lower assertion.

Proof. F is continuous and strictly decreasing, positive on $(0, 1)$, and nonnegative at one. Thus J is strictly decreasing and strictly convex, and $J(z) - z$ has opposite signs at zero and one. Its unique root is ζ . Set $a = F(\zeta) \in (0, 1)$. The convex tangent inequality gives

$$J(z) + az \geq J(\zeta) + a\zeta = (1+a)\zeta.$$

Mix the harmonic and threshold laws with weights $1/(1+a)$ and $a/(1+a)$. Lemma 4.3 gives deficiency at least ζt_e for every one-low term. This mixture is optimal for the two *scalar guarantees* $J(z)$ and z : at $z = \zeta$ every convex combination equals ζ . It does not prove optimality over all actual couplings or imply simultaneous attainability of these lower curves by an extremal polynomial. Mix this law and independence with weights proportional to $1/\zeta$ and $2/c_0$. Each term receives at least $t_e/(1/\zeta + 2/c_0)$ from its appropriate component; the other contributions are nonnegative. Proposition 2.2 and then Proposition 2.4 prove (4.12).

For the finite certificate, use $x - x^2/2 \leq 1 - e^{-x} \leq x$ for $x \geq 0$ and, for $0 < z \leq 1$,

$$\int_z^1 (1/v - \eta)^2 dv = 1/z - 1 + 2\eta \log z + \eta^2(1-z) \leq 1/z.$$

The last inequality follows from $2\eta \log z \leq 0$ and $-1 + \eta^2(1-z) \leq 0$. For $0 < A \leq \Lambda$ this gives

$$\Lambda J(A/\Lambda) \geq \log \Lambda - \log A - \eta + \eta A/\Lambda - 1/(2A) \geq \log \Lambda - \log A - \eta - 1/(2A). \quad (4.15)$$

Take $A = w - 1/(2w)$. Since $w \geq 1$, $A \geq 1/2$ and $A < \Lambda$. Using $w + \log w = \log \Lambda - \eta$,

$$A + \log A + \frac{1}{2A} - (w + \log w) = \log \left(1 - \frac{1}{2w^2}\right) + \frac{1}{4Aw^2} \leq 0.$$

The inequality follows from $\log(1 - t) \leq -t$ and $A \geq 1/2$. Hence $J(A/\Lambda) \geq A/\Lambda$; monotonicity of $J(z) - z$ proves (4.13).

For the chosen cutoff, $\Lambda = N + \log N$ and $\eta = 1/N$. The defining identity for w first gives $\log \Lambda - \eta - \log \log \Lambda \leq w \leq \log \Lambda$ for all sufficiently large N . Thus $w/\log N \rightarrow 1$, and substitution back gives $w = \log N - \log \log N + o(1)$. Since $(\log N)^2/N \rightarrow 0$,

$$\frac{N}{\Lambda} \left(w - \frac{1}{2w}\right) = \log N - \log \log N + o(1).$$

This proves (4.14). $\square$

Proof of Theorem 4.1. For its original explicit finite bound, take $\Lambda = \max\{16, 1 + \log d\}$ and $M = e^{\Lambda-1} \geq d$. For a one-low term, put $t_e = \min(u, \sum_j p_j) > 0$. If $\max_j p_j \geq t_e/\sqrt{\Lambda}$, threshold deficiency is at least $t_e/\sqrt{\Lambda} \geq t_e \log \Lambda/\Lambda$; here $\log \Lambda \leq \sqrt{\Lambda}$ follows by maximizing $(\log t)/\sqrt{t}$. Otherwise integrate the harmonic law over $[t_e/\sqrt{\Lambda}, t_e/2]$. Inactive failure mass is at most t , so active mass is at least $t_e/2$. The conditional union probability is at least $c_0 t_e/(2\Lambda t)$, using $1 - e^{-z} \geq c_0 \min(1, z)$. The harmonic deficiency is at least

$$\frac{c_0 t_e}{4\Lambda} (\log \Lambda - 2 \log 2) \geq \frac{c_0 t_e \log \Lambda}{8\Lambda}.$$

An equal mixture of independence, threshold and harmonic laws now gives (4.2), since independence supplies at least $c_0 t_e/2$ on the other two classes. Every distribution is global.

Theorem 4.4 implies $R_d \leq (1 + o(1)) \log d / \log \log d$. For the reverse inequality choose the largest L with $2^{L-1} + 1 \leq d$ in Theorem 4.2. Its degree differs from d by a factor at most two, asymptotically, and its ratio is asymptotic to $L/\log_2 L$. For dimension, $C_n \leq R_n$; choose the largest L with $2^L + L \leq n$ and add unused coordinates. Again the dimensions differ by at most a constant factor. These comparisons prove both equivalences in (4.1) for all integers tending to infinity, not merely for a dyadic subsequence. $\square$

For comparison, the earlier leading-constant mixture has a distinct explicit parameterization. Set $\Lambda = \max\{e^6, 1 + \log d\}$, $M = e^{\Lambda-1}$, $b = \log \Lambda$, $a = \Lambda/b^2$, $\beta = 1/b$, and

$$h = \frac{(1 - 1/b)(1 - 1/(2b^2))(b - 3 \log b)}{\Lambda}. \quad (4.16)$$

Here $h > 0$ because $b \geq 6$ and $b - 3 \log b > 0$. If $\max p_j \geq t_e/a$, threshold deficiency is at least t_e/a . Otherwise, on $[t_e/a, \beta t_e]$, active mass is at least $(1 - \beta)t_e$ and the lower exponent $(1 - \beta)t_e/(\Lambda t)$ is at most $1/b^2$. Integrating $1 - e^{-z} \geq z(1 - 1/(2b^2))$ gives harmonic deficiency at least $h t_e$. Weights proportional to $1/h, a, 2/c_0$ on harmonic, threshold, and independence give $R_d \leq 1/h + a + 2/c_0$, also with leading constant one. The full-curve theorem retains more of the same distributions' information and improves this certificate's lower-order loss.

A useful interpretation is simultaneous interpolation between Fréchet bounds. The global law with $Z = 1/\zeta + 2/c_0$ satisfies, for every $e \subseteq [n]$ with $2 \leq |e| \leq d$,

$$\mathbb{E} \prod_{i \in e} X_i \leq (1 - Z^{-1}) \min_{i \in e} x_i + Z^{-1} \max(0, \sum_{i \in e} x_i - |e| + 1). \quad (4.17)$$

The best fraction guaranteed for every marginal vector is asymptotic to $\log \log d / \log d$: the theorem gives the lower guarantee, and summing any proposed guarantee over the dyadic example gives the reverse order and constant. No convex-envelope evaluation algorithm is inferred.

4.3 Removing coefficients without changing the degree supremum

Theorem 4.5 (Unit coefficients and interior homogeneous examples). *For each fixed $d \geq 2$, the supremum R_d is unchanged by requiring unit coefficients, degree exactly d for every term, and strictly interior unit-cube evaluation points simultaneously. Dimension and support size are unrestricted in this assertion.*

Proof. By positive expansion it suffices to approximate any unit-cube example. Delete affine terms and homogenize with $d - 2$ padding variables as above. At padding means one the gaps are unchanged. Continuity of the finite polyhedral envelopes (Lemma 2.1) permits an interior perturbation preserving any strict target-ratio inequality. Normalize by the largest coefficient. We now have one fixed homogeneous polynomial $f = \sum_{e \in E} a_e \prod_{i \in e} x_i$, $0 < a_e \leq 1$, with n variables, s distinct supports, interior means x , and $H(f; x) > 0$.

Replace coordinate i by m clones $y_{i1}, \dots, y_{im}$, and set

$$\bar{y}_i = m^{-1} \sum_{r=1}^m y_{ir}, \quad F_m(y) = m^d f(\bar{y}), \quad y_{ir} = x_i.$$

There are sm^d distinct candidate clone monomials: each uses one clone from each of its d distinct original groups. At the repeated point,

$$\text{vex } F_m = m^d \text{vex } f, \quad \text{cav } F_m = m^d \text{cav } f, \quad T(F_m) = m^d T(f), \quad H(F_m) = m^d H(f). \quad (4.18)$$

Indeed, any binary clone law induces random group averages $q \in [0, 1]^n$ with $\mathbb{E}q = x$. Conditional independent rounding of the original variables with means q_i has expected polynomial $f(q)$, so every normalized clone expectation is an original feasible expectation. Conversely, setting all clones in a group equal to its original binary coordinate embeds every original law. This proves equality of the entire feasible expectation intervals, and hence of both envelopes. Each clone term has its original list of means, proving the termwise identity.

Independently retain each candidate term arising from e with probability a_e and give it coefficient one; call the result G_m . For N_0 independent random variables with range lengths at most one, the bounded-sum inequality of Hoeffding [14, Theorem 2] gives

$$\mathbb{P}(|\sum_j (Y_j - \mathbb{E}Y_j)| > t) \leq 2e^{-2t^2/N_0}. \quad (4.19)$$

For completeness, if $Y \in [a, b]$, the second derivative of $\log \mathbb{E}e^{\lambda Y}$ is its variance under exponential tilting, at most $(b - a)^2/4$: its squared distance from the interval midpoint is at most that quantity. Integrating twice gives $\mathbb{E}e^{\lambda(Y - \mathbb{E}Y)} \leq e^{\lambda^2(b-a)^2/8}$. Multiplication over independent variables, Markov's inequality, and optimization at $\lambda = 4t/N_0$ prove each tail in (4.19). Thus no concentration result is needed without its hypotheses or proof.

Apply this inequality with $N_0 = sm^d$ at each of the 2^{nm} clone vertices, and once to the termwise gap at the repeated point. The latter is a sum of the same retention indicators with fixed weights in $[0, 1]$. A union bound gives failure probability at most

$$2(2^{nm} + 1) \exp(-2t^2/(sm^d)) \quad (4.20)$$

for the two simultaneous assertions

$$\max_{v \in \{0,1\}^{nm}} |G_m(v) - F_m(v)| \leq t, \quad |T(G_m) - T(F_m)| \leq t.$$

Take $t = Km^{(d+1)/2}$ with $K^2 > sn \log 2/2$. The failure probability tends to zero as $m \rightarrow \infty$, with f, n, s, d fixed. Uniform vertex error bounds the error of each envelope at every marginal vector by t , by comparing expectations under all feasible laws. Thus the hull-gap error is at most $2t$. Since $t/m^d \rightarrow 0$ for $d \geq 2$, successful samples' normalized gaps converge to those of f , and their ratios converge because its hull gap is positive. This proves equality of suprema, not preservation at a fixed dimension or existence of a small sampled support. $\square$

5 Cubic gaps and the role of unequal means

The divergence in Theorem 4.1 requires increasing degree. Degree three already distinguishes positive multilinear functions from the positive bilinear factor-two theorem, however. We prove

$$\frac{1610000}{743033} \leq R_3 \leq \frac{31}{12}. \quad (5.1)$$

The lower endpoint is a bound on a supremum supplied by a sequence of certificates, not a claimed attained finite optimum. We also give exact finite hull values and a homogeneous unit-coefficient example with only 52 variables. The exact value of R_3 remains open.

5.1 Endpoint orientation and a cubic mixture

One particularly simple global law applies in every degree. Draw $U \in [0, 1]$ uniformly and independently orient each coordinate left or right with equal probability. A left coordinate is the indicator of $[0, x_i]$ and a right coordinate of $[1 - x_i, 1]$. Call this law O . Every coordinate has mean x_i .

Proposition 5.1 (Endpoint-orientation bound). *For maximum degree $d \geq 2$, on every nonnegative box,*

$$T \leq \frac{d-1}{1-2^{-(d-1)}} H. \quad (5.2)$$

In particular the constant is $8/3$ for cubics and two for bilinear terms.

Proof. On the unit cube, fix a degree- $(k+1)$ term with minimum mean u at its anchor. Conditional on the anchor orientation, a same-oriented coordinate imposes no further restriction on its success interval. An opposite coordinate excludes an interval of length $a_j = \min(u, 1 - x_j)$ at the opposite end. These excluded intervals are nested. If $a_{(1)} \geq \dots \geq a_{(k)}$, the first selected index has probability 2^{-j} of being j , and hence

$$\mathbb{E}_O D_e = \sum_{j=1}^k 2^{-j} a_{(j)} \geq \frac{1-2^{-k}}{k} \sum_j a_j \geq \frac{1-2^{-k}}{k} \min(u, \sum_j (1-x_j)).$$

The first inequality follows by summing $(2^{-i} - 2^{-j})(a_{(i)} - a_{(j)}) \geq 0$ over pairs, and the second by $\sum_j \min(u, p_j) \geq \min(u, \sum_j p_j)$. The fraction $(1-2^{-k})/k$ decreases with k , since it is the average of the first k entries of $1/2, 1/4, \dots$. Sum the simultaneous guarantees and use Proposition 2.4. $\square$

Theorem 5.2 (Three-law cubic bound). *Every positive multilinear polynomial of degree at most three on a finite nonnegative box satisfies $T \leq (31/12)H$. On the unit cube, the global law*

$$P = \frac{18}{31}O + \frac{6}{31}I + \frac{7}{31}B \quad (5.3)$$

guarantees deficiency at least $12t_e/31$ for every quadratic or cubic term. Here I is independent rounding. For B , draw one uniform U , set each low coordinate to $\mathbf{1}\{U \leq x_i\}$, and, independently conditional on U , let each high coordinate fail with probability $1/2$ when $U \leq 2(1 - x_i)$ and never otherwise. The weights are optimal among fixed mixtures of these three laws for a uniform termwise guarantee.

Proof. The definition of B gives each high coordinate failure mean $1 - x_i$; thus all three laws have the prescribed means. Consider sorted cubic means $u \leq v \leq w$, and set $t = \min(u, 2 - v - w)$. Terms with $t = 0$ need no estimate. Write D_O, D_I, D_B for their expected deficiencies.

If $u \leq v \leq 1/2$, then $t = u$. Opposite anchor and second-coordinate orientations give $D_O \geq u/2$, while $D_I = u(1 - vw) \geq u/2$. These two contributions alone give $12t/31$.

If exactly one coordinate is low, set $a = 1 - v \geq b = 1 - w$. Directly integrating the nested exclusion and failure intervals gives

$$D_O = \frac{1}{2} \min(u, a) + \frac{1}{4} \min(u, b), \quad D_B = \frac{1}{2} \min(u, 2a) + \frac{1}{4} \min(u, 2b).$$

We verify $18D_O + 7D_B \geq 12 \min(u, a + b)$ in all cases. If $a \leq u/2$, the left side is $16a + 8b \geq 12(a + b)$. If $b \geq u/2$, it is at least $18(3u/8) + 7(3u/4) = 12u$. If $b \leq u/2 \leq a \leq u$, it is $9a + 8b + 7u/2$. When $a + b \leq u$, use $3a + 4b \leq 4u - a \leq 7u/2$; when $a + b \geq u$, use $9a + 8b \geq 8u + a \geq 17u/2$. Finally, if $b \leq u/2$ and $a \geq u$, the expression is $25u/2 + 8b \geq 12u$. These cases exhaust $a \geq b \geq 0$, including their boundaries. The I contribution is nonnegative.

If all coordinates are high, put $c = 1 - u \geq a = 1 - v \geq b = 1 - w$. Integrating the B union probabilities $7/8, 3/4, 1/2$ over intervals of lengths $2b, 2(a - b), 2(c - a)$ gives total failure probability $c + a/2 + b/4$. Subtracting the anchor failure probability c yields

$$D_B = D_O = a/2 + b/4 \geq 3(a + b)/8 \geq 3t/8.$$

Independence gives $D_I = u(a + b - ab) \geq 7t/16$. Indeed, set $s = a + b \leq 1$ and use $ab \leq s^2/4$. For $s \leq u$ the normalized expression is at least $u(1 - s/4) \geq u(1 - u/4) \geq 7/16$. For $s \geq u$ it is at least $s - s^2/4 \geq u - u^2/4 \geq 7/16$. Both comparisons use $u > 1/2$ and monotonicity on the unit interval. Thus the mixture fraction is at least $(25/31)(3/8) + (6/31)(7/16) = 12/31$.

For sorted quadratic means $u \leq v$, set $t = \min(u, 1 - v)$; always $D_O = t/2$. If both means are low, $D_I \geq t/2$. If exactly one is low, $D_B = \min(u, 2(1 - v))/2 \geq t/2$. If both are high, $D_B = t/2$ and $D_I = ut \geq t/2$. The resulting mixture fractions are at least $12/31, 25/62$, and $1/2$. This proves the simultaneous guarantee. Summation and positive expansion give the claimed box bound.

To check optimality in the stated family, let the arbitrary weights of (O, I, B) be (w, r, v) with $w + r + v = 1$. The following test means give the listed exact or limiting normalized deficiencies:

means	D_O/t	D_I/t	D_B/t
$(u, 1/2), 0 < u \leq 1/2$	$1/2$	$1/2$	0
$(\varepsilon, 1 - \varepsilon^2, 1 - \varepsilon^2), \varepsilon \downarrow 0$	$3/8$	0	$3/4$
$(u, 1 - u/2, 1 - u/2), u \downarrow 1/2$ from above	$3/8$	$7/16$	$3/8$

In the second row $t = 2\varepsilon^2$ for small ε and the independent fraction is $\varepsilon - \varepsilon^3/2$. In the last row $1/2 < u < 2/3$ ensures $t = u$ and the independent fraction is $u - u^2/4$. Therefore a universal fraction α obeys

$$\alpha \leq \frac{1}{2} - \frac{v}{2}, \quad \alpha \leq \frac{3}{8} + \frac{3v}{8} - \frac{3r}{8}, \quad \alpha \leq \frac{3}{8} + \frac{r}{16}.$$

Average these inequalities with weights $3/31, 4/31, 24/31$. The variable coefficients cancel, giving $\alpha \leq 12/31$. This conclusion concerns fixed-mixture termwise guarantees; it does not identify R_3 , or exclude a stronger coefficient-dependent analysis. $\square$

5.2 A scalar certificate and the strongest limiting lower bound

For a set Q of variables, write $E_k(Q) = \sum_{S \subseteq Q, |S|=k} \prod_{i \in S} x_i$. At a binary vertex this is $\binom{K}{k}$ when K counts successes in Q . Count-polynomial identities below are vertex identities, not substitutions defining the multilinear functions at continuous points.

Lemma 5.3 (Cubic scalar minorant with positive slack). *On $[0, 1]^3$, let*

$$\begin{aligned} F(a, b, c) &= 6c^3 + 27bc^2 + 18b^2 + 30ac + 20ab + 9a^2, \\ \ell(a, b, c) &= 37a + \frac{79}{2}b + 38c - \frac{103}{3}. \end{aligned}$$

Then $F \geq \ell + \delta_$, where $\delta_* = 901/120000$.*

Proof. For fixed c , the Hessian of $h = F - \ell$ in (a, b) is $\begin{pmatrix} 18 & 20 \\ 20 & 36 \end{pmatrix}$, positive definite with determinant 248. For $0 \leq c \leq 3/10$, set $a = 1$ and $b_*(c) = 13/24 - 3c^2/4 \in (0, 1)$. There $\partial_b h = 0$ and $\partial_a h = -49/6 + 30c - 15c^2 \leq -31/60 < 0$. The convex first-order inequality proves that this is the constrained minimum on the (a, b) square: $(a - 1)\partial_a h \geq 0$ for $a \leq 1$. Its value is

$$p(c) = \frac{-972c^4 + 576c^3 + 1404c^2 - 768c + 101}{96}.$$

For $3/10 \leq c \leq 1$, minimizing over unrestricted real a, b is a valid lower bound. Completing the square gives

$$r(c) = \frac{-78732c^4 + 212256c^3 - 203796c^2 + 82032c - 11275}{2976}.$$

For example, with $v = 30c - 37$, $w = 27c^2 - 79/2$ and $q = 6c^3 - 38c + 103/3$, this value is $q - (18v^2 - 20vw + 9w^2)/248$, which verifies the elimination directly.

Table 1 supplies exact degree-four Bernstein representations. A row with interval $[l, u]$, denominator D , and numerators $v_0, \dots, v_4$ means

$$p(c) \text{ or } r(c) = D^{-1} \sum_{i=0}^4 v_i \binom{4}{i} t^i (1-t)^{4-i}, \quad t = (c-l)/(u-l).$$

These are rational polynomial identities, verified by expanding in t . The five intervals cover the required ranges. The Bernstein basis is nonnegative and sums to one, and the minimum of all 25 coefficients v_i/D is $1802/240000 = \delta_*$. Thus both lower polynomials are at least δ_* wherever used, proving the claim. $\square$

Polynomial	Interval	D	$(v_0, v_1, v_2, v_3, v_4)$
p	$[0, 1/5]$	60000	(63125, 39125, 20975, 9395, 4133)
p	$[1/5, 3/10]$	240000	(16532, 6008, 1802, 3788, 11597)
r	$[3/10, 1/2]$	7440000	(215357, 1285415, 1434125, 1250375, 1008125)
r	$[1/2, 3/4]$	190464	(25808, 18056, 7964, 9230, 15869)
r	$[3/4, 1]$	190464	(15869, 22508, 34520, 45920, 31040)

Table 1: Exact scalar positivity certificate for Lemma 5.3.

Theorem 5.4 (Analytic cubic lower family). *For each multiple $m \geq 9$ of nine, take three groups U, V, W of m variables and write A, B, C for their sums. Define*

$$f_m = 2E_3(W) + 3BE_2(W) + 2mE_2(V) + \frac{5m}{3}AC + \frac{10m}{9}AB + mE_2(U). \quad (5.4)$$

At group means $1/4, 1/2, 3/4$, respectively,

$$\frac{T(f_m)}{H(f_m)} \geq \frac{161/4 - 135/(4m) + 6/m^2}{223/12 - \delta_* + 135/(4m) + 9/m^2}. \quad (5.5)$$

Consequently $R_3 \geq 1610000/743033$.

Proof. All coefficients in (5.4) are positive integers. At binary counts put $a = A/m, b = B/m, c = C/m$. Expansion of binomial coefficients gives the exact identity

$$\frac{18f_m}{m^3} = F(a, b, c) - \frac{18c^2 + 27bc + 18b + 9a}{m} + \frac{12c}{m^2} \geq F(a, b, c) - 72/m. \quad (5.6)$$

Every feasible binary law has expected normalized counts $(1/4, 1/2, 3/4)$. Lemma 5.3 therefore gives

$$\frac{18}{m^3} \text{vex } f_m \geq 139/6 + \delta_* - 72/m.$$

Counting supports and using (2.6) yields

$$\frac{18}{m^3} \text{cav } f_m = 167/4 - 153/(4m) + 9/m^2, \quad (5.7)$$

$$\frac{18}{m^3} T(f_m) = 161/4 - 135/(4m) + 6/m^2. \quad (5.8)$$

For clarity, the six scaled upper contributions in the displayed family order are

$$\frac{9}{2} - \frac{27}{2m} + \frac{9}{m^2}, \quad \frac{27}{2} - \frac{27}{2m}, \quad 9 - \frac{9}{m}, \quad \frac{15}{2}, \quad 5, \quad \frac{9}{4} - \frac{9}{4m}.$$

Only $2E_3(W)$ has a positive termwise lower envelope; its scaled value is $3/2 - 9/(2m) + 3/m^2$. This proves the two exact formulas. Subtraction bounds the scaled hull gap by the denominator in (5.5). It is positive for $m \geq 9$, and the actual hull gap is positive as well: independence gives strictly smaller expectations than common-threshold rounding for every positive nonlinear term at interior means. Division is therefore valid. Taking arbitrarily large multiples of nine yields

$$R_3 \geq \frac{161/4}{223/12 - 901/120000} = \frac{4830000}{2229099} = \frac{1610000}{743033}.$$

Dropping the positive slack recovers the weaker $483/223$; the same certificate proves the stronger reduced fraction above. Neither minorant is asserted to be attained in expectation, and neither limiting bound asserts an attained finite gap ratio or convergence of the actual family ratios to that lower endpoint. At $m = 36$, even dropping δ_* from (5.5) gives $T/H \geq 16985/8436 > 2$; retaining the positive slack makes this finite bound strictly stronger. $\square$

5.3 Exact finite cubic witnesses

Finite examples permit a stronger type of statement: matching primal and dual certificates determine both envelopes exactly. The three-group polynomials in Table 2 have group sizes m and means $(1/4, 1/2, 3/4)$, and take the form

$$f = c_1 E_3(W) + c_2 B E_2(W) + c_3 E_2(V) + c_4 AC + c_5 AB + c_6 E_2(U). \quad (5.9)$$

Proposition 5.5 (Exact finite values). *The envelope and ratio values in Table 2 are exact. The 18- and 24-variable polynomials have 212 and 464 monomials, respectively. Complete rational primal and dual certificates, and a short executable integer/rational checker, appear in Appendix C.*

Proof. At binary counts $A, B, C \in \{0, \dots, m\}$ the polynomial is

$$\Phi_m(A, B, C) = c_1 \binom{C}{3} + c_2 B \binom{C}{2} + c_3 \binom{B}{2} + c_4 AC + c_5 AB + c_6 \binom{A}{2}.$$

The appendix certifies an affine minorant at every count state and a probability distribution of count states with means $(m/4, m/2, 3m/4)$ attaining it. Conditional uniform subsets in each group realize every individual marginal, so these give matching lower-envelope certificates. The upper and termwise lower values follow directly from the six orbit sizes and monomial bounds in the appendix. Subtracting yields the ratios. $\square$

n	$(c_1, \dots, c_6)$	$\text{cav } f$	$\sum_e \text{vex } f_e$	$\text{vex } f$	T/H
18	(2, 3, 9, 10, 7, 7)	1647/4	10	2750/13	20891/10411
24	(2, 3, 13, 12, 8, 7)	971	28	3572/7	6601/3225
192	(2, 3, 120, 105, 70, 63)	587944	20832	34172072/105	7443345/3445256

Table 2: Exact cubic witnesses, all with ratio above two.

The 24-variable example also gives a quantitative homogeneous interior variant. Write it as $f = P + Q$, where P is cubic and Q quadratic, and put $g(x, z) = P(x) + zQ(x)$ with $z = 999/1000$. All 25 coordinates are interior and all 464 monomials are cubic, with integer coefficients at most 13. Their upper and lower termwise values are unchanged, so $\text{cav } g = 971$ and $T(g) = 943$. The sum of quadratic coefficients is

$$13 \binom{8}{2} + 12 \cdot 8^2 + 8 \cdot 8^2 + 7 \binom{8}{2} = 1840.$$

Thus for any feasible law, $\mathbb{E}g \geq \text{vex } f - 1840/1000$, regardless of dependence involving Z . Consequently

$$H(g) \leq 3225/7 + 46/25 = 80947/175, \quad \frac{T(g)}{H(g)} \geq \frac{165025}{80947} > 2. \quad (5.10)$$

This is a certified bound, rather than an exact new hull value.

For completeness, it yields a finite nonconstructive unit-coefficient example through Theorem 4.5. Normalize by 13, clone each variable $m = 1000$ times, and take error allowance $t = m^3/10$. The failure probability in (4.20) is at most

$$2(2^{25m} + 1) \exp(-m^3/23200) < 1. \quad (5.11)$$

Indeed its logarithm is at most $(25m + 2) \log 2 - m^3/23200 < 0$ at $m = 1000$, even with $\log 2 < 1$. Since $943 - 2(80947/175) = 3131/175$, every successful sample satisfies

$$T(G_m) - 2H(G_m) \geq m^3 \left(\frac{3131}{2275} - \frac{1}{2} \right) > 0. \quad (5.12)$$

It is nonempty and has positive hull gap at its interior point. This proves existence in 25,000 variables with $464m^3$ candidate monomials; no successful sample is claimed to be listed. The explicit construction below is much smaller, while coefficient removal remains useful for approximating an arbitrary fixed-degree ratio.

5.4 Two means and an explicit homogeneous unit-coefficient example

Theorem 5.6 (Two-level cubic family). *For each positive multiple m of four, take two groups U, W of m variables with means $1/2, 3/4$, and define*

$$f_m = AE_2(W) + \frac{5m}{4} E_2(U), \quad A = \sum_{i \in U} x_i. \quad (5.13)$$

Then

$$\frac{243}{115}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{243}{115}. \quad (5.14)$$

For $m = 4, 8, 12, 16$ its exact ratios are, respectively, $27/16, 21/11, 99/50, 135/67$. Among positive multiples of four, this family's ratio exceeds two if and only if $m \geq 16$. There is an explicit polynomial with 52 variables and 4,320 distinct homogeneous cubic monomials, every coefficient one, at an interior point with ratio at least $2700/1343 > 2$.

Proof. For real $a, c \in [0, 1]$, set $F_2(a, c) = ac^2 + 5a^2/4$ and $\ell_2(a, c) = 11a/6 + 20c/27 - 95/108$. The identity

$$F_2 - \ell_2 = \frac{5}{4} \left(a - \frac{11 - 6c^2}{15} \right)^2 + \frac{(1 - c)(3c - 2)^2(3c + 7)}{135} \quad (5.15)$$

proves $F_2 \geq \ell_2$ on the square. Equality holds at $(1/3, 1)$ and $(5/9, 2/3)$; their mixture with weights $1/4, 3/4$ has mean $(1/2, 3/4)$ and expected value $16/27$. Thus the scalar envelope value at that mean is exactly $16/27$.

At binary counts $A = ma, C = mc$,

$$\frac{2f_m}{m^3} = F_2(a, c) - \frac{ac + (5/4)a}{m}.$$

Every feasible law has $\mathbb{E}a = 1/2$ and $\mathbb{E}c = 3/4$, and $ac \leq a$. It follows that $2 \text{vex } f_m/m^3 \geq 16/27 - 9/(8m)$. Each monomial has upper envelope $1/2$ and lower envelope zero; therefore

$$\frac{2}{m^3} \text{cav } f_m = \frac{2}{m^3} T(f_m) = \frac{9}{8}(1 - 1/m). \quad (5.16)$$

Subtraction gives $2H(f_m)/m^3 \leq 115/216$. For the reverse bound, choose one of the two equality atoms above, then sample all coordinates independently conditional on its two probabilities. Distinct variables in each term have conditional product expectation, so $2\mathbb{E}[f_m \mid a, c]/m^3 = (1 - 1/m)F_2(a, c)$. This is a feasible law and gives $2 \text{vex } f_m/m^3 \leq (1 - 1/m)16/27$. Hence

$$\frac{115}{216}(1 - 1/m) \leq \frac{2H(f_m)}{m^3} \leq \frac{115}{216}.$$

Together with (5.16) this proves (5.14), including positive denominators and convergence of the actual ratios to 243/115. For example, $m = 20$ gives the lower bound $4617/2300 > 2$ in 40 variables.

At $m = 16$, the exact integer-grid certificate is

$$A \binom{C}{2} + 20 \binom{A}{2} \geq 214A + \frac{177}{2}C - 1686 \quad (A, C \in \{0, \dots, 16\}). \quad (5.17)$$

Appendix C supplies its integer verification. Uniform success subsets of sizes 8, 12 give equality, with value 1088. The concave envelope and termwise gap are 2160, so $H = 1072$ and $T/H = 135/67$.

For $m = 4, 8, 12$, Table 6 in Appendix C gives matching affine minorants and uniform fixed-count laws, proving the three exact ratios stated above. All three are below two. For every $m \geq 20$, the lower bound in (5.14) is at least $4617/2300 > 2$, since $1 - 1/m$ is increasing. Together with $m = 16$, these facts prove the if-and-only-if assertion among positive multiples of four in (5.13); they make no minimum-dimension claim over other families.

Introduce 20 distinct coordinates z_k and define explicitly

$$g = AE_2(W) + \left(\sum_{k=1}^{20} z_k \right) E_2(U), \quad x_U = 1/2, \quad x_W = 3/4, \quad z_k = 999/1000. \quad (5.18)$$

There are $16 \binom{16}{2} + 20 \binom{16}{2} = 4320$ distinct cubic supports, all with coefficient one. Every term still has upper envelope $1/2$ and lower envelope zero, so $\text{cav } g = T(g) = 2160$. At binary vertices,

$$g = f_{16} - \left(\sum_k (1 - z_k) \right) E_2(U), \quad 0 \leq E_2(U) \leq 120.$$

Every feasible law projects to the original means, giving $\text{vex } g \geq 1088 - 120(20/1000) = 1088 - 12/5$. Thus $0 < H(g) \leq 5372/5$ and $T(g)/H(g) \geq 2700/1343$. Positivity of H follows by comparing independent and common-threshold laws at these interior means. $\square$

The two-level examples show that means in the upper half of the unit cube do not force factor two. Small upward perturbations of the $1/2$ means preserve a strict ratio above two by continuity, so all means can be strictly above $1/2$. No quantitative perturbation radius or minimal dimension claim is implied. The next result explains why at least two distinct means are needed on the unit cube.

6 Equal means and classical symmetric envelopes

The envelope of the complete elementary symmetric polynomial is classical. Sherali's equation (13) and Theorem 3 [21, pp. 252–253] give, on the entire unit cube,

$$\text{vex } E_d(x) = \max \left(0, \max_{k=d-1, \dots, n-1} \left\{ \binom{k}{d-1} \sum_i x_i - (d-1) \binom{k+1}{d} \right\} \right). \quad (6.1)$$

The following extremal gap formulation is a consequence of that symmetric-envelope machinery. We give a direct proof at equal means to make the common-law guarantee explicit.

Theorem 6.1 (Exact equal-mean formula). *Fix $n \geq 2$ and $0 < u < 1$. Put $b = \lfloor nu \rfloor$, $\theta = nu - b$, and, for $2 \leq d \leq n$,*

$$t_d(u) = \min\{u, (d-1)(1-u)\}, \quad q_{n,d}(u) = \frac{(1-\theta) \binom{b}{d} + \theta \binom{b+1}{d}}{\binom{n}{d}},$$

with $\binom{k}{d} = 0$ for integers $0 \leq k < d$. The maximum of T/H over positive multilinear polynomials with a nonzero nonlinear part at $x = (u, \dots, u)$ is

$$R_n^{\text{eq}}(u) = \max_{2 \leq d \leq n} \frac{t_d(u)}{u - q_{n,d}(u)}. \quad (6.2)$$

A maximum-degree restriction $2 \leq D \leq n$ restricts this maximum to $d \leq D$. Each candidate ratio is attained by E_d .

Proof. Draw $K = b$ or $b + 1$ with probabilities $1 - \theta, \theta$, then choose a uniformly random K -element success subset. Each mean is u and each degree- d product expectation is $q_{n,d}(u)$. This single law gives

$$H(f; u\mathbf{1}) \geq \sum_e a_e(u - q_{n,|e|}(u)), \quad T(f; u\mathbf{1}) = \sum_e a_e t_{|e|}(u).$$

The quotient is bounded by the largest term ratio, provided its denominators are positive. To see positivity and attainment, the binary value of E_d is $\binom{K}{d}$, whose first forward difference is $\binom{K}{d-1}$ and is nondecreasing. Its linear interpolation on integer counts is convex. Jensen's inequality shows that any integer-valued K with mean nu has expected value at least the adjacent-count interpolation. Our uniform-subset law attains this value with all individual means correct, establishing $\text{vex } E_d = \binom{n}{d} q_{n,d}$. Its upper envelope is $u \binom{n}{d}$. Comparing the minimum with the independent law, whose product expectation is u^d , gives $q_{n,d} \leq u^d < u$. Thus the denominators are positive and every claimed candidate is attained. The maximum is attained by a maximizing degree. $\square$

Corollary 6.2 (Dimension-free equal-mean supremum). *For $0 < u < 1$ the exact supremum over dimensions, and the limit of $R_n^{\text{eq}}(u)$ as $n \rightarrow \infty$, is*

$$R_\infty^{\text{eq}}(u) = \max_{k \geq 1} \frac{\min\{1, k(1-u)/u\}}{1 - u^k} \leq 2. \quad (6.3)$$

Writing $r_u = u/(1-u)$, a maximizing k is one of $\max(1, \lfloor r_u \rfloor)$ and $\max(1, \lceil r_u \rceil)$. For $u \leq 1/2$ the value is $1/(1-u)$. The uniform constant two is sharp.

Proof. The comparison $q_{n,d} \leq u^d$ gives the upper bound by the displayed maximum after writing $k = d - 1$. For fixed d , $q_{n,d}(u) \rightarrow u^d$ as $n \rightarrow \infty$, since both adjacent counts divided by n tend to u . Thus any fixed-degree candidate is approached by E_d in increasing dimension.

For $k \geq r_u$ the expression is $1/(1-u^k)$, decreasing with k . For $k \leq r_u$ it equals $k/(u+u^2+\dots+u^k)$, increasing with k because the average of that decreasing sequence decreases. Hence a finite maximizer occurs at one of the stated two integers. Its fixed-degree limit proves both the supremum and convergence assertions.

Set $t = k(1-u)/u$. Bernoulli's inequality gives $u^{-k} = (1 + (1-u)/u)^k \geq 1 + t$, and therefore

$$\frac{\min(1, t)}{1 - u^k} \leq \min(1, t) \frac{1+t}{t} \leq 2.$$

When $u \leq 1/2$ the maximizing integer is $k = 1$, giving $1/(1-u)$. At $u = 1/2$, the complete bilinear polynomial for even n has ratio $2(n-1)/n$, tending to two. This is the classical positive bilinear sharpness example, now viewed within the equal-mean formula. $\square$

These exact formulas concern equal normalized means on the unit cube. Positive expansion transfers the degree upper bounds to nonnegative boxes, but does not identify an original positive-box factorization with the elementary symmetric unit-cube model. In particular, the precise finite equal-mean optimization above makes no such additional box claim.

7 Incidence structure and marginal restrictions

The *incidence graph* of a specified factorization has variable vertices i , factor vertices e , and one edge ie for each $i \in e$. It is a simple bipartite graph even when two factors have identical supports. This is different from the primal variable graph, which joins two variables occurring together in a factor. All widths below concern the incidence graph. Its treewidth is the minimum, over tree decompositions, of the largest bag size minus one; a tree decomposition covers every edge and has connected bags for each vertex. Its degeneracy is the largest minimum degree of a nonempty subgraph. Its orientation parameter is the minimum possible maximum outdegree, with no acyclicity requirement. Its variable frequency ν is the largest number of factors containing one variable. A feedback variable set is a set of variable vertices whose deletion leaves a forest; write f_F for its size. Affine factors are discarded before these parameters are measured, unless a factorization is explicitly retained.

7.1 Ownership and asymmetric orientations

Write $\kappa = 2/(1 - e^{-1})$. Let $B(d)$ be any valid universal degree- d upper bound from Section 4, with $B(1) = 0$. In particular one can choose $B(d) \sim \log d / \log \log d$.

Theorem 7.1 (Incidence orientation bound). *For a positive multilinear polynomial on a unit cube, suppose an incidence orientation has variable outdegrees at most r and factor outdegrees at most s , where r, s are nonnegative integers. Then*

$$T \leq [r + \min\{s, B(s+1)\} + \kappa]H.$$

In particular frequency ν gives $\nu + \kappa$, and an orientation with all outdegrees at most k gives $k + \min\{k, B(k+1)\} + \kappa$. These conclusions transfer to boxes with zero lower bounds by scaling.

Proof. Call a coordinate low if its mean is at most $1/2$. A term with exactly one low coordinate will be called a one-low term. All other terms receive at least t_e/κ deficiency from independence by Lemma 2.3. In a one-low term the low coordinate is its anchor, of mean u_e . Write $q_i = 1 - x_i$ for the high-coordinate failure means. Partition its high variables into I_e , whose incidence edges point into the factor, and O_e , whose edges point out. Put

$$t_e^I = \min\{u_e, \sum_{i \in I_e} q_i\}, \quad t_e^O = \min\{u_e, \sum_{i \in O_e} q_i\}.$$

Capped sums are subadditive, so $t_e \leq t_e^I + t_e^O$.

For $r > 0$, independently at each high variable label its incidences into one-low factors by distinct labels in $\{1, \dots, r\}$. In round h a variable is owned by its factor with label h , if any. Thus the owned sets $J_{e,h}$ are disjoint over factors and partition I_e over rounds. Take one uniform U and give every low coordinate its success interval $[0, x_i]$. For the high variables owned by e , arrange failure sets so that their union inside $[0, u_e]$ has length $\min\{u_e, \sum_{i \in J_{e,h}} q_i\}$. This can be done by consecutive arcs of lengths $\min(q_i, u_e)$ on the circle of circumference u_e ; when $q_i > u_e$, add measure $q_i - u_e$ outside that interval. For $u_e = 0$ use arbitrary failure sets of the required measures. Unowned variables also receive arbitrary marginal-correct failure sets. Disjoint ownership makes these assignments compatible, including when several terms share one low anchor. Their full deficiencies dominate the owned failure events. A uniform mixture of rounds therefore gives

$$\mathbb{E}D_e \geq \frac{1}{r} \sum_h \min\{u_e, \sum_{i \in J_{e,h}} q_i\} \geq t_e^I/r.$$

If $r = 0$, the incoming gaps all vanish.

A different global law uses the same low success intervals and high failure intervals $[0, q_i]$. Since $|O_e| \leq s$, it gives

$$\mathbb{E}D_e \geq \min\{u_e, \max_{i \in O_e} q_i\} \geq t_e^O/s$$

when $s > 0$; empty outgoing sets have zero gap. Mixing these two laws and independence with weights proportional to r, s, κ proves the bound $r + s + \kappa$, including zero parameters by omitting their components.

For the degree alternative, form the positive polynomial with terms $a_e X_{i_e} \prod_{i \in O_e} X_i$ for one-low factors. Its degree is at most $s + 1$, its termwise gap is $\sum_e a_e t_e^O$, and its deficiency under *every* common-marginal law is at most the original total deficiency. Its hull gap is consequently at most the original hull gap. Thus the outgoing sum is at most $B(s + 1)H$, as well as at most sH . The incoming sum is at most rH by the ownership law, and the other terms' sum at most κH by independence. Adding proves the refinement. This argument needs only a polynomial-level degree bound, not an unproved per-term guarantee from that bound. Orienting all edges from variables to factors gives the frequency specialization. $\square$

Corollary 7.2 (Sharp ownership constant). *At points where every term has exactly one low coordinate, suppose each high variable occurs in at most r factors, with unrestricted low-variable frequency. The sharp supremum of T/H is r for every integer $r \geq 1$.*

The ownership proof gives the upper bound using only its r rounds. The lower bound for $r \geq 2$ follows from the next construction with $L = r$; for $r = 1$ a single nonaffine product attains one. This parameter concerns high variables at the stated point and is distinct from ordinary frequency.

7.2 Variable radix and exact width

Theorem 7.3 (Variable-radix family). *Let $L \geq 2$ and $b \geq L$ be integers, and put $m = b^L$. Index m leaves by length- L base- b words. Let $\mathcal{P}_j$ partition them by their first j digits. With separate anchors $a_1, \dots, a_L$, define*

$$f_{L,b}(a, z) = \sum_{j=1}^L \sum_{Q \in \mathcal{P}_j} a_j \prod_{i \in Q} z_i, \quad a_j = b^{-j}, \quad z_i = 1 - m^{-1}.$$

Then

$$T = L, \quad H = 1 + (L - 1)/b, \quad \text{tw}(G_{L,b}) = L. \quad (7.1)$$

Proof. A level- j block has m/b^j leaves, whose failure means sum to b^{-j} . Its local lower value is zero and upper value b^{-j} ; b^j blocks give one unit of termwise gap at each level. For an arbitrary binary law let R count failed leaves and N_j count hit level- j blocks. Then $\mathbb{E}R = 1$, $N_j \leq \min(b^j, R)$, and total deficiency is $\sum_j A_j N_j$. For a fixed distribution of R , a prescribed-mean indicator has largest correlation with the nondecreasing function $\min(b^j, R)$ by selecting its upper tail: exchanging selected probability at a smaller count with unselected probability at a larger count proves this assertion, with fractional selection at a tied count allowed. All anchors can use these nested tails simultaneously. Represent the count by its nonincreasing quantile $R(U)$ and select $A_j = \mathbf{1}\{U \leq b^{-j}\}$. If exactly l anchors are selected, then

$$\sum_{j=1}^l \min(b^j, R) \leq R + \sum_{j=1}^{l-1} b^j.$$

For $l = 0$ use $0 \leq R$. Integrating gives $H \leq 1 + \sum_{j=1}^{L-1} b^j b^{-j-1} = 1 + (L-1)/b$.

The probability of level l is $w_l = (b-1)b^{-l-1}$ for $0 \leq l < L$ and $w_L = b^{-L}$. Two count profiles are

$$R_l^{(1)} = b^l \mathbf{1}\{l \geq 1\}, \quad R_l^{(2)} = b^{l-1} \mathbf{1}\{l \geq 2\}.$$

Their means are $M_1 = [(L-1)(b-1) + b]/b$ and $M_2 = [(L-2)(b-1) + b]/b^2$. Our parameter range ensures $M_2 \leq 1 \leq M_1$, so mixing the profiles with first-profile weight $(1 - M_2)/(M_1 - M_2)$ gives mean count one. Each profile attains the preceding pointwise inequality at every level, including $l = 0, 1$. For an integer count R , fail the first R leaves in digit-reversal order. Their first j digits are the reversals of the last j digits of $0, \dots, R-1$, so they hit exactly $\min(b^j, R)$ level- j blocks. A uniform coordinatewise digit shift modulo b preserves every hit count and gives each leaf conditional failure probability R/m . The mixture therefore has all the required marginals and attains the upper bound. This argument works for composite radices as well as prime ones.

For treewidth, eliminate leaves first, filling their L factor neighbors into cliques. These neighbors form nested chains. Next eliminate factor vertices from level L down to level one. A level- j factor's remaining neighbors lie among its $j-1$ ancestor factors and anchors of levels $j, \dots, L$, at most L vertices. Induction verifies this description: eliminating a descendant joins its ancestor chain and deeper anchors, never two unrelated factors. The final L anchors also have elimination width at most L . For the reverse bound choose a level- $(L-1)$ block with b leaves and contract each deepest bilinear factor into its leaf. The resulting minor contains $K_{L,b}$, with the $L-1$ ancestor factors and a_L opposite those leaves. For $b \geq L$, this graph has a clique minor of order $L+1$: pair $L-1$ vertices on opposite sides into branch sets, then keep one unused vertex from each side. All branch sets are connected and pairwise adjacent. Minor monotonicity proves the lower bound. $\square$

The restriction $b \geq L$ is part of the exact formula just proved. In particular it cannot be replaced silently by $b = 2$ for arbitrarily many levels; Theorem 4.2 uses a different cutoff. More generally, for integers $b \geq 2, L \geq 2$, the same radix proof gives $H = s + (L-s)b^{-s}$ whenever $1 \leq s < L$ and

$$M_{s+1} \leq 1 \leq M_s, \quad M_q = \frac{(L-q)(b-1) + b}{b^q}.$$

Indeed, cap the first $l-s$ summands and bound the last s by R , giving $\sum_{j \leq l} \min(b^j, R) \leq sR + \sum_{j \leq l-s} b^j$. Its integrated constant is $(L-s)b^{-s}$. The profiles $R_l^{(q)} = b^{l-q+1} \mathbf{1}\{l \geq q\}$ for $q = s, s+1$ have means M_q and attain equality; mix them to mean one and use digit reversal as above.

Corollary 7.4 (Sharp leading structural growth). *Let W_k, D_k, P_k be the supremal unit-cube positive-monomial gap ratios under, respectively, incidence treewidth, degeneracy, or orientation parameter at most k . For every integer $k \geq 2$,*

$$W_k \geq k, \quad D_k \geq k+1, \quad P_k \geq k+1,$$

whereas

$$W_k \leq D_k \leq P_k \leq k + \min\{k, B(k+1)\} + \kappa.$$

In particular $W_k \sim D_k \sim P_k \sim k$ as $k \rightarrow \infty$.

Proof. Treewidth at most k implies degeneracy at most k : in a reduced tree decomposition take a vertex exclusive to a leaf bag and delete it, then continue on induced subgraphs. A degeneracy order oriented toward later vertices has outdegrees at most k . This proves the upper chain. Use $L = k$ in Theorem 7.3 and let $b \rightarrow \infty$ for $W_k \geq k$. For the stronger other bounds use $L = k+1$: orient each

deepest factor toward both its leaf and anchor, all other leaf incidences toward their factor, and the remaining factor incidences toward their anchor. Outdegrees are at most k , since deepest factors have outdegree two. For degeneracy delete deepest factors of degree two, their isolated anchor, then leaves of degree $L - 1 = k$, leaving stars. Letting b grow proves both lower bounds. This family still has treewidth $k + 1$. For example $L = 3, b = 5$ gives $15/7 > 2$ for degeneracy and orientation parameter two, with treewidth three. Finally $B(k + 1) = o(k)$. $\square$

For ordinary frequency, the example in Proposition 8.3 below gives a lower bound $2 - 1/\nu$ for every $\nu \geq 2$. The dyadic family has maximum frequency 2^L (at its deepest anchor), so choosing the largest power of two at most ν also gives the lower asymptotic $(1 - o(1)) \log \nu / \log \log \nu$. Frequency two has the sharper exact value $3/2$, proved in Section 8.

7.3 A lower bound on the normalized means

Let C_δ be the supremal positive-monomial gap ratio on unit cubes at points whose coordinates are all at least δ , with dimension and degree unrestricted. This restricts the *evaluation point*; the envelopes remain those on the full unit cube.

Theorem 7.5 (Marginal-floor law). *For $0 < \delta \leq 1/2$, set*

$$M = \max\{2, \log(1/\delta)\}, \quad \tau = \delta/M^2, \quad \Lambda = \log \frac{1 + \tau}{\tau}, \quad I_\delta = \int_0^1 \left(1 - e^{-1/[\Lambda(z+M^{-2})]}\right) dz.$$

Then $I_\delta > 0$ and $C_\delta \leq I_\delta^{-1} + \kappa$. For $1/2 < \delta < 1$, $C_\delta \leq C_{1/2}$. Moreover

$$C_\delta \sim \frac{\log(1/\delta)}{\log \log(1/\delta)} \quad (\delta \downarrow 0).$$

The same leading asymptotic holds with every coordinate in $[\delta, 1 - \delta]$. The upper bound transfers to nonnegative boxes when all nonfixed normalized means $(x_i - \ell_i)/(u_i - \ell_i)$ are at least δ .

Proof. The density $h(t) = 1/[\Lambda(t + \tau)]$ integrates to one on $(0, 1)$. Use low-coordinate success intervals $[0, x_i]$. For a high coordinate with failure mean $q < 1/2$ define

$$q_0(t) = \min\{1, qh(t)\}, \quad m_q = \int_0^1 q_0(t) dt, \quad q_1(t) = q_0(t) + \frac{q - m_q}{1 - m_q}(1 - q_0(t)).$$

Here $0 \leq m_q \leq q < 1$, so $q_1 \in [q_0, 1]$ and its integral is exactly q , including $q = 0$. Conditional on the common uniform variable $U = t$, sample high failures independently with probabilities $q_1(t)$. For a set of high coordinates with failure sum S , the probability of some failure is at least $1 - e^{-Sh(t)}$. If any $qh(t) \geq 1$, a failure is certain; otherwise $q_1(t) \geq qh(t)$ and the usual product bound proves it. The clipping case is needed for this conclusion.

A one-low term with anchor mean $u \geq \delta$ consequently has

$$\mathbb{E}D_e \geq \int_0^u (1 - e^{-S/[\Lambda(t+\tau)]}) dt.$$

For $S > 0$, substitute $t = uz$, $y = S/u$, and note

$$\frac{1 - e^{-ay}}{\min(1, y)} \geq 1 - e^{-a} \quad (a \geq 0, y > 0).$$

Concavity proves this for $y \leq 1$, monotonicity for $y \geq 1$. Since $\tau/u \leq M^{-2}$, division by $t_e = u \min(1, y)$ now gives $\mathbb{E}D_e/t_e \geq I_\delta$. Zero S requires no division. Mix this law and independence with weights proportional to I_δ^{-1}, κ ; Lemma 2.3 handles all other factors.

As $\delta \downarrow 0$, $\Lambda = M + 2 \log M + o(1)$ and $\Lambda/M^2 \rightarrow 0$. The inequalities $1 - e^{-a} \leq \min(1, a)$ and $1 - e^{-a} \geq a - a^2/2$ give

$$I_\delta \leq \frac{1 + \log \Lambda}{\Lambda}, \quad I_\delta \geq \frac{\log \Lambda}{\Lambda(1 + \Lambda/M^2)} - \frac{\Lambda - 1}{2\Lambda^2}.$$

For the second bound integrate only over $[1/\Lambda, 1]$, using $z + M^{-2} \leq z(1 + \Lambda/M^2)$. Hence $I_\delta \sim \log \Lambda/\Lambda$, proving the upper asymptotic. The dyadic family with $L = \lfloor \log_2(1/\delta) \rfloor$ has all means in $[\delta, 1 - \delta]$ and ratio asymptotic to $L/\log_2 L$, which gives the matching lower bound. Monotonicity gives the finite large- δ claim. Finally positive affine expansion on a nonnegative box preserves the lower restriction on normalized means, so Proposition 2.4 gives its stated extension. $\square$

Theorem 7.6 (Uniform joint asymptotic). *Let $C(n, d, k, \delta)$ be the supremal unit-cube positive-monomial ratio in dimension at most n , degree at most d , incidence treewidth at most k , and with every mean at least δ . Assume integers $n, d \geq 2$, $k \geq 1$, and $0 < \delta \leq 1/2$. Put $q = \min\{n, d, 1/\delta\} \geq e^e$, $\phi(q) = \log q / \log \log q$, and $h = \min\{k, \phi(q)\}$. Along every joint parameter sequence with $h \rightarrow \infty$,*

$$C(n, d, k, \delta) \sim h.$$

The same statement holds for the two-sided strip and with degeneracy or the orientation parameter replacing treewidth.

Proof. The separate dimension, degree, marginal-floor, and incidence estimates give $C \leq \min\{(1 + o(1))\phi(q), k + o(k)\}$. Their relative errors vanish uniformly along the stated sequences, since both q and k then diverge. For a simultaneous lower construction take

$$L = \lfloor h \rfloor, \quad b = \lfloor (q/2)^{1/L} \rfloor, \quad m = b^L.$$

Then $b \geq (1 - o(1)) \log q$, $L/b \leq (1 + o(1))/\log \log q \rightarrow 0$, and eventually $b \geq L \geq 2$. The radix family has dimension $m + L \leq q/2 + \phi(q) \leq q \leq n$, degree $m/b + 1 \leq q \leq d$, and width $L \leq k$. Its smallest anchor mean and leaf failure mean are $1/m \geq 2/q \geq \delta$, and all other means also lie in the two-sided strip. Thus its ratio $L/[1 + (L - 1)/b] \sim h$. Its degeneracy and orientation parameter are at most its treewidth, proving the alternatives. This is not an asymptotic assertion for a fixed small value of k . $\square$

8 Feedback variables and frequency two

8.1 A universal repair law

The next argument works on the original factor scopes, including on boxes with positive lower endpoints.

Lemma 8.1 (Feedback repair and gluing). *Fix singleton Bernoulli means and a feedback variable set F of size f_F . For any nonnegative local functions g_e and any prescribed-mean local laws P_e on their scopes, there is one global prescribed-mean law P' satisfying*

$$\mathbb{E}_{P'} g_e \geq 2^{-f_F} \mathbb{E}_{P_e} g_e \quad \text{for every } e.$$

Proof. Extend each local law to $F \cup e$ by independently sampling any missing feedback coordinates. Put $C = 2^{f_F}$. Construct a global reference law Q with a common uniform U : each feedback coordinate independently chooses, fairly, its success interval $[0, x_j]$ or $[1 - x_j, 1]$; every outside coordinate uses $[0, x_i]$. Write Q_F for its feedback marginal and Q_i for its (F, i) marginal. For a feedback state s and outside state b , every prescribed-mean local law satisfies

$$P(s, b) \leq m(s, b; i) := \min(\{x_j : s_j = 1\}, \{1 - x_j : s_j = 0\}, \{x_i \text{ if } b = 1; 1 - x_i \text{ if } b = 0\}).$$

The minimum includes all feedback coordinates and the outside coordinate. Choose the feedback orientations that place every desired event at the left endpoint if $b = 1$, and at the right endpoint if $b = 0$. This single orientation pattern has probability $1/C$, and its intersection has length $m(s, b; i)$. Therefore $Q_i(s, b) \geq P(s, b)/C$. The same endpoint argument gives $Q_F(s) \geq P_F(s)/C$, also when there are no outside variables. For $F = \emptyset$ its unique state has mass one.

For a fixed extended factor law P_e , define the nonnegative residuals

$$h_s = Q_F(s) - (P_e)_F(s)/C, \quad r_{i,s,b} = Q_i(s, b) - (P_e)_{F,i}(s, b)/C \quad (i \in e \setminus F).$$

For each i , $\sum_b r_{i,s,b} = h_s$. At state s with $h_s > 0$, place residual mass h_s and sample these outside coordinates independently with probabilities $r_{i,s,b}/h_s$. For $h_s = 0$ all these residuals vanish. For an empty outside scope place just the mass h_s . The resulting measure R_e has total mass $1 - 1/C$, and $P'_e = P_e/C + R_e$ has feedback marginal Q_F and (F, i) marginal Q_i , while dominating P_e/C entrywise.

Condition on any s with $Q_F(s) > 0$. The factor laws $P'_e(\cdot | s)$ have consistent outside singleton marginals $Q_i(s, b)/Q_F(s)$. Their incidence graph is a forest. Root each nontrivial component at a factor, sample its law, then traverse the tree. Each newly reached factor shares exactly one already sampled variable; sample its new variables using its local conditional law given that variable. Null conditioning states are never encountered. Isolated variables use their stated conditional means; factors with no outside variables need no sampling. This preserves every conditional factor law. Mix over s with weights $Q_F(s)$. The resulting global law has marginals P'_e on $F \cup e$, and nonnegativity gives the claimed domination of expectations. There is one loss C , regardless of the lengths of the forest paths. $\square$

Theorem 8.2 (Feedback-variable gap). *For positive monomials on any finite nonnegative box, deletion of f_F variable vertices to an incidence forest implies $T \leq 2^{f_F} H$.*

Proof. Substitute fixed coordinates first. In normalized endpoint variables Y , an original physical product is $\varphi_e(Y) = \sum_{A \subseteq e} c_A \prod_{i \in A} Y_i$ with $c_A \geq 0$. This expansion is used only inside the original scope. For each nonempty A choose an index k_A of smallest normalized mean, and let

$$\ell_e(Y) = c_\emptyset + \sum_{\emptyset \neq A \subseteq e} c_A Y_{k_A}.$$

On binary vertices $\ell_e \geq \varphi_e$, while every prescribed-mean law has $\mathbb{E}\ell_e = \text{cav } \varphi_e$, by common upper attainment. Thus $g_e = \ell_e - \varphi_e$ is a nonnegative function on the original scope whose local maximum expectation is the original factor's exact gap. Apply Lemma 8.1 to maximizing local laws, then sum with the positive coefficients. The full upper value is the sum of local upper values, proving the result. Replacing the factors by expanded hyperedges would change the incidence graph and would not justify this proof. $\square$

For $f_F = 0$ this recovers classical incidence-forest exactness; exactness of standard linearization on Berge-acyclic hypergraphs is established in Del Pia and Khajavirad [8]. The general domination

lemma is sharp for arbitrary nonnegative local payoffs: on f_F feedback bits and one outside bit, all fair, use the 2^{f_F+1} indicators of their states. Every local maximum is $1/2$, while their sum is identically one. Private variables fixed to one can make the scopes distinct. These indicators have zero literals, so they do not prove sharpness for positive monomials when $f_F > 1$.

Proposition 8.3 (One feedback variable is sharp). *For $n \geq 2$, let $f(a, x) = a \sum_{i=1}^n x_i + \prod_{i=1}^n x_i$ and evaluate at $a = 1/n$, $x_i = 1 - 1/n$ on the unit cube. Then $T = 2 - 1/n$ and $H = 1$. Its incidence graph has treewidth two and becomes a tree after deleting a .*

Proof. The bilinear terms contribute one in total and the large product contributes $(n - 1)/n$. If R counts failed leaves and A is the binary anchor, the full deficiency is $\mathbb{E}[AR] + \mathbb{P}(R \geq 1) - 1/n$, with $\mathbb{E}R = 1$ and $\mathbb{E}A = 1/n$. Pointwise $AR + \mathbf{1}\{R \geq 1\} \leq R + A$, giving the upper bound one. Exactly one uniformly chosen failed leaf and an independent anchor attain equality. Write e_0 for the large factor and e_i for ax_i . Path bags $\{a, e_0, x_i\}$ with attached bags $\{a, x_i, e_i\}$ give width two. A cycle rules out width one, and deleting a leaves a tree. $\square$

8.2 Half-integral degree slabs

For frequency at most two, form a loopless dual multigraph with factor vertices. A shared variable is an edge between its two factors; a private variable is an edge to its own new dummy vertex. Parallel edges are allowed. Unused variables may be completed independently. Dummy factors have zero weight or identically zero costs. Let g be its odd girth, with $g = \infty$ in the bipartite case.

Lemma 8.4 (Degree-slab vertices). *For a loopless multigraph and $p \in [0, 1]^E$, put $s_v(p) = \sum_{i \in \delta(v)} p_i$. Every vertex of*

$$\mathcal{S}(p) = \{z \in [0, 1]^E : \lfloor s_v(p) \rfloor \leq s_v(z) \leq \lceil s_v(p) \rceil \ (v \in V)\}$$

has fractional edges forming vertex-disjoint odd cycles, all valued $1/2$.

Proof. Let J be the fractional edges at a vertex and R the tight degree rows incident to them, counting a repeated lower/upper equality once. The restricted rows must have full column rank on J : otherwise a small perturbation in a null direction preserves every tight constraint and box bound, contradicting extremality. Hence $|J| \leq |R|$. An integer tight residual degree cannot contain exactly one fractional edge, so every such row has at least two fractional incidences. Thus $2|R| \leq 2|J|$. Equality forces each fractional edge to have both endpoints in R , and each such vertex to have fractional degree two. These are cycles; an even cycle admits an alternating perturbation and is excluded. On an odd cycle the two incident fractional values sum to the integer one, forcing all values to $1/2$. The argument excludes parallel fractional two-cycles and degree-one dummy vertices automatically. $\square$

This is the usual fractional matching rank argument, here stated for integer degree slabs; compare the fixed-degree formulation in Barrus [3, Theorem 2.1].

Theorem 8.5 (Frequency two and odd girth). *For positive monomials of frequency at most two on a unit cube,*

$$T \leq \frac{g}{g-1} H \quad (g < \infty), \quad T = H \quad (g = \infty).$$

Every finite odd-girth constant is sharp, and the unrestricted frequency-two supremum is $3/2$.

Proof. Use failure probabilities $p_i = 1 - x_i$, and define $b_v(p) = \max_{i \in \delta(v)} p_i$ and $c_v(p) = \min\{1, s_v(p)\}$, with empty values zero. For a failure edge set J , write $I_v(J)$ for its coverage indicator. Exact single-product envelopes give

$$T = \sum_v a_v(c_v - b_v), \quad H = \max_{\mathbb{P}(i \in J) = p_i} \sum_v a_v(\mathbb{E}I_v(J) - b_v). \quad (8.1)$$

In particular an approximation of unshifted coverage alone is insufficient. Decompose $p = \mathbb{E}Z$ into vertices of $\mathcal{S}(p)$. On each allowed slab c_v is affine (linear below one, constant above), so $\mathbb{E}c_v(Z) = c_v(p)$; convexity gives $\mathbb{E}b_v(Z) \geq b_v(p)$.

Given $Z = z$, keep integral edges. On each fractional odd cycle of length l , choose one of its l maximum matchings uniformly, then fairly choose that matching or its complement in the cycle. Each edge has marginal $1/2$. A matching leaves its uniformly selected vertex uncovered; its complement covers every vertex. Thus coverage probability is $1 - 1/(2l)$ at each cycle vertex. A vertex with another selected integral edge is covered surely. Otherwise its $c_v(z) = 1$ and $b_v(z) = 1/2$. All vertices off these cycles have equal integral $b_v(z) = c_v(z)$. With $\alpha = 1 - 1/g$, this proves

$$\mathbb{E}[I_v(J) \mid Z] \geq \alpha c_v(Z) + (1 - \alpha)b_v(Z).$$

Averaging retains the baseline: $\mathbb{E}I_v(J) - b_v(p) \geq \alpha[c_v(p) - b_v(p)]$. Sum this in (8.1). For a bipartite graph there are no fractional cycles, so the slab decomposition itself attains $c_v(p)$ simultaneously, proving exactness.

For sharpness take a unit-weight odd cycle of length l with all edge means $1/2$. Its termwise gap and coverage baseline both equal $l/2$. For every selected edge set J , $\sum_v I_v(J) \leq |J| + (l - 1)/2$: if $|J| \leq (l - 1)/2$ use $2|J|$, otherwise use l . Expected coverage is therefore at most $l - 1/2$, attained by the matching/complement law. The hull gap is $(l - 1)/2$. $\square$

Bipartite equality is a statement about the scalar envelopes. It does not assert compatibility of all separately feasible product-value coordinates in the full lifted multilinear polytope. Zero-lower boxes transfer by scaling; unequal positive aspect ratios need a separate analysis.

8.3 Convex cardinality factors

A discrete sequence $\varphi(0), \dots, \varphi(d)$ is convex if $\varphi(k + 2) - 2\varphi(k + 1) + \varphi(k) \geq 0$. Its cardinality factor is the unique multiaffine interpolant of the values $\varphi(\sum_i X_i)$ on binary vertices. It is generally not the literal continuous composition $\varphi(\sum_i x_i)$. Sequences may be negative, decreasing, or nonmonotone; only their discrete convexity is needed.

Lemma 8.6 (Cardinality envelopes). *Let f_e be such a factor and $\hat{\varphi}_e$ its piecewise-affine interpolation between integers. Then*

$$\text{vex } f_e(p) = \hat{\varphi}_e\left(\sum_{i \in e} p_i\right).$$

A prescribed-mean law supported on the two adjacent cardinalities attains this minimum. One common-threshold law maximizes every such factor and their sum. If $p_{(1)} \geq \dots \geq p_{(d)}$ are the local means, its upper value is

$$\varphi_e(0) + \sum_{j=1}^d [\varphi_e(j) - \varphi_e(j - 1)]p_{(j)}.$$

Proof. Jensen's inequality gives the lower bound for the convex function $\hat{\varphi}_e$. The cube slab between the two adjacent integer sums is integral: two fractional coordinates admit an opposite perturbation;

one fractional coordinate cannot make an integer sum bound tight and admits a small perturbation. Decompose p into its binary vertices to attain equality. Discrete convexity implies supermodularity of the cardinality set function. Among laws maximizing a factor, choose one maximizing also $\mathbb{E}|X|^2$. If two incomparable support sets have positive mass, replace equal mass on them by their intersection and union. Singleton means are unchanged; supermodularity cannot decrease the primary value, while the secondary value strictly increases by twice the product of the two nonempty set differences' sizes. This is impossible. The maximizer is chain-supported, and the chain law with prescribed means is precisely the common-threshold law. The same law therefore maximizes each factor individually, and summing establishes simultaneous upper attainment. Integrating its cardinality gives the displayed formula. $\square$

The single-factor principles are classical; symmetric envelope formulas appear in Sherali [21], and the common-aspect product formula in Adams et al. [1, Proposition 4.1].

Theorem 8.7 (Cardinality odd-girth bound). *For a sum of convex cardinality factors of frequency at most two, with each entire factor kept in the relaxation, the sharp constants of Theorem 8.5 hold unchanged.*

Proof. Use success probabilities in the degree slab, and decompose $p = \mathbb{E}Z$ as before. Each local lower envelope is affine on its allowed slab, so

$$\mathbb{E} \text{vex } f_v(Z) = \text{vex } f_v(p), \quad \mathbb{E}T(Z) \leq T(p),$$

where the second assertion also uses concavity of each upper envelope. At a fractional-cycle vertex v , let m count its incident integral ones. Its local lower value is $\varphi_v(m+1)$, and its upper value is $[\varphi_v(m) + \varphi_v(m+2)]/2$, since precisely two coordinates are half-valued. Matching/complement rounding on a length- l cycle selects zero or two of them with probability $1/(2l)$ each, and one otherwise. Therefore the expected local cost is exactly $\text{vex } f_v(Z) + T_v(Z)/l$. Off the cycles the local gap is zero. Averaging gives

$$\text{vex } f(p) \leq \mathbb{E}f(X) \leq \sum_v \text{vex } f_v(p) + T(p)/g.$$

Subtract from the common upper value. Bipartite slabs are integral and need no rounding. Positive bilinear odd cycles prove sharpness. $\square$

Corollary 8.8 (Common-aspect physical products). *For original positive products on boxes $[\ell_i, \rho_e \ell_i]$, $\ell_i > 0$, assume each scope e has a common coordinate aspect ratio $\rho_e > 1$. Then the frequency-two odd-girth theorem applies to the original factors. Its constant is sharp for every fixed common ratio.*

Proof. A normalized binary vertex gives the table $a_e(\prod_{i \in e} \ell_i) \rho_e^K$, which is discrete convex. Shared variables enforce ratio compatibility, so different connected components may have different ratios. No new factors are introduced. For a bilinear odd cycle on $[1, \rho]^n$, expansion gives affine terms plus $(\rho - 1)^2$ times the unit-cube bilinear polynomial. Both gaps scale by the same positive number. Fixed coordinates are substituted first, and ratio one is trivial. $\square$

8.4 Rational scalar-envelope algorithms

Proposition 8.9 (Frequency-two binary oracle). *For a rational positive polynomial of frequency at most two on a unit cube and arbitrary rational λ , one can minimize $f(X) - \lambda^\top X$ over binary X in polynomial bit time.*

Proof. With $Z = 1 - X$, the objective is

$$-\sum_i \lambda_i + \sum_i \lambda_i Z_i + \sum_v a_v \mathbf{1}\{v \text{ uncovered by } Z\}.$$

Dummy vertices have zero penalties. Include every negative-cost edge: adding it strictly lowers cost and cannot increase an uncovered penalty. Add its cost to the constant, remove it from the remaining choices, and set its endpoints' penalties to zero, retaining those vertices. Remaining costs and penalties are nonnegative. Add a hub h , its mate k , their zero-cost edge hk , and an edge vh of cost equal to each vertex v 's penalty. Any selected original edge set extends to an edge cover by paying exactly its uncovered penalties and adding hk . Conversely, retaining original edges of an augmented cover leaves uncovered only vertices whose penalty edges were paid. These two comparisons prove value equality with minimum-weight edge cover. For completeness this reduces to matching: for a nonnegative-cost graph without isolated vertices let $\mu(v)$ be its cheapest incident cost. From a matching M , add a cheapest edge at each unmatched vertex. Its cover cost is at most $\sum_v \mu(v) - \sum_{uv \in M} (\mu(u) + \mu(v) - c_{uv})$. Conversely an inclusion-minimal cover is a union of stars (a path of three edges or a cycle would have a removable edge). Choose one edge from each star as a matching. Every other edge can be charged to its unmatched leaf, with $\mu(v)$ no larger than its cost. This shows the minimum of the displayed expression is no larger than the minimum cover cost. Hence equality holds and maximum-weight matching supplies an optimal cover. Parallel edges can keep their cheapest representative. Weighted matching is polynomial-time [10]. $\square$

Proposition 8.10 (Convex-degree matching oracle). *For explicitly listed rational discrete-convex tables and arbitrary rational edge costs c_i , the binary problem*

$$\min_{J \subseteq E} \sum_v \varphi_v(\deg_J(v)) + \sum_{i \in J} c_i$$

is solvable in polynomial bit time.

Proof. At vertex v create d_v slots with nondecreasing charges $\Delta_v(j) = \varphi_v(j) - \varphi_v(j-1)$. For each edge $i = uv$, create mandatory vertices i_u, i_v , joined by an inactive zero-cost edge. Connect i_u to all slots of u with cost $\Delta_u(j) + c_i$, and i_v to all slots of v with cost $\Delta_v(j)$. A matching covering all mandatory vertices either uses the inactive edge or uses two slots; mixed activation cannot cover both endpoints. Thus it represents one binary choice per original edge. Every chosen set is representable, and the occupied slots at a vertex can be reassigned to its cheapest k slots because their available gadget neighbors are identical. Their charges telescope to $\varphi_v(k) - \varphi_v(0)$.

To enforce mandatory coverage by ordinary unconstrained matching, let W be the sum of absolute original gadget edge costs and subtract $M = 2W + 1$ per mandatory endpoint covered by each edge. The all-inactive matching covers every mandatory vertex. Missing even one such vertex loses bonus M , exceeding any possible original cost difference. Thus an optimum covers them all, also with negative increments or negative c_i . Adding $\sum_v \varphi_v(0)$ and undoing the fixed bonus recovers the desired value. The gadget has $O(|E|)$ vertices and $O(\sum_v d_v^2)$ edges with polynomial rational encoding length. This is the incremental-slot matching mechanism of convex degree optimization; compare Deza and Onn [9, Theorem 1.2 and Section 3]. $\square$

Corollary 8.11 (Exact scalar envelopes and supporting minorants). *Both preceding factor classes admit exact rational convex-envelope evaluation, a supporting affine minorant, and scalar graph-hull separation in polynomial bit time. This includes rational common-aspect physical products. No polynomial-size explicit description of the full polytope carrying all individual factor values is asserted.*

Proof. The finite vertex-law LP has dual

$$\max_{\eta, \lambda} \{ \eta + \lambda^\top p : \eta + \lambda^\top z \leq f(z) \ (z \in \{0, 1\}^n) \}.$$

Its separation oracle is precisely binary minimization of $f(z) - \lambda^\top z$, with arbitrary signed linear costs. Affine summands are absorbed into these costs and restored in the value; unused coordinates optimize independently. Let $A = \sum_v \max_j |\varphi_v(j)|$, or $A = \sum_v a_v$ for positive monomials, adding the absolute coefficient sum of separate affine terms if present. Then $|f(z)| \leq A$. The dual polyhedron contains no line, since the rows $(1, z)$ span $\mathbb{R}^{n+1}$, and its attained optimal face contains a vertex. Such a vertex solves $n + 1$ independent tight equations with a zero-one matrix. Cramer's rule bounds each coordinate by $(n + 1)!A$; its nonzero integer denominator determinant has magnitude at least one. Clearing input denominators also gives polynomial bit length for each coordinate. Thus the box of radius $\max\{1, (n + 1)!A\}$ retains an optimum and has polynomial encoding size. Rational separation-to-optimization [12, Theorem 6.4.9] gives the exact optimum and minorant, also at boundary means. Every vertex inequality extends to the whole cube by multiaffine interpolation. Sorting gives the common-threshold upper value and a supporting upper affine function; together with cube bounds these separate the scalar graph hull. Rational common-ratio products generate their convex tables by polynomially many rational multiplications. $\square$

9 The exact incidence-treewidth-two constant

Theorem 9.1 (One-sided cycle coloring). *In a finite simple bipartite graph $G = (V, F; I)$ of treewidth at most two, one can color the factor side F with two colors so that every cycle whose factor vertices all have one color contains an even number of factor vertices. Consequently every zero-one matrix with this bipartite support has a partition of its rows into two totally unimodular matrices.*

The all-cycle assertion is stronger than excluding induced odd-factor cycles (balancedness). The stronger statement will supply total unimodularity for arbitrary integer cardinality slabs.

Proof. We prove a two-terminal series-parallel invariant. A network starts with an edge; series composition identifies one terminal of each of two otherwise disjoint networks, and parallel composition identifies both terminal pairs. Intermediate networks are simple and bipartite subgraphs of the eventual graph. A coloring is good if its monochromatic-factor cycles have even factor count. A terminal path is c -monochromatic when all its factor vertices, including factor terminals, have color c . Its parity counts those factor vertices modulo two.

For each network there is a parity $\epsilon \in \{0, 1\}$ with all the following witnesses available. They are possibly different colorings. For V, V terminals and either color c , an *active* good coloring has at least one c -monochromatic terminal path, all such paths have parity ϵ , and no path is monochromatic in the other color. If $\epsilon = 0$, a *blocking* good coloring with neither type of monochromatic terminal path is also available. For mixed terminals, with either prescribed color c of the factor terminal, an active good coloring has paths of color c , all of parity ϵ ; a blocking witness with that prescribed terminal color is available whenever there is no direct terminal edge. For F, F terminals, with either prescribed common color c , the same active witness exists; if $\epsilon = 1$ a blocking witness with those colors also exists. In addition, for either assignment of distinct terminal colors, a good coloring exists. Such colors automatically block monochromatic terminal paths. Finally a network with a direct terminal edge is mixed and uses $\epsilon = 1$.

The base edge satisfies the mixed invariant with parity one. For a series composition let δ be one if the shared vertex is a factor, and zero otherwise. The output parity is $\epsilon = \epsilon_1 \oplus \epsilon_2 \oplus \delta$. Every

path concatenates component terminal paths; subtracting the shared factor once accounts for δ . Every cycle stays in a component. Active witnesses concatenate by using the same desired color and matching the shared factor's color when present. The required additional witnesses have the following complete case analysis.

1. For outer types V, V and a shared variable, use opposite active colors in the two V, V components. No monochromatic path can cross both.
2. For outer types V, V and a shared factor, blocking is required only if the component parities differ. The even mixed component has no direct edge and can block with the prescribed shared-factor color; use an active witness of that color in the other component.
3. For mixed outer types and a shared variable, use the active color opposite to the outer factor's color in the V, V component and use that prescribed factor color in the mixed component. This blocks.
4. For mixed outer types and a shared factor, give the shared factor the color opposite to the outer factor. The F, F component uses its distinct-terminal witness, and the mixed component uses a compatible active witness. This again blocks.
5. For outer types F, F of common color and a shared variable, blocking is required only for output parity one, so the two mixed component parities differ. Block the even component, which has no direct edge, and activate the other, respecting both outer colors.
6. For outer types F, F of common color and a shared factor, color the shared factor oppositely. Both F, F components have their required distinct-terminal good witnesses and block.
7. For outer F, F terminals of distinct colors and a shared variable, activate each mixed component in its outer terminal's color. If instead the shared vertex is a factor, give it either color and use in each component the equal-color active or distinct-color witness as appropriate.

These cases include reversed mixed types and exhaust all triples of terminal types. A nontrivial series composition has no direct outer edge, so its mixed blocking requirement has also been checked.

In a parallel composition every new simple cycle consists of one terminal path from each component. If the paths are monochromatic in one color, its factor parity is their parity XOR plus τ , the number of factor terminals modulo two. Thus $\tau = 0$ for equal terminal types and $\tau = 1$ for mixed types. We check the invariant as follows. For V, V terminals with equal component parities, activate both in the same color. If both are even, both may also block. For different parities, activate the odd component and block the even component. Output parity is $\epsilon_1 \vee \epsilon_2$; whenever both components contribute paths their equal parities make new cycles even. For F, F terminals of common prescribed color, activate both when parities agree; if both are odd, both may also block. If parities differ, activate the even component and block the odd one. Output parity is $\epsilon_1 \wedge \epsilon_2$. With distinct prescribed terminal colors, use their good witnesses in both components; a crossing monochromatic cycle would contain both differently colored terminals and is impossible. For mixed terminals with no direct edge, both components can block in the prescribed factor color. Activate either one and block the other for an active witness, choosing its parity as output, or block both for the blocking witness. For mixed terminals with a direct edge, simplicity implies exactly one component contains it. Activate its parity-one witness and block the edge-free component. No blocking output is required. These mixed choices create no crossing monochromatic cycle. The induction is complete.

A treewidth-two graph has no K_4 minor, hence no subdivision of K_4 . The classical block characterization supplies a two-terminal series-parallel representation for each nontrivial block [13, Theorem 3.1]. Apply the invariant to each block. Root the block-cut forest. If a child block attaches at a factor, interchange its two colors if needed to match the articulation color; if it attaches at a variable no matching is needed. Bridges use the base edge and isolated factors receive arbitrary colors. Every cycle lies in one block, proving the graph assertion.

For the matrix assertion use Camion's criterion: a $0, \pm 1$ matrix is totally unimodular exactly when every Eulerian submatrix has entry sum divisible by four [6, Theorem 6.5]. In either color class an Eulerian submatrix has even row and column degrees. Its bipartite support decomposes into edge-disjoint simple cycles. Each has an even number of factor vertices, hence length divisible by four. Its total number of ones is consequently divisible by four, as required. The argument covers rectangular Eulerian submatrices too. $\square$

Theorem 9.2 (Sharp width-two gap). *A sum of discrete-convex cardinality factors with incidence treewidth at most two satisfies $T \leq 2H$. This includes positive monomials on unit cubes and zero-lower boxes, and original positive products with a common aspect ratio within each scope. The sharp supremum is two, including for every fixed common positive aspect ratio $\rho > 1$.*

Proof. Partition factor rows into two TU classes by Theorem 9.1. For a class C and prescribed means p , the polytope

$$0 \leq z \leq 1, \quad \left\lfloor \sum_{i \in e} p_i \right\rfloor \leq \sum_{i \in e} z_i \leq \left\lceil \sum_{i \in e} p_i \right\rceil \quad (e \in C)$$

is integral and contains p . Indeed row sign changes, duplicate rows, and unit rows preserve total unimodularity; each basic solution with integer right-hand side is integral by its determinant ± 1 . A binary-vertex decomposition of p attains all class lower envelopes by Lemma 8.6. Under that law every other factor has nonnegative expected deficiency from its own upper value. The equal mixture of the two class laws therefore attains at least half the sum of local gaps. Their common upper extremizer makes this a lower bound for half the full termwise gap in the scalar hull gap.

For positive unit-cube monomials one can equivalently use failure rows $\sum_{i \in e} z_i \leq 1$ when their mean sum is at most one and $\sum_{i \in e} z_i \geq 1$ otherwise. A classwise integral decomposition attains the failure union bound in every row. The mixed unit-right-hand-side integrality theorem for balanced matrices suffices for this particular argument [6, Theorem 6.13]; it would not justify arbitrary integer cardinality slabs without the stronger TU conclusion. Common-aspect products are cardinality factors by Corollary 8.8. Proposition 8.3 proves the unit-cube lower supremum two.

For sharpness at a fixed positive ratio, put $\epsilon = 1/\rho$ and $\alpha = 1 - \epsilon$. On $[\epsilon, 1]^{n+1}$ use

$$f_n(a, x) = \alpha^{-1} a \sum_{i=1}^n x_i + \prod_{i=1}^n x_i, \quad \mathbb{E}A = 1/n, \quad \mathbb{E}X_i = 1 - 1/n,$$

where $a = \epsilon + \alpha A$ and $x_i = \epsilon + \alpha X_i$ are endpoint variables. The normalized nonlinear bilinear terms are $\alpha A X_i$, so their gaps sum to α . The large product is ϵ^R for failure count R , with $\mathbb{E}R = 1$. Its lower value is ϵ by adjacent-cardinality attainment, and its upper value $1 - (1 - \epsilon^n)/n$. Hence

$$T_n = 2\alpha - \frac{1 - \epsilon^n}{n}, \quad H_n = \max \mathbb{E}[\alpha AR + 1 - \epsilon^R] - \frac{1 - \epsilon^n}{n}.$$

For any $M > 0$, $1 - \epsilon^r \leq \alpha r$ on integer $r \geq 0$ gives

$$\mathbb{E}[1 - \epsilon^R] \leq \alpha(1 - \mathbb{E}[R; R > M]) + 1/M, \quad \alpha \mathbb{E}[AR] \leq \alpha M/n + \alpha \mathbb{E}[R; R > M].$$

Markov's inequality was used in the first bound. Thus $H_n \leq \alpha + \alpha M/n + 1/M$. Exactly one uniformly failed leaf and an independent anchor give $H_n \geq \alpha + \alpha/n - (1 - \epsilon^n)/n$. Taking $M = \sqrt{n}$ yields $H_n \rightarrow \alpha$ and $T_n/H_n \rightarrow 2$. The incidence graph is the same width-two graph as before; all means are interior. Scaling physical coordinates by ρ maps the box to $[1, \rho]^{n+1}$ and preserves the gaps of the correspondingly rescaled positive polynomial. $\square$

9.1 What fails for general nonnegative factors

Example 9.3 (An exact width ratio of three). Use six fair variables in pairs AB, BC, CA , and put

$$E(u, v) = 1 - u - v + 2uv, \quad N(u, v) = u + v - 2uv.$$

These are the multiaffine equality and inequality indicators, both in $[0, 1]$ on the square. Define

$$g_A = E(AB)E(CA), \quad g_B = N(AB)E(BC), \quad g_C = N(BC)N(CA).$$

Each local upper envelope is one at the center: mix a satisfying assignment and its complement. Each local lower envelope is zero, by the analogous mixture of a falsifying assignment. No two factors can equal one on a binary assignment, because adjacent factors demand opposite parities on their common pair. Thus the sum's upper envelope is at most one, attained by the all-zero/all-one mixture, which satisfies g_A . The pair assignment $AB = 01, BC = 01, CA = 00$ and its complement make all factors zero, proving the sum's lower envelope zero. Therefore $T = 3$ and $H = 1$ for these three entire factors.

A central bag $\{A, B, C\}$ and six attached bags $\{A, B, X_{AB,j}\}$, $\{B, C, X_{BC,j}\}$, $\{C, A, X_{CA,j}\}$, $j = 1, 2$, form a width-two decomposition. A four-cycle proves the width is exactly two. Every variable has frequency two. The factors are nonnegative on the full cube, but their monomial coefficients have mixed signs, so neither the positive-monomial nor the convex-cardinality factor theorem applies.

Proposition 9.4 (Independent-set payoff family). *For a simple graph G with nonnegative vertex weights w_v , place two fair bits on each edge and assign its endpoints opposite demanded parities. Give vertex v payoff w_v if all its incident demanded parities hold, and zero otherwise. The sum of separate maximum expected payoffs is $\sum_v w_v$, while the global maximum is $\alpha_w(G)$, the maximum independent-set weight. With all constructed factor scopes retained, including zero-weight factors, a graph with an edge has incidence treewidth $\max\{2, \text{tw}(G)\}$.*

Proof. Each local satisfaction pattern and its complement attain w_v with fair means. Satisfied vertices form an independent set, since the two endpoints of an edge demand opposite parities. Conversely fix a maximum independent set, give each incident edge its selected endpoint's demand, and choose remaining parities arbitrarily. Complement-pair mixing satisfies all selected factors with fair means, proving the maximum $\alpha_w(G)$; any extra satisfied factors still form an independent set and cannot improve that maximum. To a decomposition of G attach two size-three bags $\{u, v, X_{uv,j}\}$ for every edge at a bag containing u, v . This proves the treewidth upper bound. Contracting one length-two path per edge and deleting the other gives G as a minor; each edge also creates a four-cycle, proving both lower bounds. $\square$

Thus K_{k+1} with unit vertex weights gives maximum-payoff ratio $k + 1$ at incidence width k for $k \geq 2$. A general payoff maximum ratio is not an envelope-width ratio: the minima must also be computed. Example 9.3 does compute them and gives the claimed exact width obstruction.

9.2 Limits of the coloring method

A distinct open question at incidence treewidth two asks whether the factors admit two classes whose incidence subgraphs are each chordal bipartite, meaning they contain no induced cycle of length at least six. The proved all-cycle parity property permits induced eight-cycles and does not establish this target. The recorded experiment found such a partition for all 5,000 generated supports; this is finite evidence only.

The all-cycle coloring invariant cannot extend with a bounded number of colors to width three. In $K_{3,m}$, $m \geq 3$, with its three-vertex side interpreted as variables, every three factors lie on a six-cycle. Each permitted color class therefore contains at most two factors, requiring $\lceil m/2 \rceil$ colors, although the graph has treewidth three. Its all-ones matrix is already TU: determinants of order at least two vanish. The six-cycles have chords and do not obstruct balancedness. This separates the stronger coloring mechanism from a TU-row-partition target.

Finite searches found three balanced classes in 3,000 generated width-at-most-three supports (212 required more than two), and three TU classes in 300 further supports. Their accepted widths were certified by particular elimination orders; rejected greedy orders did not certify larger width. The TU tests enumerated column signings by the Ghouila-Houri criterion and used exact coloring backtracking. These searches are finite evidence, not proofs of universal partitions. The exact W_k for $k \geq 3$, k -class balanced or TU partitions, and sharp planar constants remain open here. A planar bipartite incidence graph does admit an orientation of maximum outdegree at most two: Hall's theorem assigns every edge to one of two slots at an endpoint, since every edge set uses at most twice as many edges as incident vertices. Orient from the assigned endpoint. Thus the general orientation theorem gives a finite planar bound, without settling its sharp value. Auxiliary reductions and the simpler forest-coloring obstruction are recorded in Appendix D.

10 Positive boxes and coefficient regularity

Let $C_{\text{box}}(\rho_B)$ be the supremal gap ratio for positive monomial factorizations on strictly positive boxes with every coordinate aspect ratio $u_i/\ell_i \leq \rho_B$, over arbitrary dimensions and evaluation points with positive hull gap. Unlike C_δ , this restricts the box, not the normalized evaluation point. Fixed coordinates are removed.

10.1 A coefficient inequality with one ambient orientation law

Fix an ambient dimension $N \geq 2$. Choose a uniform subset $J \subseteq [N]$ of size $\lfloor N/2 \rfloor$ and an independent uniform $U \in [0, 1]$. Coordinate i of mean p_i succeeds on $[0, p_i]$ if $i \in J$ and on $[1 - p_i, 1]$ otherwise. Call this balanced orientation law O . It has all prescribed means, though for odd N its individual orientation choices are not fair. The probability that a fixed pair has opposite orientations is

$$q_N = \frac{2\lfloor N/2 \rfloor \lceil N/2 \rceil}{N(N-1)}, \quad \beta_N = q_N^{-1} = \begin{cases} 2 - 2/N, & N \text{ even,} \\ 2 - 2/(N+1), & N \text{ odd.} \end{cases}$$

All local supports and smaller-dimensional induction steps below use restrictions of this one law. Deleting a deterministic coordinate neither conditions nor resamples the orientation subset.

For a support S of size $d \leq N$, let $K = \sum_{i \in S} X_i$. Write C_j, P_j, O_j for $\mathbb{E} \binom{K}{j}$ under common-threshold, independent, and ambient orientation rounding. With $s = \sum_{i \in S} p_i$, $k = \lfloor s \rfloor$, and $\theta = s - k$, put

$$V_j = (1 - \theta) \binom{k}{j} + \theta \binom{k+1}{j}.$$

Degree-zero moments are one; negative orders and orders above support size are zero. By Lemma 8.6, C_j and V_j are the true upper and lower envelope values of the elementary symmetric polynomial of degree j .

Lemma 10.1 (Finite spreading inequality). *For every support $S \subseteq [N]$, every mean vector, and every integer $j \geq 0$,*

$$P_j - V_j \leq \beta_N(C_j - O_j) + (C_{j-1} - P_{j-1}). \quad (10.1)$$

The same statement holds with β_N replaced by 2 and independently fair endpoint orientations in every coordinate.

Proof. Write β for the indicated constant and set $F_{S,j} = \beta C_j + V_j - P_j - \beta O_j + C_{j-1} - P_{j-1}$. Orders zero and one vanish; order $d+1$ is $C_d - P_d \geq 0$, and higher orders vanish. For order two let $L_2 = \sum_{a < b} \max(0, p_a + p_b - 1)$. Equal orientations give a pair's upper Fréchet value; opposite orientations give its lower value. Their opposite probability is $1/\beta$, including under restriction. Thus $O_2 = (1 - 1/\beta)C_2 + L_2/\beta$, and

$$F_{S,2} = (C_2 - P_2) + (V_2 - L_2) \geq 0.$$

The first inequality follows from upper optimality, the second because every law, including an adjacent-cardinality law, respects every pair lower bound.

We induct on support size. A mean-zero coordinate is deterministic and its deletion gives $F_{S,j} = F_{S \setminus \{i\},j}$. A mean-one coordinate and Pascal's identity give $F_{S,j} = F_{S \setminus \{i\},j} + F_{S \setminus \{i\},j-1}$. The adjacent-cardinality values obey the same identity after shifting the sum by one. These are restrictions of the ambient law, not newly sampled balanced laws, so its correlated coins cause no exception. All boundary points follow by induction; dimension one is immediate.

For an interior vector and $3 \leq j \leq d$, select distinct coordinates of global minimum x and global maximum y . Move them to $x - h, y + h$ with $0 \leq h \leq \min(x, 1 - y)$. The sum stays fixed, so $\Delta V_j = 0$. In increasing order of means,

$$C_j = \sum_{i=1}^d p_{(i)} \binom{d-i}{j-1}.$$

Putting the selected minimum first and maximum last also handles ties. Hence, with $B = \binom{d-1}{j-1}$ and $D = \binom{d-1}{j-2}$, $\Delta C_j = -Bh$ and $\Delta C_{j-1} = -Dh$; the last coordinate has zero coefficient since $j \geq 3$. If E_r is the elementary symmetric polynomial in the other $d-2$ means, the unchanged sum and product change $\Delta(xy) = -h(y - x + h)$ give exactly

$$\Delta(-P_j - P_{j-1}) = h(y - x + h)(E_{j-2} + E_{j-3}) \leq Dh.$$

Here $0 \leq y - x + h \leq 1$ and $E_r \leq \binom{d-2}{r}$. Finally couple both orientation vectors using the same U and coins. Increasing a single mean by h changes only that bit, monotonically, on an event of probability h . The increase in $\binom{K}{j}$ is then $\binom{K_{\text{rest}}}{j-1} \leq B$. Decreasing the minimum first and increasing the maximum second therefore gives $\Delta O_j \geq -Bh$, regardless of dependence among orientation coins. Combining the finite changes yields $\Delta F_{S,j} \leq -\beta Bh - Dh + Dh + \beta Bh = 0$. Take $h = \min(x, 1 - y)$ to reach a boundary point, whose coefficient is nonnegative by induction. The original coefficient is at least that boundary value. No differentiability assumption is used. $\square$

The choice of global extrema in this proof matters. For independently fair orientations, the stronger Schur-concavity assertion is false:

$$\begin{aligned} F_{[4],3}(2/5, 7/10, 24/25, 97/100) &= 6201/12500, \\ F_{[4],3}(2/5, 7/10, 959/1000, 971/1000) &= 4961031/10000000. \end{aligned} \quad (10.2)$$

Spreading the last two coordinates by $1/1000$ increases the coefficient by $231/10000000$. These rational values follow by averaging the sixteen orientation patterns, integrating between their interval endpoints, and evaluating the displayed symmetric moments. They refute arbitrary pair spreading, without affecting the global-minimum/global-maximum comparison.

Theorem 10.2 (Positive-box upper bounds). *For $N \geq 2$ nonfixed coordinates and $\rho_B > 1$, every positive multilinear polynomial on a strictly positive box of aspect ratios at most ρ_B satisfies*

$$T \leq (\rho_B + \beta_N)H \leq (\rho_B + 2)H.$$

The common mixture of independence and balanced orientation with weights $\rho_B/(\rho_B + \beta_N)$ and $\beta_N/(\rho_B + \beta_N)$ captures this fraction of each original monomial's exact gap simultaneously. For zero or one nonfixed coordinate all gaps vanish.

Proof. First consider a support on $[1, \rho_B]^N$ and put $t = \rho_B - 1$. Its binary physical value is $(1 + t)^K$, so its upper, independent, orientation, and lower values are $C(t) = \sum_j C_j t^j$, $P(t)$, $O(t)$, and $V(t) = \sum_j V_j t^j$. The last identity follows from a single adjacent-cardinality law attaining every order together. The coefficient of t^j in $(\beta_N + t)C + V - (1 + t)P - \beta_N O$ is $F_{S,j} \geq 0$. Consequently the true local gap satisfies

$$C - V \leq \rho_B(C - P) + \beta_N(C - O). \quad (10.3)$$

The same mixture works on all supports in the ambient N coordinates. Summing with nonnegative coefficients and using common upper attainment proves the common-aspect conclusion. Fair independent orientation coins and $\beta = 2$ prove $\rho_B + 2$ directly in every dimension.

For unequal endpoints $0 < \ell_i < u_i$ put

$$s_i = \frac{u_i/\ell_i - 1}{\rho_B - 1} \in (0, 1], \quad z_i = \ell_i[(1 - s_i) + s_i y_i], \quad y_i \in [1, \rho_B].$$

This is an affine bijection onto the original box. Each original monomial becomes a positive sum of physical y -monomials. Its true factor gap is at most the sum of their gaps, by (2.3), while the full hull gap is affine invariant. Apply the common-aspect bound to the expansion, always using restrictions of the same ambient orientation law. Moreover the original factor's upper value is the sum of the expanded upper values by common-threshold attainment. Its deficiency under either law is thus the sum of expanded deficiencies, proving the claimed per-original-factor guarantee. The certificate can be sampled without forming the expansion: the endpoint laws pull back to the original endpoints. Box inclusion alone would not prove any of these comparisons. $\square$

Theorem 10.3 (Coefficient-regularity cardinality bound). *In ambient dimension $N \geq 2$, suppose each entire local factor is the multiaffine interpolant of*

$$\psi_e(k) = \sum_{j=0}^{d_e} a_{e,j} \binom{k}{j}, \quad a_{e,j} \geq 0, \quad a_{e,j+1} \leq L a_{e,j} \quad (j \geq 2),$$

where $L \geq 0$ is common to all factors and coefficients beyond the degree are zero. Then

$$T \leq (L + 1 + \beta_N)H.$$

Separate affine terms may be added freely. Fair independent endpoint orientations give the dimension-free bound $L + 3$. These are bounds for the gaps of the entire cardinality factors, not the gaps obtained by separately relaxing their expanded monomials.

Proof. The nonnegative binomial coefficients make each sequence discrete convex: its second difference is $\sum_{j \geq 2} a_{e,j} \binom{k}{j-2} \geq 0$. Lemma 8.6 supplies a common upper extremizer, and within each factor one adjacent-cardinality law attains every V_j together. Denote that factor's four values by $C_\psi, V_\psi, P_\psi, O_\psi$. Multiply (10.1) by its a_j and sum. Since $C_j - P_j \geq 0$ and the order-zero and order-one deficiencies vanish, reindexing gives

$$\begin{aligned} P_\psi - V_\psi &\leq \beta_N(C_\psi - O_\psi) + \sum_{j \geq 2} a_{j+1}(C_j - P_j) \\ &\leq \beta_N(C_\psi - O_\psi) + L(C_\psi - P_\psi). \end{aligned}$$

Add $C_\psi - P_\psi$ to obtain

$$C_\psi - V_\psi \leq (L+1)(C_\psi - P_\psi) + \beta_N(C_\psi - O_\psi).$$

One global mixture with weights proportional to $L+1$ and β_N therefore captures the required fraction for all factors. Summing and using common upper attainment proves the theorem. This also handles $L=0$, deterministic means, and affine factors. With at most one nonfixed coordinate the interpolants are affine and all gaps vanish. The product sequence ρ_B^k has $a_j = (\rho_B - 1)^j$, so $L = \rho_B - 1$ recovers the common-aspect product inequality. $\square$

This refinement is a direct consequence of the coefficient mechanism; no optimality or independent priority is asserted. Distinct earlier low/high and asymmetric proofs, and the precise limitation of fixed independence/orientation mixtures, appear in Appendix E.

10.2 A lower family on each fixed positive box

Theorem 10.4 (Aspect-ratio lower bound). *For every $\rho_B > 1$,*

$$\max\{2, \rho_B\} \leq C_{\text{box}}(\rho_B) \leq \rho_B + 2.$$

Thus $C_{\text{box}}(\rho_B) = \rho_B + O(1)$ as $\rho_B \rightarrow \infty$. The exact finite-aspect-ratio constant is not determined.

Proof. The lower bound two already holds at width two by Theorem 9.2. For the lower bound ρ_B , fix $\epsilon = 1/\rho_B \in (0, 1)$ and put $\alpha = 1 - \epsilon$. Use the radix supports with $L \geq 2$, $b = L^2$, $m = b^L$, now on $[\epsilon, 1]^{m+L}$ with normalized means $\mathbb{E}A_j = b^{-j}$ and $\mathbb{E}Z_i = 1 - m^{-1}$. A level- j term has $k = m/b^j$ leaves and total failure mean $(1 - b^{-j}) + k/m = 1$. Its vertex value is ϵ^{R_e} , so its exact lower value is ϵ , attained by exactly one failure with the prescribed categorical probabilities. Common-threshold rounding gives its upper value

$$C_{j,Q} = \epsilon + \alpha b^{-j} - \frac{\epsilon}{m}(1 - \epsilon^k).$$

Indeed its successive threshold states have product values $1, \epsilon, \epsilon^{k+1}$ with interval lengths $b^{-j}, 1 - m^{-1} - b^{-j}, m^{-1}$. Therefore

$$T_L = \alpha L - D_L, \quad D_L = \epsilon \sum_{t=0}^{L-1} \frac{1 - \epsilon^{b^t}}{b^t}, \quad 0 \leq D_L \leq \frac{\epsilon b}{b-1}. \quad (10.4)$$

For any common endpoint law let R_Q count failed leaves in block Q and R count them in total. Since $\mathbb{E}R = 1$, subtracting expected physical products from their common upper value gives exactly

$$H_L = \max \mathbb{E} \left[\alpha \sum_j A_j \sum_{Q \in \mathcal{P}_j} (1 - \epsilon^{R_Q}) + \epsilon \sum_j \sum_{Q \in \mathcal{P}_j} (1 - \epsilon^{R_Q}) \right] - D_L. \quad (10.5)$$

For its first term use $1 - \epsilon^{R_Q} \leq \mathbf{1}\{R_Q > 0\}$ and the unit-cube radix hull upper bound. For its second use $1 - \epsilon^r \leq \alpha r$ and $\sum_{Q \in \mathcal{P}_j} R_Q = R$. This proves

$$H_L \leq \epsilon \alpha L + \alpha[1 + (L - 1)/b] - D_L.$$

Exactly one uniformly failed leaf, with independent anchors of the required means, instead gives

$$H_L \geq \epsilon \alpha L + \alpha^2 \sum_{j=1}^L b^{-j} - D_L.$$

The same geometric inequality gives $D_L \leq \epsilon \alpha L$, so this last bound is strictly positive for every finite L . The bounded correction in (10.4) now implies $T_L/L \rightarrow \alpha$, $H_L/L \rightarrow \epsilon \alpha$, and their ratio tends to ρ_B . Scale coordinates by ρ_B to get positive coefficients on $[1, \rho_B]^{m+L}$; both gaps are preserved. This is a limit in dimension for each fixed aspect ratio, requiring no exchange of those limits. The upper bound is Theorem 10.2. $\square$

10.3 Unequal aspect ratios at frequency two

Example 10.5 (A rational bipartite ratio 7/6). For $f(x, y, z) = xy + xyz$ on $[1, 2] \times [1, 2] \times [1, 3]$, evaluate at $(5/4, 5/4, 5/2)$. Put $x = 1 + a, y = 1 + b, z = 1 + 2c$, so the normalized mean is $(1/4, 1/4, 3/4)$. The following valid affine bounds give exact values:

$$xy \geq 1 + a + b, \quad xyz \geq 2a + 2b + 3c, \quad f \geq 4a + 4b + 4c, \quad f \leq 2 + 8a + 4b + 2c.$$

Validity follows by substitution at all eight binary vertices and multiaffine interpolation. The respective attaining distributions are

xy lower	000, 001, 011, 101 each of mass 1/4
xyz lower	001 of mass 3/4; 110 of mass 1/4
f lower	001 of mass 1/2; 011, 100 each of mass 1/4
f upper	000 of mass 1/4; 001 of mass 1/2; 111 of mass 1/4.

Each row has the required means. The separate upper bounds $xy \leq 1 + 2a + b$ and $xyz \leq 1 + 6a + 3b + 2c$ are attained by the last law. Thus $T = 13/2 - 19/4 = 7/4$, while $H = 13/2 - 5 = 3/2$. The dual multigraph is bipartite: its two factor vertices share two parallel edges, with a dummy leaf for z . This disproves unequal-box bipartite scalar exactness, while leaving the common-aspect theorem intact.

Proposition 10.6 (Bipartite positive-box boundary). *For arbitrary nonnegative boxes with a bipartite dual graph, positive monomial gaps satisfy $T \leq 2H$. Their worst ratio over strictly positive boxes is at least 3/2, already with two factors and three variables. Its exact value between 3/2 and 2 remains open here, as does a universal 3/2 bound for unequal-box frequency-two instances.*

Proof. More generally a proper coloring of the term conflict graph into q classes gives $T \leq qH$: terms in one class have disjoint scopes, so join their minimizing laws independently and complete unused means. They attain their entire class gap, and other factors have nonnegative expected deficiency from their upper envelopes. Since those upper values are commonly attained, H is at least each class's total gap. Sum these inequalities. This also proves a bound two for any two positive terms regardless of their boxes or frequency.

For the lower family let $0 < \epsilon \leq 2$ and use $f_\epsilon = (2/\epsilon)xy + xyz$, with $x, y \in [1, 1 + \epsilon]$, $z \in [1, 3]$, at normalized means $(a, b, c) = (1/4, 1/4, 3/4)$. After removing affine parts the two factors are

$$A = 2\epsilon ab, \quad B = \epsilon^2 ab + 2\epsilon ac + 2\epsilon bc + 2\epsilon^2 abc, \quad C = A + B.$$

Their lower bounds at the point are attained at the first three laws of Example 10.5, respectively. The affine certificates are $A \geq 0$, $B \geq \epsilon^2(a + b + c - 1)$, and $C \geq 2\epsilon(a + b + c - 1)$. In binary order 000, 001, 010, 011, 100, 101, 110, 111, the last two residual lists are

$$\begin{aligned} &(\epsilon^2, 0, 0, \epsilon(2 - \epsilon), 0, \epsilon(2 - \epsilon), 0, \epsilon(\epsilon + 4)), \\ &(2\epsilon, 0, 0, 0, 0, 0, \epsilon^2, \epsilon(3\epsilon + 2)), \end{aligned}$$

which are nonnegative throughout the stated range. The local lower values sum to $\epsilon^2/4$ and the full lower value is $\epsilon/2$. The same common upper law as before gives $3\epsilon/2 + 3\epsilon^2/4$. Hence

$$T = \epsilon(3/2 + \epsilon/2), \quad H = \epsilon(1 + 3\epsilon/4), \quad \frac{T}{H} = \frac{2(\epsilon + 3)}{3\epsilon + 4} \rightarrow 3/2.$$

To obtain unit coefficients divide the polynomial by $2/\epsilon$ and put $z' = (\epsilon/2)z$. It becomes $xy + xyz'$ on $[1, 1 + \epsilon]^2 \times [\epsilon/2, 3\epsilon/2]$, with the same ratio. $\square$

For frequency-two products on unequal positive boxes, normalizing upper endpoints to one gives a binary failure cost $\prod_{i \in J \cap e} \alpha_i = \exp(-\sum_{i \in J \cap e} w_i)$, where $\alpha_i = \ell_i/u_i$ and $w_i = -\log \alpha_i$. Each edge has the *same* weight w_i at both incident factors. This is a weighted convex degree potential, whereas the matching theorem above uses unweighted cardinalities. Finite floating-point searches approaching $3/2$ from below do not supply an upper bound. If the dual multigraph is a forest, conditional gluing actually preserves all local minima and maxima even for signed multiaffine factors on arbitrary boxes; Appendix D gives the elementary argument.

11 Exact complexity of a single physical product

A ratio bound can be small even when evaluating an individual factor's exact envelope is difficult. For one factor the ratio is identically one wherever its gap is positive. The next result concerns computing that factor's envelope, not loss caused by separating interacting terms. All complexity statements use rational binary encodings and bit time.

Theorem 11.1 (Midpoint decision on narrow unequal boxes). *Given rational $u_i > 1$ and q , deciding whether*

$$\text{vex}_B m(\bar{x}) \leq q, \quad m(x) = \prod_{i=1}^n x_i, \quad B = \prod_i [1, u_i], \quad \bar{x}_i = (1 + u_i)/2,$$

is NP-complete. For every fixed rational $\eta > 0$ it remains NP-complete when $u_i \leq 1 + \eta$ for all i .

Proof. Reduce from PARTITION, the classical NP-complete integer partition problem [16]. Let its positive integers be $a_1, \dots, a_n$ with total A . Doubling them permits us to assume A even and at least two. Fix an integer $K \geq 1$ and put

$$\epsilon = (16KA^3)^{-1}, \quad u_i = 1 + \epsilon a_i.$$

These boxes satisfy $1 < u_i \leq 1 + 1/(16KA^2) < 2$. In normalized endpoint variables $X_i = 1 + \epsilon a_i Z_i$, the midpoint means are $\mathbb{E}Z_i = 1/2$. Write $S(z) = \sum_i a_i z_i$. On a binary vertex,

$$\prod_i (1 + \epsilon a_i z_i) = 1 + \epsilon S(z) + \frac{\epsilon^2}{2} \left(S(z)^2 - \sum_i a_i^2 z_i \right) + R(z). \quad (11.1)$$

All remainder coefficients are nonnegative. Its order- j elementary symmetric sum is at most A^j , and $\epsilon A \leq 1/16$, so

$$0 \leq R(z) \leq \frac{(\epsilon A)^3}{1 - \epsilon A} \leq 2\epsilon^3 A^3 = \epsilon^2/(8K) \leq \epsilon^2/8. \quad (11.2)$$

Every admissible law has $\mathbb{E}S = A/2$, whence

$$\mathbb{E}m(X) = b + \frac{\epsilon^2}{2} \text{Var}(S) + \mathbb{E}R(Z), \quad b = 1 + \frac{\epsilon A}{2} + \epsilon^2 \left(\frac{A^2}{8} - \frac{1}{4} \sum_i a_i^2 \right).$$

A partition state and its complement, with equal probabilities, have all means $1/2$ and $S = A/2$ deterministically. Thus a YES instance has lower envelope at most $b + \epsilon^2/8$. In a NO instance every binary state has integer $S \neq A/2$, so $\text{Var}(S) \geq 1$ under every feasible law. Its lower envelope is at least $b + \epsilon^2/2$. The rational threshold $q = b + \epsilon^2/4$ separates the cases. All numbers have polynomial encoding length, since $\log A$ is polynomial in the original input length. For a fixed $\eta > 0$, choose fixed K with $1/(16K) \leq \eta$ to enforce the narrower interval.

For NP membership, the vertex-law LP has equality rows $1, z_1, \dots, z_n$ and right-hand side $(1, 1/2, \dots, 1/2)$. An optimal basic solution has at most $n + 1$ positive probabilities. It provides that many binary states and rational weights of polynomial bit length: a nonsingular subsystem has zero-one entries and half-integer right-hand side, so determinant bounds apply. Each vertex product has polynomial bit length as a product of at most n input rationals. Exact arithmetic verifies nonnegative weights, total mass, all means, and expected product at most q in polynomial time. $\square$

The NO case also has an explicit affine certificate in normalized coordinates:

$$\ell(z) = 1 - \epsilon^2 A^2/8 + \epsilon^2/2 + \sum_i \left[\epsilon a_i + \frac{\epsilon^2}{2} (A a_i - a_i^2) \right] z_i. \quad (11.3)$$

Subtracting it from (11.1) gives $\epsilon^2[(S - A/2)^2 - 1]/2 + R(z) \geq 0$ at every binary vertex of a NO instance, hence everywhere by multiaffine interpolation. Its midpoint value is $b + \epsilon^2/2$. The reduction is from weakly NP-complete PARTITION; it establishes ordinary, not strong, NP-hardness. It concerns varying unequal boxes contained in a narrow common box, not the envelope on that one common box. The separating accuracy can be exponentially small in input length.

Corollary 11.2 (Graph-hull membership). *Deciding whether $(\bar{x}, q)$ belongs to the scalar graph hull of the same single monomial is NP-complete under the same narrow-box restrictions.*

Proof. At the midpoint its common-threshold upper value is $C = (1 + \prod_i u_i)/2$. Retaining the product's nonnegative quadratic terms gives

$$C \geq 1 + \epsilon A/2 + \epsilon^2(A^2 - \sum_i a_i^2)/4, \quad C - b \geq \epsilon^2 A^2/8 \geq \epsilon^2/2 > q - b.$$

Thus the upper-hull condition always holds on the reduced instances, and membership is exactly the lower-envelope threshold question. Membership lies in NP: a basic distribution for the normalization, coordinate, and value equations has at most $n + 2$ positive weights. These equations have rational polynomial-bit entries and determinant bounds give polynomial-bit weights. Exact verification proves membership. $\square$

Proposition 11.3 (A fixed number of ratio types). *For one product on a rational strictly positive box having at most k distinct coordinate aspect ratios, its exact scalar envelope and a supporting affine minorant can be computed in $n^{O(k)}$ times a polynomial in the input bit length. In particular fixed k is tractable.*

Proof. After substituting fixed coordinates and normalizing endpoints, a binary product with count h_j in ratio group j has value $(\prod_i \ell_i) \prod_{j=1}^k r_j^{h_j}$. To minimize this value minus $\sum_i \pi_i z_i$ for arbitrary dual coefficients, sort the coefficients within each group. For fixed counts choose the h_j largest coefficients, and use prefix sums. Enumerate all $\prod_j (n_j + 1) \leq (n + 1)^k$ count tuples. Rational multiplication and sorting have polynomial bit cost; this is an exact vertex oracle. The determinant-bounded dual argument of Corollary 8.11 then computes the lower envelope and its minorant. The common-threshold upper envelope is explicit. Thus mere closeness of the ratios does not substitute for a bound on their number of distinct values. $\square$

11.1 Rank-one transport and logarithmic accuracy dependence

Multimarginal optimal transport with specified finite marginals minimizes expected cost over their joint laws. With n fair binary marginals and factored cost tensor

$$C = \bigotimes_{i=1}^n (1, 1 + \epsilon a_i), \quad C_z = \prod_i (1 + \epsilon a_i z_i),$$

it is exactly the midpoint envelope above. This tensor is strictly positive and rank one, with no sparse correction, and

$$1 \leq C_z \leq \prod_i (1 + \epsilon a_i) \leq \sum_{j \geq 0} (\epsilon A)^j \leq 16/15.$$

Corollary 11.4 (Rank-one value and weighted dual accuracy). *Exact threshold evaluation for these rank-one MOT instances is NP-complete. Unless $P = NP$, neither their values nor their weighted vertex minima admit a deterministic additive- δ approximation algorithm polynomial in rational input length and $\log(C_{\max}/\delta)$. For randomized bounded-error algorithms the corresponding consequence requires $NP \not\subseteq BPP$.*

Proof. Take $\delta = \epsilon^2/32$ in the YES and NO intervals of Theorem 11.1. Their approximation intervals remain disjoint and $\log(1/\delta)$ is polynomial in the PARTITION input length. Such an algorithm would decide PARTITION. The weighted vertex oracle is hard directly: set

$$\pi_i = \epsilon a_i + \frac{\epsilon^2}{2} (A a_i - a_i^2), \quad d_0 = 1 - \epsilon^2 A^2/8.$$

Equation (11.1) gives

$$C_z - \pi^\top z = d_0 + \frac{\epsilon^2}{2} (S(z) - A/2)^2 + R(z).$$

Its minimum is at most $d_0 + \epsilon^2/8$ in a YES instance and at least $d_0 + \epsilon^2/2$ in a NO instance. The same approximation margin and polynomial encoding bounds apply. Exact MOT decision has the NP certificates already given. If an input convention requires bit lengths polynomial in the number of marginals, append a number of dummy binary factors $(1, 1)$ linear in the original input bit length. Give them fair marginals and zero dual potentials. They change neither rank, cost range, value, nor oracle minimum, and meet that convention. $\square$

The published structured-MOT paper of Altschuler and Boix-Adserà [2, Section 7.2, after Theorem 7.4] asks whether constant-rank approximate dual minimization can have logarithmic inverse-accuracy dependence, leading to exact algorithms. The preceding reduction rules out that dependence in the rational bit model under the stated complexity assumption, already at rank one. Their Corollary 7.5 has polynomial dependence on inverse accuracy and is consistent with it. This comparison identifies the historical question; it is not a claim that no subsequent work addressed it. Nor is it a lower bound in an unrestricted unit-cost arithmetic model. The unweighted smallest entry of this tensor is simply one; arbitrary dual weights are essential. None of these arguments proves hardness at fixed additive error.

11.2 A boundary of multilinearity

Example 11.5 (Two positive rational-power terms). On the fixed interval $[1, 3]$, let $f_\epsilon(x) = x^{1-\epsilon} + x^{1+\epsilon}$, $0 < \epsilon < 1$. The first term is concave and the second convex, while

$$f''_\epsilon(x) = \epsilon x^{-1-\epsilon}[(1+\epsilon)x^{2\epsilon} - (1-\epsilon)] > 0.$$

At $x = 2$ their separate widths and full width are consequently

$$T_\epsilon = 3 \sinh(\epsilon \log 3) - 4 \sinh(\epsilon \log 2), \quad H_\epsilon = 1 + 3 \cosh(\epsilon \log 3) - 4 \cosh(\epsilon \log 2).$$

These follow by subtracting each function from its endpoint secant, with the opposite subtraction for the concave term. Taylor expansion gives

$$T_\epsilon = (3 \log 3 - 4 \log 2)\epsilon + O(\epsilon^3), \quad H_\epsilon = \frac{1}{2}[3(\log 3)^2 - 4(\log 2)^2]\epsilon^2 + O(\epsilon^4).$$

The first leading coefficient is $\log(27/16) > 0$; the second is positive by strict convexity of $x(\log x)^2$ on $[1, 3]$. Thus their ratio diverges like a positive constant times $1/\epsilon$. Taking $\epsilon = 1/k$, $k \geq 2$, keeps rational exponents in $[1/2, 3/2]$ with only two positive unit coefficients and one variable. Under $x = e^y$, however, both terms become convex exponentials on $[0, \log 3]$. Their lower envelopes are exact and endpoint secants add, so the termwise and full widths coincide in that representation. The example concerns the original coordinate and does not preclude useful logarithmic reformulation.

12 Fractional cardinality and spatial certificates

The following optimization problems have explicit solutions. Their role is to distinguish the size of certificates produced by specified spatial oracles from the difficulty of finding an optimizer. The obstruction uses a concave quadratic penalty and an affine balance; it is a separate construction from the positive multilinear gap examples above.

Fix integers $n \geq 2$ and $0 \leq k \leq n-1$, and write

$$K = k + \frac{1}{2}, \quad \mathcal{F}_{n,k} = \{x \in [0, 1]^n : \sum_i x_i = K\}, \quad F(x) = \sum_{i=1}^n x_i(1 - x_i). \quad (12.1)$$

The dimension parameter n counts original coordinates. A vertex of $\mathcal{F}_{n,k}$ has exactly k coordinates equal to one, one equal to $1/2$, and the others zero: two strictly fractional coordinates would admit a small opposite perturbation, whereas a point with at most one fractional coordinate is determined by its active bounds and the balance. Concavity and compactness therefore give

$$\min_{\mathcal{F}_{n,k}} F = \frac{1}{4}. \quad (12.2)$$

For each oracle below, a certified region is either infeasible or has oracle bound at least T_* . A certificate is a finite cover of $\mathcal{F}_{n,k}$ by such regions, as in Section 2.3. For absolute tolerance $\epsilon \geq 0$ we grant the best incumbent and set $T_* = 1/4 - \epsilon$. Pruning at $U - \epsilon$, for any feasible incumbent $U \geq 1/4$, implies this condition. Infeasible regions contain no witness and do not help cover the feasible set. All lower bounds also hold for a weaker valid oracle: certifying a region with that oracle certifies it with the stronger oracle used in the proof.

There are close precedents in other certificate models. Jarre’s binary knapsack construction retains an exponential branch-and-bound obstruction with a max-cut semidefinite bound [15, Sections 2–3]. Coniglio’s hyperplane-clustering analysis gives a related spatial obstruction under midpoint branching on symmetric coordinate domains [5, Assumption 1, Propositions 2–4, Appendix C in the review version]. Our arguments concern arbitrary coordinate restrictions and a fixed objective gap under the explicitly defined node oracles. We make no priority claim for exponential spatial lower bounds in general. The fractional-cardinality moments used later are classical; the additional argument places them in spatial regions and counts how many witnesses a certified region can contain.

12.1 Chords and endpoint avoidance

For a box $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, including degenerate intervals, define

$$\ell_i(x_i) = a_i b_i + (1 - a_i - b_i)x_i, \quad x_i(1 - x_i) - \ell_i(x_i) = (x_i - a_i)(b_i - x_i). \quad (12.3)$$

Its chord bound is

$$\text{LB}_{\text{ch}}(B) = \min_{x \in B \cap \mathcal{F}_{n,k}} \sum_i \ell_i(x_i),$$

with $+\infty$ for an empty feasible intersection. If g_i is any convex underestimator of $x_i(1 - x_i)$ on $[a_i, b_i]$, convexity between the endpoints gives $g_i \leq \ell_i$. This also holds on a singleton interval. Thus the infimum for any separable convex underestimator $\sum_i g_i$ is at most $\text{LB}_{\text{ch}}(B)$; an infimum is used because arbitrary convex underestimators need not attain their infima on a closed interval without lower semicontinuity at its endpoints.

For clarity, two common formulations yield exactly these chords. For $y_i = 1 - x_i \in [1 - b_i, 1 - a_i]$, both McCormick lower planes for $w_i = x_i y_i$ become $w_i \geq \ell_i(x_i)$. The upper planes remain compatible: the exact product lies above the chord and below each upper plane. The separable αBB expression $x_i(1 - x_i) - \alpha(x_i - a_i)(b_i - x_i)$ has second derivative $2(\alpha - 1)$, so the smallest convexifying value $\alpha = 1$ gives the chord. Larger valid values give weaker underestimators. These statements concern the specified separable formulations, without additional coupled cuts or nonlinear subproblem solution rules.

We isolate the counting argument so that later oracle changes do not obscure it. Let $n = k + m + z$, where $k, z \geq 0$ and $m \geq 1$ are integers. For each ordered partition (H, M, Z) of $[n]$ with these sizes, put

$$p = \frac{1}{2m}, \quad w = \mathbf{1}_H + p\mathbf{1}_M. \quad (12.4)$$

These $n!/(k!m!z!)$ distinct witnesses belong to $\mathcal{F}_{n,k}$. For an arbitrary product $S = \prod_i S_i$, with nonempty $S_i \subseteq [0, 1]$, define

$$A(S) = \{i : 0 \notin S_i\}, \quad D(S) = \{i : 1 \notin S_i\}, \quad R(S) = A(S) \cup D(S). \quad (12.5)$$

For boxes these are $A = \{i : a_i > 0\}$ and $D = \{i : b_i < 1\}$.

Lemma 12.1 (Endpoint-avoidance count). *Suppose each certified product domain that contains a witness has $|R(S)| \geq q$, where $q \geq 0$ is real. Any certified cover of $\mathcal{F}_{n,k}$ contains at least*

$$\mathcal{N}(q) := \min \left\{ \left(\frac{n}{n-k} \right)^{q/2}, \left(\frac{n}{n-z} \right)^{q/2} \right\} \quad (12.6)$$

regions. The same conclusion follows from the stronger hypothesis $|R(S)| > q$. Both denominators are positive, and $\mathcal{N}(0) = 1$.

Proof. Choose the ordered partition uniformly. A contained witness must have $Z \cap A = \emptyset$ and $H \cap D = \emptyset$. For a fixed set of s excluded positions, a uniformly chosen z -subset avoids it with probability

$$\frac{\binom{n-s}{z}}{\binom{n}{z}} = \prod_{j=0}^{s-1} \frac{n-z-j}{n-j} \leq \left(\frac{n-z}{n} \right)^s \quad (s \leq n-z).$$

For $s > n-z$ the probability is zero, so the upper bound still holds; for $s = 0$ the product is one. The analogous inequality holds for H . No independence between the two avoidance events is required. A fixed region's witness fraction is at most

$$\min \left\{ \left(\frac{n-z}{n} \right)^{|A|}, \left(\frac{n-k}{n} \right)^{|D|} \right\}.$$

Since $|A| + |D| \geq |R| \geq q$, at least one exponent is at least $q/2$. The displayed fraction is consequently at most $1/\mathcal{N}(q)$. Regions containing no witness have fraction zero. The union bound over a cover proves the assertion, whether or not its regions overlap. Bases equal to one when $k = 0$ or $z = 0$ cause no exception. $\square$

Theorem 12.2 (Separable spatial certificates). *Let $n = k + m + z$, with integers $k, z \geq 0$, $m \geq 1$, $n \geq 2$, and $0 \leq \epsilon < 1/4$. For problem (12.1), every cover by boxes certified by the chord oracle at $T_* = 1/4 - \epsilon$ has at least $\mathcal{N}(h)$ members, where*

$$h = m(1/2 - 2\epsilon). \quad (12.7)$$

This includes arbitrary real coordinate split points and all separable convex underestimators dominated by the chord oracle. For $k = m = z = t$, $n = 3t$, the bound is $(3/2)^{t(1/4-\epsilon)}$, hence $(3/2)^{n/24}$ at $\epsilon = 1/8$.

Proof. In a box containing w , the chord values at its zero and unit witness coordinates vanish. On $U = [n] \setminus R$ the interval is $[0, 1]$, whose chord vanishes identically. Consequently

$$T_* \leq \text{LB}_{\text{ch}}(B) \leq \sum_{i \in M \cap R} \ell_i(p) \leq |M \cap R| p(1-p) < \frac{|M \cap R|}{2m}.$$

The last strict inequality is applicable because $T_* > 0$ forces $M \cap R \neq \emptyset$. Thus $|R| > 2mT_* = h$, and Lemma 12.1 applies. A complete coordinate split tree provides such a cover, regardless of split locations or branching order. $\square$

12.2 Upper certificates, tightening, and tolerances

Proposition 12.3 (Explicit upper certificates). *For $0 < \epsilon < 1/4$, a chord certificate for (12.1) has at most $2^{n+2} - 3$ nodes and $2^{n+1} - 1$ leaves. For $\epsilon = 0$, a midpoint certificate has exactly $\binom{n+1}{k+1}$ leaves if only the stopping rules in the proof are used. In particular its node count is $O(n^{\min\{k+1, n-k\}})$ when that exponent is fixed.*

Proof. Choose $0 < \alpha < 1/2$ with $\alpha(1-\alpha) = 1/4 - \epsilon$. Process coordinates in order. At each surviving node split the next coordinate at α , then split its upper child at $1 - \alpha$. The middle interval has constant chord $\alpha(1 - \alpha)$, and all other chords are nonnegative, so the middle child is certified. Continue on the low and high intervals. At a final corner box the chord sum is $(1 - \alpha)\|x - v\|_1$, where $v \in \{0, 1\}^n$ is its corner. On the demand slice, $\|x - v\|_1 \geq |K - \sum_i v_i| \geq 1/2$. This gives a bound $(1 - \alpha)/2 > 1/4 > T_*$. Processing coordinate i adds four nodes per surviving prefix, of which there are 2^{i-1} . The stated counts follow. For a strict pruning rule choose a slightly larger middle-chord target still below $1/4$.

For exact certification split at $1/2$ instead. If a prefix has c high intervals and d low intervals, its chord bound is at least each of

$$\frac{c - K}{2}, \quad \frac{K - (n - d)}{2}.$$

Indeed, the sum on high coordinates is at most K , whereas the sum on low coordinates is at least $K - (n - d)$; all other chord contributions are nonnegative. Stop at $c = k + 1$ or $d = n - k$. No path survives depth n . With remaining high and low stopping counts a, b , the number of leaves satisfies $L(a, b) = L(a - 1, b) + L(a, b - 1)$ and $L(0, b) = L(a, 0) = 1$, hence $L(a, b) = \binom{a+b}{a}$. Start at $(a, b) = (k + 1, n - k)$. A full binary tree has twice as many leaves minus one nodes. This proves the exact count and polynomial assertion. $\square$

The balanced fixed-tolerance family therefore has certificate size $2^{\Theta(n)}$ under the chord model. If both k and $n - k$ are proportional to n , choose m, z proportional to $n - k$ to obtain the same exponential order. Fixed demand level, or fixed complementary demand level, admits the polynomial certificate just proved. At $\epsilon \geq 1/4$, the root chord bound zero already meets the target; no positive exponent is asserted there.

The chord error is genuinely quadratic in interval width:

$$0 \leq F(x) - \sum_i \ell_i(x_i) \leq \frac{1}{4} \sum_i (b_i - a_i)^2 \leq \frac{n}{4} \max_i (b_i - a_i)^2. \quad (12.8)$$

In particular, the difference between the exact node optimum and the chord bound is at most this quantity, by evaluating F at a chord minimizer. Thus the fixed-tolerance lower bound persists despite this second-order error estimate; it does not require a first-order local approximation error.

Proposition 12.4 (Charged reductions and equality tolerance). *Suppose a run keeps feasible points covered by its final boxes together with discarded boxes certified by the same oracle at T_* . If it processes N nodes and discards at most a such boxes per node, a cover lower bound L implies $N \geq L/(a + 1)$. Pure feasibility reductions need not be charged when they discard no feasible point.*

For a demand feasibility tolerance $0 \leq \delta < 1/2$, replace $\mathcal{F}_{n,k}$ by $\mathcal{F}_{n,k}^\delta = \{x \in [0, 1]^n : |\sum_i x_i - K| \leq \delta\}$. Its exact optimum is $1/4 - \delta^2$. If $\epsilon + \delta^2 < 1/4$, a chord certificate at absolute tolerance ϵ for this enlarged set satisfies Theorem 12.2 with ϵ replaced by $\epsilon + \delta^2$.

Proof. The augmented cover has at most $(a+1)N$ members. A discarded open coordinate slab may be charged by its closed box only when that box is certified; this condition is part of the assertion, not an unproved closure assumption. Subsequent product-domain results can charge the removed coordinate set itself.

For a fixed sum $s \in [k + 1/2 - \delta, k + 1/2 + \delta] \subset (k, k+1)$, the same vertex argument as above gives minimum penalty $(s-k)(1-s+k)$. Its minimum on this interval is $1/4 - \delta^2$, attained by k ones and a remaining coordinate $1/2 - \delta$ or $1/2 + \delta$. A tolerance-feasible incumbent is therefore at least this value. The exact-balance witnesses remain feasible for the enlarged problem, and the chord bound over the enlarged feasible set is at most its value at each witness. The proof of Theorem 12.2 applies at the positive target $1/4 - (\epsilon + \delta^2)$. $\square$

The restriction $\delta < 1/2$ is material: once the tolerance interval contains an integer sum, the optimum is zero. An unrestricted formula $1/4 - \delta^2$ would then be false. Relative pruning in the convention $\text{LB} \geq (1 - \theta)U$, $0 \leq \theta < 1$, on the original problem implies the absolute target with $\epsilon = \theta/4$.

12.3 Full second moments and box RLT

For $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, let $\text{LB}_{\text{SDP}}(B)$ be the infimum of $\sum_i (\mu_i - X_{ii})$ over symmetric X and $\mu \in \mathbb{R}^n$ satisfying

$$a \leq \mu \leq b, \quad \sum_i \mu_i = K, \quad \begin{pmatrix} 1 & \mu^T \\ \mu & X \end{pmatrix} \succeq 0, \quad \sum_j X_{ij} = K\mu_i \quad (i \in [n]), \quad (12.9)$$

and all pairwise box inequalities

$$\begin{aligned} X_{ij} - a_i\mu_j - a_j\mu_i + a_ia_j &\geq 0, \\ b_j\mu_i - X_{ij} - a_ib_j + a_i\mu_j &\geq 0, \\ b_i\mu_j - X_{ij} - b_ia_j + a_j\mu_i &\geq 0, \\ b_ib_j - b_i\mu_j - b_j\mu_i + X_{ij} &\geq 0 \quad (i, j \in [n]). \end{aligned} \quad (12.10)$$

Indices may coincide. These are the linearizations of all products of $x_i - a_i$ and $b_i - x_i$. Exact evaluation $(\mu, X) = (x, xx^T)$ shows validity. As before the infimum of an empty relaxation is $+\infty$, and a region is certified when its bound is at least $T_* = 1/4 - \epsilon$ (or it is infeasible).

Lemma 12.5 (Restricted fractional second moments). *Let $k, m, z \geq 1$, and let B contain a witness (12.4). If $|R(B)| < \min\{k, z\}$, the model (12.9)–(12.10) has a feasible point of value $|M \cap R|p(1-p)$.*

Proof. Fix $x_i = w_i$ deterministically on R and put

$$U = [n] \setminus R, \quad s = |U|, \quad t = K - \sum_{i \in R} w_i = |H \cap U| + p|M \cap U|.$$

At least one unit and one zero witness coordinate remain, so $s \geq 2$ and $1 \leq t \leq s-1$. On U set

$$c = t/s, \quad d = t(t-1)/(s(s-1)), \quad \mu_i = c, \quad X_{ii} = c, \quad X_{ij} = d \quad (i \neq j).$$

On R set $\mu_i = w_i$, and set $X_{ij} = \mu_i\mu_j$ whenever at least one index lies in R . First moments meet the box bounds because the intervals on U are $[0, 1]$ and $w \in B$. Their sum is K . The covariance is zero on rows and columns in R , and on U it is

$$X_{UU} - \mu_U \mu_U^T = (c-d)(I - J/s), \quad c-d = \frac{t(s-t)}{s(s-1)} \geq 0.$$

It is positive semidefinite, proving augmented PSD by the Schur complement of the entry one. For $i \in U$, $c + (s - 1)d = tc$ gives $\sum_j X_{ij} = Kc$; for $i \in R$, the equality is $\sum_j X_{ij} = w_i \sum_j \mu_j = Kw_i$.

For distinct $i, j \in U$, the four slack-product expectations are

$$d, \quad c - d, \quad c - d, \quad 1 - 2c + d = \frac{(s - t)(s - t - 1)}{s(s - 1)},$$

all nonnegative. For $i = j \in U$ they are $c, 0, 0, 1 - c$. If at least one index is in R , the expectation factors into a nonnegative deterministic slack and a nonnegative expected slack; this also covers repeated restricted indices and degenerate intervals. Finally $\mu_i - X_{ii} = 0$ on U , and on R it is $w_i(1 - w_i)$. $\square$

Theorem 12.6 (SDP–RLT cover bound). *Let $n = k + m + z$, where $k, m, z \geq 1$ are integers, and let $0 \leq \epsilon < 1/4$. Every cover for (12.1) certified by (12.9)–(12.10) at $T_* = 1/4 - \epsilon$ has at least $\mathcal{N}(q)$ members, where $q = \min\{k, z, m(1/2 - 2\epsilon)\}$. In the balanced case $k = m = z = t$, this is the chord lower bound in Theorem 12.2.*

Proof. If a containing box has $|R| < q$, Lemma 12.5 gives a feasible objective at most $|R|p(1 - p) < q/(2m) \leq T_*$, where the strict comparison holds also when $R = \emptyset$ because $q > 0$. The box cannot be certified. Therefore $|R| \geq q$ on every certified box containing a witness, and Lemma 12.1 applies. The diagonal mixed inequality gives $X_{ii} \leq (a_i + b_i)\mu_i - a_i b_i$, so the SDP bound dominates the chord bound. The upper certificates of Proposition 12.3 remain valid. $\square$

This resolves the behavior under spatial branching for this SDP–RLT model; it is not merely a calculation of its root value.

One can also grant all products of two affine inequalities valid on $B \cap \mathcal{F}_{n,k}$ without strengthening this oracle. Indeed, linear-programming duality represents any such nonnegative affine function, on a nonempty polytope, as a nonnegative constant plus a nonnegative combination of box slacks plus a multiple of $e(x) = \sum_i x_i - K$. The representation is an identity of affine polynomials, including for degenerate boxes. Expanding the product of two representations leaves only existing nonnegative slack moments and terms $L[ev] = 0$ with v affine. This assertion concerns affine inequalities in the original x coordinates. It does not grant all valid affine cuts in the lifted (μ, X) coordinates.

13 Full preorderings and fractional-cardinality moments

An order- r oracle below acts on polynomials of total degree at most $2r$, where $r \geq 1$ is an integer. On a box $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, it consists of linear functionals $L : \mathbb{R}[x]_{\leq 2r} \rightarrow \mathbb{R}$ satisfying

$$L[1] = 1, \tag{13.1}$$

$$L[ev] = 0 \quad (\deg v \leq 2r - 1), \quad e = \sum_i x_i - K, \tag{13.2}$$

$$L[gP^2] \geq 0 \quad \left(\sum_{j=1}^a \deg g_j + 2 \deg P \leq 2r, \quad g = \prod_{j=1}^a g_j \right), \tag{13.3}$$

where each g_j is a coordinate-bound slack $x_i - a_i$ or $b_i - x_i$. The empty product and repeated slack factors are included. The square polynomial P may couple every coordinate. Zero polynomials impose no condition; constants have degree zero. The node bound $\text{LB}_r(B)$ is the infimum of $L[F]$ over these functionals. This is the full truncated box preordering, including all allowed products

of slacks, not just the separate single-generator localizers. Evaluation at a feasible point proves validity.

At $r = 1$ this model is exactly (12.9)–(12.10): positivity on affine squares is augmented PSD, products of at most two slacks give box RLT, single slacks give first-moment bounds, and the balance times affine polynomials gives its first and second moment equations. At every order $r \geq 1$ the model dominates that degree-two model and hence the chord oracle. Certification again means an infeasible region or $\text{LB}_r(B) \geq T_* = 1/4 - \epsilon$ for problem (12.1). Original dimension remains n .

13.1 A self-contained positivity proof

The moments below are the classical fractional-cardinality moments associated with Grigoriev’s knapsack obstruction [11]. Potechin gives the explicit construction and its classical provenance [20, Theorem 1, Example 18, Theorem 44]. We prove the precise sufficient range required for the present full preordering; no external positivity theorem is needed.

For real v and integer $a \geq 0$, write $(v)_a = v(v-1)\cdots(v-a+1)$, with $(v)_0 = 1$. In s formal Boolean variables $u_1, \dots, u_s$, let red replace every positive power of u_i by u_i . This is the canonical multilinear representative modulo $u_i^2 - u_i$; it respects sums and products modulo these identities and cannot increase degree. For $u_S = \prod_{i \in S} u_i$ define

$$E_{s,t}[u_S] = \frac{(t)_{|S|}}{(s)_{|S|}}, \quad E_{s,t}[P] = E_{s,t}[\text{red } P]. \quad (13.4)$$

All subset sizes are at most s , so denominators are positive. The functional is defined on the Boolean quotient and is used only through the degree stated in each assertion; no positivity on all polynomials is assumed. In particular $E_{s,t}[1] = 1$. For $s = 0$, deterministic evaluation of constants will be used instead of a fractional-cardinality functional.

Lemma 13.1 (Cardinality identities and moment positivity). *Let $d \geq 1$ be an integer, $s \geq 2d$ an integer, and $t, s - t \geq 2d - 1$. Then*

$$E_{s,t} \left[\left(\sum_i u_i - t \right) v \right] = 0 \quad (\deg v \leq 2d - 1), \quad (13.5)$$

and $E_{s,t}[P^2] \geq 0$ for every $\deg P \leq d$.

Proof. For squarefree u_S of degree $a \leq 2d - 1 \leq s - 1$,

$$E_{s,t} \left[\left(\sum_i u_i \right) u_S \right] = a \frac{(t)_a}{(s)_a} + (s - a) \frac{(t)_{a+1}}{(s)_{a+1}} = t \frac{(t)_a}{(s)_a}.$$

Boolean reduction and linearity prove (13.5) for every allowed multiplier.

To reduce positivity to homogeneous polynomials, for $|S| = a \leq d$ put

$$H_S(u) = \frac{\sum_{T \supseteq S, |T|=d} u_T}{(t-a)_{d-a}}. \quad (13.6)$$

The denominator is positive under $t \geq 2d - 1$ (and is one for $a = d$). In the Boolean quotient the numerator equals $u_S \left(\sum_{d-a} u_i^{u_i-a} \right)$: at a Boolean vector it counts the d -subsets containing S when

all coordinates of S equal one, and both sides vanish otherwise. The polynomial $\binom{z-a}{d-a} - \binom{t-a}{d-a}$ is divisible by $z - t$ with quotient degree at most $d - a - 1$ when $a < d$. Consequently

$$u_S - H_S = \left(\sum_i u_i - t \right) Q_S \quad \text{in the Boolean quotient,} \quad \deg Q_S \leq d - 1.$$

Replace each monomial of red P by its H_S , obtaining a homogeneous squarefree polynomial H of degree d . Then $P - H = (\sum_i u_i - t)Q$ modulo Boolean identities with $\deg Q \leq d - 1$. Since $\deg(Q(P + H)) \leq 2d - 1$, the cardinality identity gives $E_{s,t}[P^2] = E_{s,t}[H^2]$. Every expression used here has degree at most $2d$ before Boolean reduction.

Index the homogeneous moment matrix N by d -subsets I, J of $[s]$:

$$N(I, J) = \frac{\binom{t}{2d-|I \cap J|}}{\binom{s}{2d-|I \cap J|}}.$$

Let $P_j(I, J) = \binom{|I \cap J|}{j}$ for $0 \leq j \leq d$. This is a Gram matrix: associate with I the vector indexed by j -subsets A , with entry $\mathbf{1}\{A \subseteq I\}$. The exact decomposition is

$$N = \sum_{j=0}^d \frac{\binom{t}{2d-j} \binom{s-t}{j}}{\binom{s}{2d}} P_j. \quad (13.7)$$

Indeed, in an entry with $|I \cap J| = \ell$, the numerator on the right is

$$\begin{aligned} \sum_{j=0}^{\ell} \binom{\ell}{j} \binom{t}{2d-j} \binom{s-t}{j} &= \binom{t}{2d-\ell} \sum_{j=0}^{\ell} \binom{\ell}{j} (t - 2d + \ell)_{\ell-j} (s - t)_j \\ &= \binom{t}{2d-\ell} (s - 2d + \ell)_{\ell}. \end{aligned}$$

The last equality is the falling-factorial Vandermonde identity, obtained by counting ordered choices from two disjoint finite sets for nonnegative integer arguments and then by polynomial identity for real arguments. Division by $\binom{s}{2d}$ gives the claimed entry. No division by $\binom{t}{2d-\ell}$ was used, so zero coefficients cause no exception. All coefficients in (13.7) are nonnegative because $t, s - t \geq 2d - 1$, and $\binom{s}{2d} > 0$. Thus $N \succeq 0$ and $E_{s,t}[H^2] \geq 0$. $\square$

Lemma 13.2 (Assignment indicators and every slack product). *Suppose $r \geq 1$, $s \geq 2r$, and $t, s - t \geq 2r - 1$. For every product g of u_i and $1 - u_i$, including repetitions, and every polynomial P with $\deg g + 2 \deg P \leq 2r$, one has $E_{s,t}[gP^2] \geq 0$.*

Proof. Boolean reduction makes g zero if both u_i and $1 - u_i$ occur for some i . Otherwise it becomes an assignment indicator

$$I_{A,B} = \prod_{i \in A} u_i \prod_{j \in B} (1 - u_j), \quad A \cap B = \emptyset.$$

Put $a = |A|$, $b = |B|$, and $v = a + b \leq \deg g$. For a squarefree monomial u_C on the remaining coordinates, direct expansion gives

$$E_{s,t}[I_{A,B} u_C] = \frac{\binom{t}{a+|C|} \binom{s-t}{b}}{\binom{s}{v+|C|}} \quad (v + |C| \leq s). \quad (13.8)$$

One verification is induction on b : inserting an additional factor $1 - u_j$ subtracts the expression with j also fixed to one, and putting the two terms over their common denominator supplies the next factor $s - t - b$. The case $b = 0$ is (13.4). In particular the indicator weight is

$$\pi = E_{s,t}[I_{A,B}] = \frac{(t)_a(s-t)_b}{(s)_v} \geq 0.$$

If P is constant, the desired inequality is $\pi P^2 \geq 0$. If P is nonconstant, $v \leq \deg g \leq 2r - 2$, so all factors in this weight are strictly positive. Write $\hat{P} = P(1_A, 0_B, u_{[s] \setminus (A \cup B)})$. For $v < s$, cancellation in (13.8) proves

$$E_{s,t}[I_{A,B}P^2] = \pi E_{s-v,t-a}[\hat{P}^2]. \quad (13.9)$$

This identity is used only when its remaining moments are defined. If $\hat{P}$ is constant, its value is simply $\pi \hat{P}^2$; the same direct evaluation applies when $v = s$, without defining a zero-variable conditional functional. Otherwise let $d = \deg \hat{P} \geq 1$. Then

$$v + 2d \leq 2r, \quad s - v \geq 2d, \quad t - a \geq 2r - 1 - a \geq 2d - 1, \quad s - t - b \geq 2d - 1.$$

Lemma 13.1 applies to $E_{s-v,t-a}$. This proves positivity, including for arbitrary squares that couple all unassigned variables and for repeated original slacks. $\square$

The sufficient degree range in Lemma 13.1 is not the sharp classical range for the moment matrix alone. For the *full preordering* and this particular functional, the slack products explain the two-sided restriction: if $0 < t < s$ is noninteger, $s \geq 2r$, and $t < 2r - 1$, a product of $\lfloor t \rfloor + 2 \leq 2r$ distinct lower slacks has a negative falling-factorial moment. Applying the same argument to upper slacks treats $s - t < 2r - 1$. Integer t is different: (13.4) is then expectation under the uniform distribution of t -subsets. Sharper moment-matrix positivity alone therefore does not extend this full-preordering construction past the noninteger slack-product obstruction.

13.2 The order versus region tradeoff

Theorem 13.3 (Full-preordering spatial cover bound). *For problem (12.1), let $n = k + m + z$ with positive integers k, m, z , let $r \geq 1$ be an integer, and let $0 \leq \epsilon < 1/4$. Put*

$$q_r = \min\{k - 2r + 2, z - 2r + 2, m(1/2 - 2\epsilon)\}. \quad (13.10)$$

If $q_r > 0$, every box cover certified by (13.1)–(13.3) at $T_ = 1/4 - \epsilon$ has at least $\mathcal{N}(q_r)$ members. If $q_r \leq 0$, only the trivial bound of one region is asserted.*

Proof. Take a box containing a witness (12.4) and suppose $|R| < q_r$. Since $|R|$ and $k - 2r + 2$ are integers, $|R| < k - 2r + 2$ implies $|H \cap U| \geq k - |R| \geq 2r - 1$; likewise $|Z \cap U| \geq 2r - 1$. Set

$$s = |U|, \quad t = |H \cap U| + p|M \cap U|.$$

Then $t, s - t \geq 2r - 1$. Also $s \geq 2r$: for $r = 1$ a unit and a zero coordinate remain, and for $r \geq 2$ we have $s \geq 4r - 2 \geq 2r$. Define L by fixing $x_R = w_R$ and applying $E_{s,t}$ to $x_U = u$. This degree-preserving substitution is normalized. It changes e into $\sum_{i \in U} u_i - t$, whose products with every substituted multiplier of degree at most $2r - 1$ vanish by Lemma 13.1. Bound slacks on R are nonnegative constants, and those on U are Boolean slacks. Removing constant factors cannot increase degrees, so Lemma 13.2 proves every required positivity constraint.

The Boolean penalty on U vanishes, leaving

$$L[F] = |M \cap R|p(1-p) \leq |R|p(1-p) < \frac{q_r}{2m} \leq \frac{1}{4} - \epsilon. \quad (13.11)$$

The strict inequality follows from $|R| < q_r$ and $q_r > 0$, including $R = \emptyset$. This feasible functional precludes certification. Thus every certified box containing a witness has $|R| \geq q_r$, and Lemma 12.1 proves the bound. In particular non-strict pruning at the target is covered: the constructed value is strictly below it. No integer rounding of $q_r/2$ is necessary. $\square$

For balanced parts $k = m = z = t$, positivity of q_r is precisely $r < (t+2)/2$ together with $\epsilon < 1/4$. A uniform exponential bound as $n = 3t \rightarrow \infty$ follows whenever $\epsilon < 1/4$ is fixed and $r \leq ct$ for a fixed $c < 1/2$. At $\epsilon = 1/8$, the exponent $n/24$ survives whenever

$$t - 2r + 2 \geq t/4, \quad \text{equivalently} \quad r \leq 3t/8 + 1.$$

An order approaching the positivity boundary can instead give only a constant q_r ; positivity alone is not an exponential-size assertion. When $\epsilon = 1/4$, the root has an available certificate because the mixed slacks give $L[F] \geq 0$. We do not divide by T_* or by a vanishing q_r in any of these cases. Increasing r may itself make the oracle costly; the theorem grants that oracle and counts regions only.

The affine-inequality closure following Theorem 12.6 extends to this degree. Expand a product of valid affine inequalities in their nonnegative box-slack and balance representations and multiply by an allowed square. Terms with a balance factor vanish within degree $2r$, and all remaining terms are existing preordering inequalities. This still grants no arbitrary coupled nonlinear valid cut, no localizer of an objective cutoff, and no analytic solution of the coupled problem. The charged-cover convention of Proposition 12.4 applies to objective-based spatial reductions.

13.3 Unique minimizers and increasing concave costs

Corollary 13.4 (Distinct perturbations). *Let $0 < d_1 < \dots < d_n$ and $\sum_i d_i \leq \eta$, where $\eta > 0$. On $\mathcal{F}_{n,k}$ replace F by $F_d = F + \sum_i d_i x_i$. Its unique optimizer has*

$$x_i^* = 1 \ (i \leq k), \quad x_{k+1}^* = 1/2, \quad x_i^* = 0 \ (i \geq k+2), \quad (13.12)$$

with value $F_d^ = 1/4 + \sum_{i \leq k} d_i + d_{k+1}/2$. For the cover assertion take $n = k + m + z$ with positive integers k, m, z and integer $r \geq 1$, as in Theorem 13.3. Under the order- r box oracle, a cover at $T_* = F_d^* - \epsilon$ has at least $\mathcal{N}(q_{r,\eta})$ members provided $0 \leq \epsilon$, $\epsilon + \eta < 1/4$, and*

$$q_{r,\eta} = \min\{k - 2r + 2, z - 2r + 2, m(1/2 - 2\epsilon - 2\eta)\} > 0. \quad (13.13)$$

There is no nonidentity coordinate permutation preserving both feasible set and objective, even when objective equality is required only on the demand slice.

Proof. The Hessian of F_d is $-2I$, so a nonvertex, lying strictly between two distinct feasible points, cannot minimize this strictly concave function. All vertices have base value $1/4$. Among their linear costs, an exchange puts the largest coordinate value at the smallest coefficient: if $i < j$ and $x_i < x_j$, exchanging them reduces cost by $(d_j - d_i)(x_j - x_i) > 0$. Hence the unique minimizing vertex is (13.12). It also uniquely minimizes the linear cost over the whole slice: if $i < j$, $x_i < 1$, and $x_j > 0$, transferring $\min\{1 - x_i, x_j\} > 0$ from coordinate j to coordinate i strictly decreases that cost. At a linear-cost minimizer no such pair remains, so the balance forces exactly (13.12).

The node constraints themselves impose $0 \leq L[x_i] \leq 1$, since x_i and $1 - x_i$ are nonnegative combinations of node slacks and constants. The constructed functional therefore gives linear cost at most η . If $|R| < q_{r,\eta}$, its total value is strictly below

$$\frac{m(1/2 - 2\epsilon - 2\eta)}{2m} + \eta = 1/4 - \epsilon \leq F_d^* - \epsilon,$$

so the previous count applies.

The slice has an interior point relative to its balance hyperplane, namely $x_i = K/n \in (0, 1)$. A linear form constant on this slice has coefficient vector proportional to the all-ones vector. If a coordinate permutation preserves the objective there, the permuted coefficient vector differs from d by a constant vector. Comparing sums forces this constant to be zero, and distinctness forces the identity permutation. $\square$

An explicit choice is $d_i = 2\eta i/[n(n+1)]$, whose sum is η . For $n = 3t$, $\epsilon = 1/8$, and $\eta = 1/32$, the lower bound is $(3/2)^{n/32}$ whenever $r \leq 13t/32 + 1$. If $\eta < \epsilon$, the chord certificate at tolerance $\epsilon - \eta$ has base bound at least $1/4 + \eta - \epsilon \geq F_d^* - \epsilon$; the nonnegative linear term preserves it. Thus this explicit perturbed family again has $2^{\Theta(n)}$ certificate size under the specified oracles.

Adding the conserved flow gives increasing strictly concave coordinate costs:

$$C_d(x) = \sum_i [x_i(2 - x_i) + d_i x_i] = F_d(x) + K \quad (x \in \mathcal{F}_{n,k}). \quad (13.14)$$

Each derivative is $2 - 2x_i + d_i > 0$ on $[0, 1]$ and each second derivative is -2 . Every equality-preserving functional adds exactly K , so the optimizer and all absolute-gap conclusions are unchanged. When k is proportional to n , this raises the optimum to order n , while the unshifted obstruction has constant size. A fixed relative tolerance cannot be inferred from that absolute bound. Section 15 uses fixed-dimensional blocks to address precisely this distinction.

14 Arbitrary coordinate domains and local graph lifts

The endpoint argument uses membership, not interval geometry. We first strengthen the information available about each original coordinate, then show that adding arbitrarily many auxiliaries depending on that coordinate does not cost degree in this particular moment construction.

14.1 All valid local polynomial information

Let $S = \prod_i S_i$, where each S_i is an arbitrary nonempty subset of $[0, 1]$. Closedness, connectedness, semialgebraicity, and a finite description are not required. Define the order- r product-domain oracle by normalized functionals on $\mathbb{R}[x]_{\leq 2r}$, all balance products (13.2), and all inequalities

$$L \left[\left(\prod_{j=1}^a g_j \right) P^2 \right] \geq 0, \quad \sum_j \deg g_j + 2 \deg P \leq 2r, \quad (14.1)$$

where each g_j is any univariate polynomial nonnegative on its coordinate set S_i . The empty product is allowed, factors may repeat, and P may couple every coordinate. We also grant

$$L[hv] = 0 \quad (\deg h + \deg v \leq 2r) \quad (14.2)$$

for every univariate polynomial h vanishing on its S_i and every polynomial multiplier v in the original coordinates. In particular this model grants all local information through degree, not merely

a chosen list of input generators. Its representation need not be finite or computationally efficient. The node bound is the infimum of $L[F]$ or $L[F_d]$, as appropriate. A certified cover has original feasible set $\mathcal{F}_{n,k}$, original dimension n , and target $1/4 - \epsilon$ or $F_d^* - \epsilon$ respectively.

Theorem 14.1 (Product-domain certificates). *Under the respective parameter assumptions of Theorem 13.3 and Corollary 13.4, the cover bounds $\mathcal{N}(q_r)$ and $\mathcal{N}(q_{r,\eta})$ hold for the product-domain oracle (14.1)–(14.2).*

Proof. For a domain containing a witness, use the exclusions (12.5). If $|R| < q_r$, or $|R| < q_{r,\eta}$ in the perturbed case, fix $x_R = w_R$ and use $E_{s,t}$ on U exactly as in Theorem 13.3. Its parameter and balance calculations are unchanged.

A local inequality on a restricted coordinate becomes a nonnegative constant because $w_i \in S_i$. For $i \in U$, both endpoints belong to S_i , and its Boolean reduction is

$$\text{red } g(u_i) = g(0)(1 - u_i) + g(1)u_i, \quad g(0), g(1) \geq 0. \quad (14.3)$$

Group factors belonging to the same coordinate and expand their product in endpoint indicators. After removing constant factors, let J be the coordinates with nonconstant reduced factors. The expansion has the form

$$\sum_{\sigma \in \{0,1\}^J} c_\sigma \prod_{i \in J: \sigma_i=1} u_i \prod_{i \in J: \sigma_i=0} (1 - u_i), \quad c_\sigma \geq 0.$$

Every such coordinate requires at least one original degree, so $|J| \leq \sum_j \deg g_j$. If $\hat{P}$ is the substituted square polynomial, $\deg \hat{P} \leq \deg P$ and $|J| + 2 \deg \hat{P} \leq 2r$. Lemma 13.2 gives nonnegativity of each indicator times $\hat{P}^2$, hence every inequality in (14.1). This includes repeated factors and squares coupling restricted and unrestricted coordinates.

A locally vanishing h becomes zero at w_i on R , or has zero values at both endpoints on U . Its Boolean reduction is zero. This remains true after multiplication by every allowed, possibly globally coupled, v , proving (14.2). The objective value and perturbation estimate are exactly (13.11) and Corollary 13.4. Thus certification forces the same restriction count, and Lemma 12.1 finishes the proof. $\square$

The constructed first moment need not itself belong to a disconnected S_i ; it is a moment satisfying the stipulated oracle. The proof uses only endpoint and witness membership. It invokes no compact-domain hull theorem and takes no closure of S_i . A branch on $\psi_i(x_i) \leq \beta$ versus $\psi_i(x_i) \geq \beta$ changes only S_i , for any real-valued univariate function ψ_i . In particular, branching on $x_i(1 - x_i)$, with its disconnected endpoint-side region, is covered. For an objective-based reduction $S_i \rightarrow T_i \subset S_i$, the removed product region has coordinate set $S_i \setminus T_i$. If nonempty and certified at the target, it can be charged directly under Proposition 12.4, without a closure convention.

14.2 Endpoint interpolation in lifted coordinates

For each original coordinate i , introduce finitely many auxiliaries

$$y_{i,j} = \psi_{i,j}(x_i), \quad j = 1, \dots, a_i, \quad \psi_{i,j} : [0, 1] \rightarrow \mathbb{R}. \quad (14.4)$$

Every function is defined and finite on the entire interval. Neither continuity nor polynomiality is required. Write $\Gamma_i(v) = (v, \psi_{i,1}(v), \dots, \psi_{i,a_i}(v))$. A node restricts each local graph independently to $\Gamma_i(S_i)$ for arbitrary nonempty $S_i \subseteq [0, 1]$. Thus restrictions in the lifted variables are measured by

their exact preimages. Original coordinates x_i are retained and the original balances remain affine. The number of auxiliaries is finite for each instance but otherwise unrestricted.

The lifted order- r oracle acts on polynomials in all retained and auxiliary coordinates of total *lifted* degree at most $2r$. It imposes normalization; all multiples of the original affine balance through that degree; every local polynomial equality vanishing on $\Gamma_i(S_i)$ times every allowed global polynomial multiplier; and

$$L \left[\left(\prod_j g_j \right) P^2 \right] \geq 0, \quad \sum_j \deg g_j + 2 \deg P \leq 2r, \quad (14.5)$$

where each g_j is a polynomial in one local graph block, nonnegative on that block's restricted graph. Degrees in this display are measured before substitution. Squares can couple every lifted block, and factors may repeat. Graph equations are included whenever their lifted degrees allow them. Arbitrary coupled valid inequalities are not supplied.

Let $\tilde{F}(x, y)$ be a polynomial of lifted degree at most $2r$ that agrees exactly with $F(x)$, or with $F_d(x)$, on the *full* product graph $\prod_i \Gamma_i([0, 1])$. Agreement merely on the balance-feasible graph is not the stated assumption. The node bound is $\inf L[\tilde{F}]$. A graph-region cover means that its preimages cover the original compact set $\mathcal{F}_{n,k}$. Full-graph objective agreement preserves its optimum and an attaining optimizer, even if the lifted graph itself is noncompact because some $\psi_{i,j}$ is noncontinuous or unbounded near a point. Targets are still $T_* = 1/4 - \epsilon$ or $T_* = F_d^* - \epsilon$.

Theorem 14.2 (No order loss for coordinatewise graph lifts). *For the lifted model just defined, under the respective assumptions of Theorem 13.3 and Corollary 13.4, every certified cover has at least $\mathcal{N}(q_r)$ or $\mathcal{N}(q_{r,\eta})$ regions. The parameter is the lifted order r itself, independent of the number or polynomial degrees of the univariate auxiliaries. The dimension in the bound is the number n of original coordinates.*

Proof. Fix a node containing a graph witness $\Gamma(w)$ and form A, D, R, U from its preimage sets. Suppose $|R|$ is below the appropriate threshold. The cardinality parameters s, t satisfy the same hypotheses as before. Define an algebraic substitution Φ by

$$\begin{aligned} \Phi(x_i) &= w_i, & \Phi(y_{i,j}) &= \psi_{i,j}(w_i) & (i \in R), \\ \Phi(x_i) &= u_i, & \Phi(y_{i,j}) &= \psi_{i,j}(0) + (\psi_{i,j}(1) - \psi_{i,j}(0))u_i & (i \in U). \end{aligned} \quad (14.6)$$

Every image is affine or constant, so $\deg \Phi(P) \leq \deg P$ for every lifted polynomial. Define

$$L[P] = E_{s,t}[\Phi(P)] \quad (\deg P \leq 2r).$$

This is well-defined and normalized. The original affine balance maps to $\sum_{i \in U} u_i - t$, and any allowed multiplier maps to degree at most $2r - 1$. Lemma 13.1 proves all balance products.

A local valid inequality g maps to a nonnegative constant on R . For $i \in U$, the two values of $\Phi(g)$ are $g(\Gamma_i(0))$ and $g(\Gamma_i(1))$. Both are nonnegative because both endpoint graph tuples belong to the restricted local graph. Thus its Boolean reduction has the nonnegative endpoint form (14.3). Grouping and expanding all local factors gives nonnegative combinations of assignment indicators on at most $\sum_j \deg g_j$ distinct Boolean coordinates. Moreover $\deg \Phi(P) \leq \deg P$, so each indicator times $\Phi(P)^2$ lies in the degree range of Lemma 13.2. This proves every inequality (14.5), including squares coupling all original and auxiliary variables.

For a local equality h , $\Phi(h)$ vanishes at the deterministic witness tuple on R or at both endpoint graph tuples on U . Its Boolean reduction is zero. For any allowed multiplier v , $\text{red}(\Phi(h)\Phi(v)) = 0$,

and the degree bound ensures $L[hv]$ is defined. Thus every granted local equality product holds. The same reasoning would also respect any polynomial identity valid on the full product graph, even a coupled identity, and its allowed multiples: at every Boolean assignment the image is an actual graph point. This observation does not grant coupled inequalities that use the balances or objective cutoff.

At each Boolean assignment of u_U , (14.6) is an actual full-graph tuple with $x_R = w_R$. Therefore $\Phi(\tilde{F})$ agrees at every Boolean point with the polynomial obtained by substituting $x_R = w_R, x_U = u$ in F or F_d . Two polynomials agreeing on the Boolean cube have the same multilinear reduction: successively evaluating at zero and one in each coordinate proves uniqueness of that representation. Both polynomials have degree at most $2r$. Their functional values are consequently identical. The penalty is $|M \cap R|p(1-p)$ and any perturbation contributes at most η . The strict below-target estimate and the endpoint count from the preceding proofs finish the theorem. $\square$

The substitution Φ constructs a moment functional. It is not an identity $\psi_i(x_i) = \psi_i(0) + (\psi_i(1) - \psi_i(0))x_i$ for continuous x_i . The proof checks true graph identities only at deterministic or Boolean endpoint tuples. For example, it covers a square-root auxiliary $y_i = \sqrt{x_i}$ with its polynomial identity $y_i^2 = x_i$, and a penalty auxiliary $y_i = x_i(1 - x_i)$ with linear objective $\sum_i y_i$. Both maps are finite on all of $[0, 1]$ and their objective agreement, when used to encode the intended objective, is checked on the full graph. It also covers finite-valued nonpolynomial maps that have few polynomial identities; the oracle still grants every local polynomial inequality valid on the restricted graph.

The theorem requires independent local restrictions and retained original coordinates. Eliminating x_i through a nonlinear inverse, branching on functions of several original coordinates, granting arbitrary coupled valid cuts, or replacing the oracle by the exact hull of the coupled feasible graph changes the model. Arbitrarily many auxiliaries also mean that an exponential bound in original dimension need not be exponential in lifted dimension.

Proposition 14.3 (Literal polynomial substitution as a coarser bound). *If all $\psi_{i,j}$ are polynomials of degree at most an integer $D \geq 1$, with the originals counted as degree one, literal graph substitution also proves the lifted cover bounds with effective order rD in place of r . This is a valid weaker conclusion of the same type.*

Proof. Let Ψ replace each auxiliary by its defining polynomial and fix no coordinates yet. Then $\deg \Psi(P) \leq D \deg P$. Take the product-domain functional of order rD from Theorem 14.1 and set $\tilde{L}[P] = L[\Psi(P)]$. A lifted local inequality pulls back to a univariate polynomial valid on S_i , and

$$\sum_j \deg \Psi(g_j) + 2 \deg \Psi(P) \leq D \left(\sum_j \deg g_j + 2 \deg P \right) \leq 2rD.$$

Thus every localizer holds; local equalities pull back to equalities vanishing on S_i , and graph identities vanish identically. The original balance has degree one after substitution, so for a lifted multiplier v of degree at most $2r - 1$,

$$\deg(e\Psi(v)) \leq 1 + D(2r - 1) \leq 2rD.$$

All equality products are therefore available. Objective agreement on the full graph means the substituted objective equals F or F_d on $[0, 1]^n$. Since both are polynomials and the cube has interior, this is a polynomial identity. Hence its evaluated objective is the required one. The product-domain count at order rD applies. $\square$

For quadratic penalty auxiliaries the literal argument uses $\min\{k - 4r + 2, z - 4r + 2, m(1/2 - 2\epsilon)\}$. Theorem 14.2 improves this to q_r by using endpoint interpolation instead of literal substitution. It is this specific construction, not invariance of polynomial hierarchies under arbitrary nonlinear reformulations, that removes the degree loss.

15 Relative tolerance, tensor blocks, and escape cuts

A fixed relative-gap obstruction for increasing concave costs follows by using many independent blocks of fixed size. The node oracle is still global: its square multipliers can couple variables from every block. The independence is a feature of the optimization problem, and also gives a short alternative certificate outside the single-cover model.

Fix integers $r \geq 1$, $t \geq 2r - 1$, and $G \geq 1$. There are G blocks, each with $\ell = 3t$ original coordinates and its own balance:

$$\begin{aligned} n &= 3tG, & K &= t + \frac{1}{2}, & \sum_{i=1}^{3t} x_{b,i} &= K \quad (b = 1, \dots, G), & 0 \leq x_{b,i} \leq 1, \\ C(x) &= \sum_{b=1}^G \sum_{i=1}^{3t} [x_{b,i}(2 - x_{b,i}) + d_{b,i}x_{b,i}]. \end{aligned} \tag{15.1}$$

Assume $d_{b,1} < \dots < d_{b,3t}$ are positive and $\sum_i d_{b,i} \leq \eta$ for every block, with $\eta > 0$. Every coordinate cost is strictly increasing and strictly concave, by its derivatives $2 - 2x + d_{b,i} > 0$ and -2 . Corollary 13.4 applied blockwise gives the exact optimum

$$C^* = G(K + \frac{1}{4}) + \sum_{b=1}^G \left(\sum_{i \leq t} d_{b,i} + \frac{1}{2}d_{b,t+1} \right) \geq G(K + \frac{1}{4}). \tag{15.2}$$

On an arbitrary coordinate product domain, grant the order- r oracle (14.1)–(14.2) with all G balance equalities and every polynomial multiplier through *total* degree $2r$. Squares may involve all n variables. This includes the box preordering and, for boxes at $r = 1$, full global SDP–RLT with every balance product. Let $\text{LB}_r(S)$ be the infimum of $L[C]$. A relative certificate at tolerance $0 < \theta < 1$ is a finite cover of the feasible set by regions that are infeasible or satisfy

$$\text{LB}_r(S) \geq T_* = (1 - \theta)C^*. \tag{15.3}$$

Pruning at $(1 - \theta)U$ for any feasible incumbent $U \geq C^*$ implies this condition. The dimension in all ensuing bounds is original $n = 3tG$.

Lemma 15.1 (Tensor positivity with a total-degree truncation). *For each block b , suppose L_b is a normalized functional on block polynomials through degree $2r$, satisfying the block balance products, local equalities, and every block preordering inequality. Define L on global monomials of degree at most $2r$ by*

$$L \left[\prod_b x_b^{\alpha_b} \right] = \prod_b L_b[x_b^{\alpha_b}], \tag{15.4}$$

and extend linearly. Then L satisfies the corresponding global oracle, including every globally coupled square multiplier. The same assertion holds when a block contains coordinatewise graph auxiliaries and degree is measured in the lifted variables.

Proof. Each block degree is at most the global degree, so the definition is well-defined and normalized. To test a balance equality from block b times a global monomial, factor the latter across blocks. Its expectation is the block- b balance product, of degree at most $2r$, times moments of the other blocks, so it vanishes. Linearity treats every multiplier. The argument for local equalities is identical and also covers products involving several equalities within the total degree bound.

For a preordering expression let $g = \prod_b g_b$ group all local inequality factors according to their block, and let $d = \deg P$ with $\sum_b \deg g_b + 2d \leq 2r$. For every b , form the matrix

$$M_b(\alpha, \beta) = L_b[g_b x_b^\alpha x_b^\beta], \quad |\alpha|, |\beta| \leq d.$$

It is defined and positive semidefinite: the degree bound $\deg g_b + 2d \leq 2r$ allows the block preordering inequality for every polynomial in this index span. Their tensor product is positive semidefinite, as follows for example by tensoring Gram representations. Keep only tuples of indices whose total degree across blocks is at most d . The corresponding principal submatrix has entries exactly $L[gx^\alpha x^\beta]$ by (15.4); each such entry has available global degree at most $2r$. Its quadratic form on the coefficient vector of P is $L[gP^2] \geq 0$. The unused entries of the larger tensor matrix need not represent globally available moments: it is only a Gram construction whose required principal submatrix has the stated total-degree interpretation. Nothing in this proof uses that the coordinates within a block were unlifted. $\square$

Theorem 15.2 (A fixed relative-gap cover bound). *For (15.1) and the global product-domain oracle above, set*

$$q_0 = t - 2r + 2 \geq 1, \quad \tau = \frac{1}{4} - \theta(K + \frac{1}{4}) - \eta. \quad (15.5)$$

If $\tau > 0$, every certificate at target (15.3) has at least

$$(3/2)^{q_0 G \tau} = (3/2)^{q_0 \tau n / (3t)} \quad (15.6)$$

regions. For fixed r, t, θ, η satisfying these conditions, this is exponential in n . If $\tau \leq 0$, only the trivial bound one is asserted. The same conclusions hold for coordinatewise graph lifts satisfying the local oracle and full-graph objective-agreement assumptions of Theorem 14.2, retaining all original balances; there is no auxiliary-degree loss in r .

Proof. Choose independently in each block a uniform ordered partition (H_b, M_b, Z_b) into three sets of size t , and set $w_b = \mathbf{1}_{H_b} + p\mathbf{1}_{M_b}$ with $p = 1/(2t)$. Every combined witness is feasible. Fix a product domain containing one and define A_b, D_b, R_b by endpoint exclusion within block b . If $|R_b| < q_0$, the construction of Theorem 14.1 gives a feasible block functional whose penalty is at most

$$|R_b|p(1-p) \leq \frac{|R_b|}{2t} \leq \frac{|R_b|}{2q_0},$$

since $1 \leq q_0 \leq t$. No absolute pruning target is needed to construct this functional: only the endpoint restriction count is used. If $|R_b| \geq q_0$, instead use exact evaluation at the actual block witness. Its penalty is $tp(1-p) = 1/2 - 1/(4t) \leq 1/2 \leq |R_b|/(2q_0)$. Thus in either case the block functional is feasible through degree $2r$, its first moments lie in $[0, 1]$, and its penalty has the same upper bound. Lemma 15.1 gives a feasible global functional of objective at most

$$GK + \frac{1}{2q_0} \sum_b |R_b| + G\eta. \quad (15.7)$$

A certified region containing this witness must therefore satisfy

$$GK + \frac{1}{2q_0} \sum_b |R_b| + G\eta \geq \text{LB}_r(S) \geq (1 - \theta)C^* \geq (1 - \theta)G(K + \frac{1}{4}).$$

It follows that $\sum_b |R_b| \geq 2q_0 G\tau$.

For a fixed domain, witness membership factors across blocks. In block b , endpoint avoidance bounds its witness fraction by

$$\min\{(2/3)^{|A_b|}, (2/3)^{|D_b|}\} \leq (2/3)^{|R_b|/2}.$$

Multiplying over independent blocks bounds the full witness fraction in a certified region by $(2/3)^{q_0 G\tau}$. The union bound over a cover proves (15.6). The restriction counts belong to the region, not to the selected witness, so this is a uniform bound for every certified region with at least one witness.

For lifted regions work with the exact original-coordinate preimages. In a lightly restricted block use the affine endpoint substitution (14.6); in a heavily restricted block use evaluation at its true graph witness. These block functionals satisfy all lifted localizers and equality products by the proof of Theorem 14.2, with the original order r . Lemma 15.1 proves every global lifted localizer, including squares coupling auxiliaries from all blocks. It also proves all balance and local equality products.

For completeness, a lifted polynomial objective agreeing with C on the full product graph need not be written as a sum of block polynomials. After the block substitutions, it agrees with the substituted C at every Boolean assignment of all remaining formal variables; the heavily restricted blocks are actual fixed graph tuples. Affine substitution preserves its total degree at most $2r$, and uniqueness of Boolean reduction shows the two polynomials have the same value under the tensor functional. That functional annihilates Boolean identities in every remaining coordinate because each factor does. Hence (15.7) still holds, and the preimage witness count is identical. This proves the lifted assertion without assuming blockwise separability of the written lifted objective. $\square$

For the explicit parameters

$$r = 1, \quad t = 2, \quad \theta = 1/32, \quad \eta = 1/32, \quad K = 5/2, \quad q_0 = 2, \quad \tau = 17/128, \quad (15.8)$$

the lower bound is $(3/2)^{17n/384}$, even for full global SDP–RLT. For any other fixed order choose fixed $t \geq 2r - 1$, $0 < \eta < 1/4$, and $0 < \theta < (1/4 - \eta)/(K + 1/4)$. There is no assertion of a common positive relative tolerance while r grows without bound in this fixed-block construction.

Proposition 15.3 (Uniqueness and symmetry modulo the balances). *Problem (15.1) has a unique optimizer: in every block its first t coordinates equal one, coordinate $t + 1$ equals $1/2$, and the rest equal zero. For the explicit choice*

$$d_{b,i} = \frac{\eta((b-1)3t+i)^2}{3t n^2}, \quad b \in [G], \quad i \in [3t], \quad (15.9)$$

no nonidentity permutation of the original coordinates preserves both the feasible set and the objective restricted to that set.

Proof. Each independent block has the unique optimizer from Corollary 13.4; their concatenation is therefore the unique global optimizer. The coefficients in (15.9) are positive, globally distinct, and increasing in each block. Each is at most $\eta/(3t)$, so its block sum is at most η as required.

The point with every coordinate $K/(3t)$ is strictly between zero and one. Thus the affine hull of the feasible set is exactly the displayed balance space. Its normal row space consists of vectors constant on each block. A feasible-set-preserving coordinate permutation preserves this row space. The image of a block indicator is a zero-one vector in it with precisely $3t$ ones, so it is the indicator of one entire block. Hence such a permutation maps whole blocks to whole blocks.

The unperturbed quadratic and linear parts are invariant under every coordinate permutation. If the perturbed objective is preserved on the feasible set, its coefficient difference is constant within each block; this necessary condition follows by varying coordinates within their balance hyperplane. Consequently the coefficient multiset of each block must equal the coefficient multiset of its image block up to translation. Put $c = \eta/(3t n^2)$ and $\ell = 3t$. The sorted successive differences in block b are

$$c(2((b-1)\ell + i) + 1), \quad i = 1, \dots, \ell - 1.$$

Their first entries distinguish all blocks. Translation preserves these differences, so every block is fixed. Within a fixed block, comparison of the minimum shows that a finite coefficient set cannot equal a nonzero translate of itself. The translation is zero, and its distinct entries force the identity permutation. $\square$

Distinctness alone would not prove the statement modulo several balances: two blocks with coefficient lists differing by additive offsets could be exchanged without changing the objective on their equal-demand slices. The squared-index choice explicitly excludes this possibility. The symmetry assertion is about the original optimization problem; redundant auxiliary variables can of course introduce symmetries of their own.

15.1 A short alternative certificate and the scope of the obstruction

The blocks have fixed dimension. Solving each separately and summing its exact certified value gives a certificate of size proportional to G , if reuse and addition of component certificates is allowed. Even spatial certification of each separate block by Proposition 12.3 has fixed cost for fixed t . Theorem 15.2 counts regions in one global cover with the stated node oracle. It does not constrain a method that retains separate component certificates and adds their bounds. In this family the explicit sorted optimizer is an even simpler way to solve each component.

There is also a classical coupled cut that closes every root under study. Its low written degree does not mean that its nonnegativity follows from a low-degree local preordering.

Proposition 15.4 (A Boolean-quadric clique cut closes the root). *For integers $1 \leq k \leq n - 1$, the inequality*

$$2 \sum_{i < j} X_{ij} - 2k \sum_i \mu_i + k(k+1) \geq 0 \quad (15.10)$$

is valid on the continuous quadratic graph $(\mu, X) = (x, xx^T)$ for $x \in [0, 1]^n$. Adding it to the root equality-product relaxation yields the exact lower bound $1/4$ for F , and the exact root bound for F_d and C_d . One such inequality per block makes the root bound for (15.1) exact.

Proof. This is Padberg's clique inequality [19, Lemma 2, equation (17)], with the whole coordinate set and parameter k , after multiplying by two and changing sides. Its continuous validity also has a direct proof. After substituting $X_{ij} = x_i x_j$, the left side is multiaffine. Successively choosing endpoints can only decrease its value, so its minimum on the cube is attained at a Boolean vertex. At a vertex with s ones its value is $(s - k)(s - k - 1) \geq 0$, since $s - k$ is an integer.

The balance and its linearized products imply $\sum_{i,j} X_{ij} = K \sum_i \mu_i = K^2$. Equation (15.10) therefore gives

$$\sum_i X_{ii} \leq K^2 - 2kK + k(k+1) = k + \frac{1}{4}, \quad \sum_i (\mu_i - X_{ii}) \geq \frac{1}{4}.$$

For the perturbed objective, first moments meet the unit bounds and balance, and their linear cost is at least $\sum_{i \leq k} d_i + d_{k+1}/2$, the linear-program minimum proved in Corollary 13.4. Adding the

two bounds gives F_d^* . The actual optimizer supplies an attaining exact moment matrix, so the root bound is exact. Adding the conserved K treats C_d , and applying the argument separately to all block balances treats C . The deduction does not use PSD. For the optional boundary case $k = 0$, the same polynomial is $2 \sum_{i < j} x_i x_j \geq 0$ and the algebra still applies directly. $\square$

Thus these easy-to-optimize families exhibit limits of particular spatial certificate systems, including full node preorderings at the specified orders, while known global cuts avoid the obstruction. In a penalty lift the coupled affine inequality $\sum_i y_i \geq 1/4$ would also close the unperturbed root; its validity is precisely the global conclusion, so it is not supplied as a local graph inequality. These escape certificates motivate seeking different families when stronger coupled cuts are granted. In particular, a later signed cubic XOR construction serves a different spatial purpose and must not be identified with the positive multilinear gap examples of the earlier sections.

A Exhaustive verification of the finite signing constants

This appendix gives the finite computational proof used in Proposition 3.6. Label vertices $0, \dots, n-1$. For vertex signs t_i , switching replaces a_{ij} by $a'_{ij} = a_{ij} t_i t_j$. The identity $Q_{a'}(s) = Q_a(t \cdot s)$ shows that switching preserves both extrema of Q , and hence R_V . Taking $t_0 = 1$ and $t_j = a_{0j}$ makes every coefficient incident to zero positive. This normalized representative is unique within each switching class: a switching that keeps those coefficients positive has $t_j = t_0$ for all j , so changes no edge. Thus it suffices to enumerate the $2^{\binom{n-1}{2}}$ choices of negative edges among vertices $1, \dots, n-1$. There are respectively 2, 8, 64, 1024, 32768 representatives for $n = 3, \dots, 7$.

Encode a cut by the subset on side one, always leaving vertex zero on side zero. Complementation changes no cut, so the 2^{n-1} such subsets include every cut. If a cut has c crossing edges, of which h are negative, its weight is exactly $c - 2h$. The following complete Python 3.10+ program computes the first row of Proposition 3.6 using unbounded integers. The operation `bit_count()` counts set bits in an integer; each free-edge mask records which edges cross the cut. No floating-point arithmetic or solver is used.

```
from itertools import combinations
def min_range(n):
    edges = list(combinations(range(n), 2))
    free = [(i, j) for i, j in edges if i != 0]
    cuts = []
    for side in range(1 << (n - 1)):
        side <= 1 # Vertex zero stays on side zero.
        cross = lambda i, j: ((side >> i) ^ (side >> j)) & 1
        size = sum(cross(i, j) for i, j in edges)
        mask = sum(cross(i, j) << k
                    for k, (i, j) in enumerate(free))
        cuts.append((size, mask))
    best = len(edges) + 1
    for negative in range(1 << len(free)):
        weights = [size - 2 * (negative & mask).bit_count()
                    for size, mask in cuts]
        best = min(best, max(weights) - min(weights))
    return best
result = [min_range(n) for n in range(2, 8)]
assert result == [1, 2, 4, 4, 5, 8]
print(result)
```

Every signed cut weight lies between $-\binom{n}{2}$ and $\binom{n}{2}$; its range is at most $\binom{n}{2}$, because two cut weights differ by a sum of at most that many coefficients in $\{-1, 1\}$. The initial value `len(edges) + 1` therefore exceeds every candidate range. The two loops cover all normalized signings and all cuts, so their minimum is exactly $\min_a R_V(a)$, rather than merely an upper bound supplied by the best witness found. This is an exhaustive finite proof, not an analytic formula for general n .

n	Negative edges; all other edges have coefficient +1	$\min Q$	$\max Q$
3	$\emptyset$	-1	3
4	$\emptyset$	-2	6
5	12, 14, 23	-4	4
6	14, 15, 23, 25, 34	-5	5
7	12, 13, 16, 23, 25, 34	-7	9

Table 3: Attaining full-center signings, with vertices numbered from zero. An entry ij denotes the unordered pair $\{i, j\}$.

The extrema in Table 3 can be checked by direct evaluation of Q on the 2^{n-1} vertex-sign vectors with $s_0 = 1$. They give $R_V = (\max Q - \min Q)/2$ equal to the computed minimum in each case. Extending the listed K_6 witness by vertex six and positive new edges gives $\min Q = -9$, $\max Q = 11$ at the full center, hence ratio $21/10$; its original six-vertex face retains ratio three.

The accompanying script `verification/check_complete_signings.py` implements the same integer enumeration, records a histogram of all candidate ranges, and checks the displayed witnesses by direct evaluation of Q on every induced face. Running it with Python 3.10+ writes `verification/complete_signings.json`, including the witnesses, counts, exact rational ratios, and the extended six-vertex example.

B Distinct multiscale arguments and finite refinements

B.1 Arbitrary partitions and the dense predecessor

The exact dyadic formula in Theorem 4.2 relies on a simultaneous prefix-hitting construction. A separate logarithmic estimate works for arbitrary partitions in (4.5).

Proposition B.1 (Partition-independent estimate). *For the polynomial and means in (4.5), with any choice of equal-size level partitions,*

$$0 < H \leq 5 + \log_2(1 + (L - 1) \log 2), \quad T = L.$$

Proof. Use the quantile surrogate from (4.6), writing $r(t)$ for the nonincreasing failure-count quantile and $\delta = 2^{-L}$. The interval $0 < t \leq \delta$ contributes at most $\delta \sum_{j=1}^L 2^j < 2$, and $t > 1/2$ contributes zero. For $\delta < t \leq 1/2$, let l be the largest integer with $t \leq 2^{-l}$. Write $\log_2^+ v = \max\{0, \log_2 v\}$ for $v > 0$. Then $2^l \leq 1/t$, and for $r > 0$,

$$\sum_{j=1}^l \min(2^j, r) \leq r \left(3 + \log_2^+ \frac{1}{tr} \right). \quad (\text{B.1})$$

If $r \geq 2^l$, the sum is less than $2r$. If $r < 2^l$, it is lr and $l \leq 1 + \log_2(2^l/r)$. Otherwise, with $h = \lfloor \log_2 r \rfloor$, the first h capped terms sum to less than $2r$ and the others sum to $(l - h)r$; use $h > \log_2 r - 1$. This proves (B.1), interpreted as zero at $r = 0$.

Put $D = (\delta, 1/2)$ and $M_0 = \int_D r(t) dt \leq 1$. As $\log_2^+ v \leq \log(1+v)/\log 2$, the bound becomes

$$H \leq 2 + 3M_0 + \frac{1}{\log 2} \int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt.$$

For $M_0 > 0$, Jensen's inequality under the probability measure $r(t) dt/M_0$ gives

$$\int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt \leq M_0 \log \left(1 + \frac{(L-1) \log 2}{M_0} \right) \leq \log(1 + (L-1) \log 2).$$

The final step uses that $M \log(1+B/M)$ increases for $M > 0$, since $\log(1+s) \geq s/(1+s)$. At $M_0 = 0$ the integral is zero. This proves the upper bound. Independent rounding and common-threshold rounding have distinct expected polynomial values at these interior means, so $H > 0$. $\square$

A dense symmetric construction gives a different exact reduction. With $m = 2^L$, $u_j = 2^{-j}$, $k_j = mu_j$, define

$$\tilde{f}_L(a, z) = \sum_{j=1}^L \frac{1}{u_j \binom{m}{k_j}} \sum_{|K|=k_j} a_j \prod_{i \in K} z_i, \quad a_j = u_j, \quad z_i = 1 - 1/m. \quad (\text{B.2})$$

Again $T = \text{cav } \tilde{f}_L = L$. Conditional on r failed leaves, the fraction of size- k_j subsets hit is

$$g_j(r) = 1 - \frac{\binom{m-r}{k_j}}{\binom{m}{k_j}} \leq \min(1, u_j r).$$

The inequality is the union bound: a uniform size- k_j subset contains each specified failed leaf with probability $k_j/m = u_j$. Thus the deficiency is $\sum_j A_j g_j(R)/u_j$, bounded by the same surrogate $\sum_j A_j \min(2^j, R)$ and consequently by Proposition B.1.

In this dense model the hull gap itself is exactly the optimum of the finite rational program

$$\begin{aligned} \max \quad & \sum_{j=1}^L \sum_{r=0}^m z_{jr} g_j(r) / u_j \\ \text{subject to} \quad & p_r \geq 0, \quad 0 \leq z_{jr} \leq p_r, \\ & \sum_{r=0}^m p_r = 1, \quad \sum_{r=0}^m r p_r = 1, \quad \sum_{r=0}^m z_{jr} = u_j \quad (j = 1, \dots, L). \end{aligned} \quad (\text{B.3})$$

Every feasible binary law supplies $p_r = \mathbb{P}(R = r)$ and $z_{jr} = \mathbb{P}(A_j = 1, R = r)$. Conversely draw R with probabilities p_r , choose a uniform r -subset of failed leaves, and, conditional on $R = r$, choose anchors independently with probabilities z_{jr}/p_r when $p_r > 0$. These choices realize all means and the objective; zero-mass states can be ignored. Thus (B.3) is an exact reduction, not a formula for the sparse model. No floating-point values from this program are needed for the divergent bound.

B.2 Dyadic conditional-failure rounding and its tail lemma

The following proof uses finitely many dyadic scales rather than a harmonic kernel. Though its $O(\log d)$ bound is weaker, its tail argument is distinct.

Lemma B.2 (Tail integral). *For a finite collection $p_j \geq 0$, let $N(s) = |\{j : p_j \geq s\}|$ and $S = \sum_j p_j$. For $u > 0$,*

$$\int_0^u \min\{N(s), u/s\} ds \geq \min(u, S). \quad (\text{B.4})$$

Proof. If $S \leq u$, then every $p_j \leq u$ and $\int_0^u N(s) ds = S$. Monotonicity gives $sN(s) \leq \int_0^s N(t) dt \leq S \leq u$, so the integrand equals $N(s)$ throughout. If $S > u$, then $\int_0^u N(s) ds \geq u$: either a $p_j \geq u$, or that integral equals S . By continuity choose $v \leq u$ with $\int_0^v N(s) ds = u$. For $s \leq v$ the same monotonicity bound gives $sN(s) \leq u$. Integrating on $(0, v)$ therefore gives u . $\square$

Proposition B.3 (Dyadic degree bound). *Let $K = 1 + \lfloor \log_2(d-1) \rfloor$ for $d \geq 2$. Every positive multilinear polynomial of maximum degree at most d on a nonnegative box satisfies $T \leq 2(K+1)H/c_0$.*

Proof. For each scale $B \in \{1, 2, \dots, 2^{K-1}\}$ restricted to powers of two, draw a common uniform U . Low coordinates succeed when $U \leq x_i$. A high coordinate with failure mean $p > 0$ has $h = \min(Bp, 1)$ and, conditional on U , fails independently with probability p/h when $U \leq h$, and never otherwise. Coordinates with $p = 0$ never fail. The marginal failure is $h(p/h) = p$ and the probabilities are valid because $B \geq 1$.

For a one-low term with anchor mean $u > 0$, let $p_1, \dots, p_r$ be its high failure means. At $U = t \leq u$, each $p_j \geq t/B$ is active and has conditional failure probability at least $1/B$. Conditional independence and $1 - e^{-z} \geq c_0 \min(1, z)$ give deficiency

$$G_B \geq c_0 \int_0^u \min(1, N(t/B)/B) dt = c_0 \int_0^{u/B} \min(B, N(s)) ds.$$

For $0 < s \leq u$ with $N(s) > 0$, choose the largest power of two $B \leq \min(N(s), u/s)$. It is in the allowed set, since $N(s) \leq r \leq d-1$, and exceeds half this minimum. Therefore

$$\sum_B \mathbf{1}\{s \leq u/B\} \min(B, N(s)) \geq \frac{1}{2} \min(N(s), u/s).$$

Integrating and invoking Lemma B.2 yields $\sum_B G_B \geq c_0 \min(u, S)/2 = c_0 t_e/2$. Terms with $u = 0$ have zero gap. Independence supplies the same guarantee for terms with zero or at least two low coordinates by Lemma 2.3. The uniform mixture of the K scale laws and independence thus guarantees $c_0 t_e/(2(K+1))$ for every term. Sum and apply Proposition 2.4. $\square$

B.3 The simplest harmonic cutoff and its reciprocal correction

The general finite theorem permits $M = d-1$, so $\eta = 1$ and $\Lambda = 1 + \log(d-1)$. Let z_Λ solve $J_{\Lambda,1}(z_\Lambda) = z_\Lambda$ and put $\alpha = \Lambda z_\Lambda$, $\ell = \log \Lambda$. For $\ell \geq 4$,

$$z_\Lambda \geq \frac{\ell - \log \ell - 2}{\Lambda}. \quad (\text{B.5})$$

Indeed $A = \ell - \log \ell - 2 \geq 1/2$ and $A \leq \ell < \Lambda$. Equation (4.15) with $\eta = 1$ gives $\Lambda J(A/\Lambda) \geq \ell - \log \ell - 1 - 1/(2A) \geq A$. Monotonicity of $J(z) - z$ proves the finite bound.

The integrated linear upper bound and quadratic lower bound imply

$$\ell - \log \alpha - 1 - \frac{1}{2\alpha} \leq \alpha \leq \ell - \log \alpha - 1 + \frac{\alpha}{\Lambda}.$$

Together with (B.5), these show $\alpha/\ell \rightarrow 1$: once $\alpha \geq 1$, the upper bound gives $\alpha \leq \ell/(1 - 1/\Lambda)$, while the finite lower bound is $\ell - \log \ell - 2$. Substitution yields

$$\Lambda z_\Lambda = \log \Lambda - \log \log \Lambda - 1 + o(1). \quad (\text{B.6})$$

The constant -1 belongs to this fixed cutoff, whereas the tuned cutoff in (4.14) removes that constant after expressing the bound with numerator N .

There is also a reciprocal refinement. If $we^w = \Lambda/e$, then

$$\alpha = w - \frac{1}{2w} + O(w^{-2}) \quad (\Lambda \rightarrow \infty). \quad (\text{B.7})$$

To prove it, integrate the cubic Taylor bound for $1 - e^{-x}$ with $x = (1 - v)/(\Lambda v)$. After multiplication by Λ , the error beyond the quadratic term is at most

$$\Lambda \int_{\alpha/\Lambda}^1 \frac{x^3}{6} dv \leq \frac{1}{12\alpha^2}.$$

The exact quadratic integral is $\int_z^1 (1/v - 1)^2 dv = 1/z + 2 \log z - z$. Its omitted terms, and the linear endpoint term, are $O(\log \Lambda/\Lambda) = o(\alpha^{-2})$ because $\alpha \sim \ell$. Consequently

$$\alpha + \log \alpha = \log \Lambda - 1 - \frac{1}{2\alpha} + O(\alpha^{-2}).$$

Compare this with $w + \log w = \log \Lambda - 1$. Since $\alpha \sim w$, the mean value theorem for $t + \log t$ first gives $\alpha - w = O(1/w)$ and, after substitution, gives (B.7). This refines the scalar certificate only; it is not a second-order lower bound on R_d .

B.4 A second rational two-level parameter choice

A related finite variant illustrates that the two-level mechanism is not tied to the particular coefficients in Theorem 5.6. For real $a, c \in [0, 1]$, direct expansion gives

$$\begin{aligned} ac^2 + \frac{24}{25}a^2 - \left(\frac{41}{25}a + \frac{4}{5}c - \frac{68}{75} \right) &= \frac{24}{25} \left(a - \frac{41 - 25c^2}{48} \right)^2 \\ &+ \frac{(1 - c)(5c - 3)^2(5c + 11)}{480}. \end{aligned} \quad (\text{B.8})$$

Both terms on the right are nonnegative. The equality atoms $(1/3, 1)$ and $(2/3, 3/5)$, each with probability $1/2$, have mean $(1/2, 4/5)$ and scalar envelope value $83/150$. For $25 \mid m$, use groups of m variables with these two means and $f_m = AE_2(W) + (24m/25)E_2(U)$. At binary normalized counts,

$$\frac{2f_m}{m^3} = ac^2 + \frac{24}{25}a^2 - \frac{ac + (24/25)a}{m}.$$

The expected correction is at most $49/(50m)$ because $ac \leq a$. The scaled concave envelope and termwise gap are, respectively, $(49/50)(1 - 1/m)$ and $(22/25)(1 - 1/m)$; cubic terms have lower envelope $1/10$, while quadratic terms have lower envelope zero. The affine minorant gives scaled hull gap at most $49/50 - 83/150 = 32/75$. Conditional independent sampling from the equality atoms, as in Theorem 5.6, gives scaled hull gap at least $(32/75)(1 - 1/m)$. Thus

$$\frac{33}{16}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{33}{16}.$$

At $m = 50$ the lower certificate is $1617/800 > 2$. This variant has a smaller limiting ratio than the two-level family in the main text.

C Exact finite cubic certificates

The following certificates use success counts, so their verification does not enumerate all 2^{192} binary vectors. Within each group, every binary vector with the same count has the same polynomial value. The finite enumeration therefore covers every vertex, including every possible dependence pattern relevant to the envelope.

For the six-term form (5.9), let $\Phi_m(A, B, C)$ be its binary count polynomial as defined in the proof of Proposition 5.5. Table 4 gives integers D, b_A, b_B, b_C, b_0 such that

$$D\Phi_m(A, B, C) \geq b_AA + b_BB + b_CC + b_0 \quad (0 \leq A, B, C \leq m \text{ integers}). \quad (\text{C.1})$$

Table 5 gives attaining laws. Each listed weight is nonnegative, the weights sum to one, and their mean count vector is $(m/4, m/2, 3m/4)$. Uniform success subsets conditional on these counts therefore give the required singleton marginals. The checker below verifies the inequality at every integer count triple and equality of primal and dual expectations in rational arithmetic.

m	D	b_A	b_B	b_C	b_0
6	13	962	778	842	−4816
8	7	793	770	798	−5882
64	105	871710	900446	899046	−51743768

Table 4: Integer affine minorants for the three-group witnesses.

m	Count states (A, B, C)	Corresponding probabilities
6	$(1, 4, 4), (2, 1, 6), (6, 2, 1)$	$17/26, 4/13, 1/26$
8	$(1, 2, 8), (1, 5, 6), (8, 4, 2)$	$2/7, 4/7, 1/7$
64	$(4, 19, 64), (5, 19, 64), (16, 40, 43), (64, 31, 17)$	$241/735, 12/735, 419/735, 63/735$

Table 5: Primal laws attaining the affine bounds.

For $m = 8$, minimizing the residual in (C.1) over A, B for each $C = 0, \dots, 8$ gives

$$168, 42, 0, 30, 22, 0, 0, 16, 0.$$

The three primal states have values 405, 507, 734 respectively, with expected value $3572/7$. In all three examples the orbit sizes, upper values per unweighted monomial, and lower values per unweighted monomial are respectively

$$s = \left(\binom{m}{3}, m \binom{m}{2}, \binom{m}{2}, m^2, m^2, \binom{m}{2} \right),$$

$$u = (3/4, 1/2, 1/2, 1/4, 1/4, 1/4), \quad l = (1/4, 0, 0, 0, 0, 0).$$

Thus $\text{cav } f = \sum_i c_i s_i u_i$ and $\sum_e \text{vex } f_e = \sum_i c_i s_i l_i$. These formulas prove the remaining columns of Table 2. The exact hull gaps for dimensions 18, 24, 192 are $10411/52, 3225/7, 27562048/105$, respectively.

For the two-level $m = 16$ certificate (5.17), the integer residual is

$$2 \left[A \binom{C}{2} + 20 \binom{A}{2} \right] - 428A - 177C + 3372.$$

Its minimum over $C = 0, \dots, 16$ for successive $A = 0, \dots, 16$ is

$$540, 352, 204, 96, 28, 0, 9, 7, 0, \\ 0, 19, 63, 132, 235, 369, 540, 742.$$

At counts $(8, 12)$ the value and its affine bound both equal 1088.

For the smaller members of (5.13), write $\Phi_m(A, C) = A \binom{C}{2} + (5m/4) \binom{A}{2}$. Table 6 gives affine minorants valid at every integer pair $0 \leq A, C \leq m$. At the listed counts each minorant equals Φ_m . Independently choosing uniform success subsets of those fixed sizes in the two groups gives every singleton mean $1/2, 3/4$ and attains the bound. Taking expectations of the minorant under any feasible law therefore proves the stated exact convex-envelope value. Every term has lower envelope zero and upper envelope $1/2$, so $\text{cav } f_m = T(f_m) = (9m/8) \binom{m}{2}$; subtraction gives the exact ratios. The checker verifies all $5^2 + 9^2 + 13^2$ minorant inequalities and these attainment and envelope equalities in rational arithmetic.

m	Affine minorant	Counts (A, C)	$\text{vex } f_m$	$\text{cav } f_m = T$	T/H
4	$9A + 4C - 19$	$(2, 3)$	11	27	$27/16$
8	$47A + 20C - 188$	$(4, 6)$	120	252	$21/11$
12	$(231/2)A + 48C - 684$	$(6, 9)$	441	891	$99/50$

Table 6: Exact smaller two-level certificates and attaining counts.

Here is a complete Python 3 checker using only the standard library. It enumerates $7^3 + 9^3 + 65^3$ three-group states and $5^2 + 9^2 + 13^2 + 17^2$ two-group states. The data are exactly those in the manuscript tables; no numerical optimization or proprietary solver is involved. The same executable text is supplied as `verification/check_stage02_finite.py`. Concatenate the two code blocks in order to obtain that file.

```
from fractions import Fraction as Q
from itertools import product
from math import comb

# m, coefficients, denominator, integer affine coefficients,
# probability denominator, (state, probability numerator), ratio
cases = [
    (6, (2,3,9,10,7,7), 13, (962,778,842,-4816), 26,
     [((1,4,4),17), ((2,1,6),8), ((6,2,1),1)],
     Q(20891,10411)),
    (8, (2,3,13,12,8,7), 7, (793,770,798,-5882), 7,
     [((1,2,8),2), ((1,5,6),4), ((8,4,2),1)],
     Q(6601,3225)),
    (64, (2,3,120,105,70,63), 105,
     (871710,900446,899046,-51743768), 735,
     [((4,19,64),241), ((5,19,64),12),
      ((16,40,43),419), ((64,31,17),63)],
     Q(7443345,3445256))]

def phi(a, b, c, coef):
    values = (comb(c,3), b*comb(c,2), comb(b,2),
              a*c, a*b, comb(a,2))
    return sum(v*w for v,w in zip(values,coef))
```

```

for m,coef,D,dual,P,atoms,ratio in cases:
    for a,b,c in product(range(m+1), repeat=3):
        rhs = sum(v*w for v,w in zip((a,b,c,1),dual))
        assert D*phi(a,b,c,coef) >= rhs
        assert all(p >= 0 for _,p in atoms)
        assert sum(p for _,p in atoms) == P
        means = [sum(Q(p,P)*state[j] for state,p in atoms)
                  for j in range(3)]
        assert means == [Q(m,4), Q(m,2), Q(3*m,4)]
        primal = sum(Q(p,P)*phi(*state,coef) for state,p in atoms)
        lower = sum(Q(v,D)*w for v,w in zip(dual,means+[1]))
        assert primal == lower
        sizes = (comb(m,3), m*comb(m,2), comb(m,2),
                  m*m, m*m, comb(m,2))
        upper = (Q(3,4), Q(1,2), Q(1,2), Q(1,4), Q(1,4), Q(1,4))
        cav = sum(c*s*u for c,s,u in zip(coef,sizes,upper))
        term_lower = Q(coef[0]*comb(m,3),4)
        assert cav > primal
        assert (cav-term_lower)/(cav-primal) == ratio
        assert ratio > 2
        print(3*m, cav, term_lower, primal, ratio)

# m, affine coefficients (A,C,1), attaining value, cav=T, ratio
small = [
    (4, (9,4,-19), 11, 27, Q(27,16)),
    (8, (47,20,-188), 120, 252, Q(21,11)),
    (12, (Q(231,2),48,-684), 441, 891, Q(99,50))]
for m,dual,vex,cav,ratio in small:
    def value(a,c):
        return a*comb(c,2)+Q(5*m,4)*comb(a,2)
    slacks = [value(a,c)-dual[0]*a-dual[1]*c-dual[2]
              for a,c in product(range(m+1), repeat=2)]
    assert min(slacks) == 0
    a,c = m//2,3*m//4
    assert Q(a,m) == Q(1,2) and Q(c,m) == Q(3,4)
    assert value(a,c) == dual[0]*a+dual[1]*c+dual[2] == vex
    assert (m+Q(5*m,4))*comb(m,2)/2 == cav
    assert cav > vex and Q(cav,cav-vex) == ratio < 2
    print(2*m, cav, vex, ratio)

mins = []
for a in range(17):
    residuals = [2*(a*comb(c,2)+20*comb(a,2))
                 -428*a-177*c+3372 for c in range(17)]
    mins.append(min(residuals))
assert mins == [540,352,204,96,28,0,9,7,0,0,19,63,
                132,235,369,540,742]
assert 8*comb(12,2)+20*comb(8,2) == 1088
assert 214*8+Q(177,2)*12-1686 == 1088
assert Q(2160,2160-1088) == Q(135,67)
assert Q(135,67) > 2
assert Q(243,115)*(1-Q(1,20)) == Q(4617,2300) > 2
print("All finite cubic certificates passed.")

```

D Auxiliary structural reductions

The following arguments are useful independently of the width-two proof. They concern the stated local laws or reductions and make no claim about arbitrary larger-width extensions.

Lemma D.1 (Canonical pair laws on a forest). *On a forest of binary variables of prescribed means p_i , let Q_{ij} be the midpoint of the two Fréchet endpoint pair laws on each edge:*

$$Q_{ij}(1, 1) = \frac{\min(p_i, p_j) + \max(0, p_i + p_j - 1)}{2}.$$

For any target set S there is a global law with these pair marginals and target union probability at least $[\min(1, \sum_{i \in S} p_i) + \max_{i \in S} p_i]/2$, with both expressions zero for $S = \emptyset$. For any nonnegative target payoff, such a law also attains at least half its unconstrained singleton-feasible maximum. The resulting law may depend on the target.

Proof. Every pair cell is affine in its $(1, 1)$ probability and nonnegative on the Fréchet interval. Its midpoint value is therefore at least half every feasible cell value. Choose a singleton-feasible law P maximizing the target union, using consecutive circular event intervals to attain $c = \min(1, \sum_S p_i)$. The edge measures $R_{ij} = 2Q_{ij} - P_{ij}$ are nonnegative probability measures with the prescribed singleton means. They glue on the forest by rooting each component and sampling each child conditionally on its parent. Null conditioning states are irrelevant. Thus a law R with these edge marginals exists. The mixture $(P + R)/2$ has exactly the desired Q marginals, and its target union is at least $(c + b)/2$ since every law has union at least $b = \max_S p_i$. For a nonnegative payoff use its maximizing law P and $\mathbb{E}_{Rg} \geq 0$ in the same construction. $\square$

This residual construction cannot simply be applied on a triangle: for three fair bits, take P equally on the two constant states. Each canonical pair is uniform, so $2Q_{ij} - P_{ij}$ requires perfect anticorrelation. Three bits cannot be pairwise anticorrelated. This is a failure of that residual-gluing step, not a counterexample to a positive-monomial width-two bound.

Lemma D.2 (Identical-neighborhood compression). *In a positive unit-cube monomial factorization, suppose a nonempty variable block B occurs in every incident factor in its entirety. Replace it by one bit W of mean $w = \max(0, \sum_{i \in B} x_i - |B| + 1)$ in those factors. The full lower envelope and sum of local lower envelopes equal those of the reduced instance. For some $\Delta \geq 0$,*

$$T_{\text{original}} = T_{\text{reduced}} + \Delta, \quad H_{\text{original}} = H_{\text{reduced}} + \Delta.$$

Thus every reduced gap inequality with constant at least one transfers.

Proof. For any original binary law, the product $V = \prod_{i \in B} X_i$ has mean $q \geq w$. If $q > 0$, independently retain its ones with probability w/q , producing $W \leq V$ of mean w ; if $q = 0$ use $W = 0$. The reduced objective is nondecreasing in W , so this operation cannot increase it. Conversely choose a local law on B attaining product mean w with all the original means. Attach it to a reduced law by conditioning on its product value W . This lifts the law with the same objective and original means. These two operations prove full lower equality, including the endpoint cases. For a factor with scope $B \cup C$, its original local lower value is

$$\max(0, \sum_B x_i + \sum_C x_i - |B| - |C| + 1) = \max(0, w + \sum_C x_i - |C|),$$

where if $\sum_B x_i - |B| + 1 < 0$ both sides vanish. The upper value can only decrease under compression because $w \leq \min_B x_i$; this also holds when C is empty and the reduced factor is affine. Common upper attainment permits summing these decreases into one $\Delta \geq 0$. The two gap identities follow, and $T_{\text{reduced}} + \Delta \leq c(H_{\text{reduced}} + \Delta)$ for $c \geq 1$ proves the transfer without dividing by a gap. $\square$

Proposition D.3 (Signed forest gluing on arbitrary boxes). *For any multiaffine local factors on finite boxes whose incidence graph is a forest, both upper and lower envelopes add. In particular this holds for frequency-two factors whose dual multigraph is a forest.*

Proof. Choose an optimizing vertex law for each factor and normalize its coordinate endpoints. Root each incidence-tree component at a factor and traverse it, sampling each new factor conditionally on its single already sampled variable. Consistent singleton means ensure consistency at that separator; every other variable is new, or there would be an incidence cycle. Null separator states have zero probability. This simultaneously preserves all chosen optimizing laws. Use minimizing laws for the lower identity and maximizing laws for the upper identity. Different components and unused variables can be sampled independently. Fixed coordinates are substituted first. Signs do not enter the proof. $\square$

A factor partition into q incidence forests would give a q bound by mixing their exact laws for positive monomials, but treewidth does not bound such a partition number. In $K_{2,m}$, $m \geq 2$, with the two-vertex side interpreted as variables, any two factors create a four-cycle; all m factors need distinct forest colors, although treewidth is two. Adding one private variable to each factor makes their monomial scopes distinct and preserves the obstruction. This explains why the stronger matrix argument is useful.

For the radix construction, the general cutoff following Theorem 7.3 also records a fixed-radix limitation. For integers $b \geq 2, L \geq 2$ its profile capacities satisfy

$$M_s = \frac{(L-s)(b-1)+b}{b^s}, \quad \sum_{q=s+1}^L M_q = \frac{L-s}{b^s}.$$

The sum identity follows by telescoping $M_q = (L-q+1)b^{1-q} - (L-q)b^{-q}$. If s brackets one between M_s and M_{s+1} , the exact hull is $s + (L-s)b^{-s}$, as proved by its two attaining profiles. For fixed b , $s = \log_b L + O_b(1)$: the inequality $M_s \geq 1$ gives $s \leq \log_b((b-1)L+b)$, hence $L-s \sim L$; the reverse bracket then gives $b^{s+1} \geq (b-1)(L-s-1)+b$. The remainder term is $O_b(1)$, so the hull is $\log_b L + O_b(1)$. This is a limitation of this fixed-radix construction, not a sharp second-order lower theorem for all positive polynomials.

E Distinct positive-box comparisons

These proofs are superseded as upper bounds by Theorem 10.2, but use different estimates. Throughout this appendix $\rho > 1$, $t = \rho - 1$, $\alpha = 1/\rho$, and $\eta = 1 - \alpha$. Work first on $[1, \rho]^n$. The law I is independent endpoint rounding and O is independently fair endpoint orientation. For one product write C, V, P, O for its upper, lower, independent, and orientation values, and $D_I = C - P, D_O = C - O$.

E.1 The symmetric low/high argument

Proposition E.1 (A coarse finite constant). *For positive monomials on a box whose aspect ratios are at most ρ ,*

$$T \leq K(\rho)H, \quad K(\rho) = 2(1 + 1/\rho) + \max\{4(1 + \rho), 8\eta^{-3}\}.$$

In particular, with $R = \max\{4, \max_i u_i/\ell_i\}$, $T \leq (4R + 6 + 2/R)H$, which is $45/2$ for aspect ratios at most four. Direct use of $\rho = 2$ instead gives the older constant 67.

Proof. Classify a normalized mean as low when it is at most $1/2$. Put $q_i = p_i$ for lows and $q_i = 1 - p_i$ for highs, so $q_i \leq 1/2$. Divide the product's values by ρ^h , where h is the number of high coordinates. Let $L(s), H(s)$ count the respective deviations at least s , and let Q_L, Q_H be their sums. Integrals in this paragraph are over $[0, 1/2]$. Define

$$\begin{aligned} C_L &= 1 + \int (\rho^{L(s)} - 1) ds, & P_L &= \prod_{\text{low}} (1 + tq_i), \\ C_H &= 1 + \int (\alpha^{H(s)} - 1) ds, & P_H &= \prod_{\text{high}} (1 - \eta q_i). \end{aligned}$$

Common-threshold low successes and high failures occupy opposite ends of the interval. Thus $C = C_L + C_H - 1$, $P = P_L P_H$, and

$$D_I = (C_L - P_L) + (C_H - P_H) + J_I, \quad J_I = (P_L - 1)(1 - P_H) \geq 0. \quad (\text{E.1})$$

All three terms are nonnegative. Set $A = C_L - 1 - tQ_L$, $B = C_H - 1 + \eta Q_H$, and let q_L, q_H be the respective largest deviations, zero for empty groups. Integer-count inequalities give

$$A \geq t^2(Q_L - q_L), \quad B \geq \eta^2(Q_H - q_H). \quad (\text{E.2})$$

For the first, use $(1+t)^l - 1 - tl \geq t^2(l-1)_+$; for the second, $\alpha^l - 1 + \eta l$ has increments $\eta(1 - \alpha^l) \geq \eta^2$ for $l \geq 1$. Integrating proves both. Conditional orientation expectations at either end of the interval are $b_L = (1+t/2)^{L(s)}$ and $b_H = (1-\eta/2)^{H(s)}$, so

$$O = 1 + 2 \int (b_L b_H - 1), \quad D_O = D_{OL} + D_{OH} + J_O, \quad J_O = 2 \int (b_L - 1)(1 - b_H).$$

Here $D_{OL}, D_{OH} \geq 0$ are the separate-group deficiencies. Expansion in powers of t gives $(1+t)^l - 2(1+t/2)^l + 1 \geq [(1+t)^l - 1 - tl]/2$, since each order- $j \geq 2$ coefficient is multiplied by $1 - 2^{1-j} \geq 1/2$. Therefore

$$D_O \geq A/2 + J_O, \quad J_O \geq (t\eta/2) \min(q_L, q_H). \quad (\text{E.3})$$

The high separate-group deficiency is also nonnegative directly from convexity of the integer power at the midpoint of α and one.

Write $D_{IH} = C_H - P_H$. If $Q_H \leq 1$, put $q = q_H$ and $R' = Q_H - q$. Bonferroni's bound gives

$$D_{IH} \geq B - \eta^2 \sum_{i < j} q_i q_j \geq B - \eta^2 (qR' + R'^2/2) \geq (1 - q - R'/2)B \geq B/4.$$

Here $B \geq \eta^2 R'$, $q \leq 1/2$, and $q + R' \leq 1$; no division by R' is needed. If $Q_H \geq 1$, order deviations decreasingly. At a fixed sum, the linear common-threshold high value is minimized by filling the largest-weight coordinates to their cap $1/2$ and at most one remainder. Convex interpolation then gives

$$C_H \geq \frac{1}{2} + \frac{1}{2} \alpha^{2Q_H}, \quad P_H \leq e^{-\eta Q_H}.$$

The difference $g(Q) = 1/2 + \alpha^{2Q}/2 - e^{-\eta Q}$ is nondecreasing for $Q \geq 1$. Indeed $-\log \alpha \leq \eta/\alpha$ and $\alpha^{2Q-1} \leq e^{-\eta Q}$ imply $g'(Q) \geq \eta[e^{-\eta Q} - \alpha^{2Q-1}] \geq 0$. The fourth-order exponential remainder yields

$$D_{IH} \geq g(1) = 1 - \eta + \eta^2/2 - e^{-\eta} \geq \eta^3/6 - \eta^4/24 \geq \eta^3/8.$$

Let ϕ_ρ interpolate ρ^k between every consecutive integer, including negative integers. The exact normalized lower envelope is $V = \phi_\rho(Q_L - Q_H)$, because the original count sum is shifted by integer h . Supporting slopes at zero are η and t , so

$$C - V \leq A + B + t\eta \min(Q_L, Q_H). \quad (\text{E.4})$$

For $Q_H \leq 1$, use $\min(Q_L, Q_H) \leq \min(q_L, q_H) + A/t^2 + B/\eta^2$ to obtain

$$T \leq (1 + 1/\rho)A + (1 + \rho)B + 2J_O \leq 2(1 + 1/\rho)D_O + 4(1 + \rho)D_I.$$

For $Q_H \geq 1$, $T \leq C_L = 1 + A + tQ_L$ and $J_I \geq tQ_L(1 - e^{-\eta}) \geq t\eta Q_L/2$. The preceding bounds therefore imply

$$T \leq 2D_O + 8\eta^{-3}D_{IH} + 2\eta^{-1}J_I \leq 2D_O + 8\eta^{-3}D_I.$$

Combining the cases proves the per-product inequality with the two coefficients in $K(\rho)$. Mix the global laws proportionally, undo the product's normalization, and sum to prove the common-aspect bound. The positive affine substitution of Theorem 10.2 transfers it to unequal aspects. For $R \geq 4$, $8(1 - 1/R)^{-3} \leq 512/27 < 20 \leq 4(1 + R)$, giving $K(R) = 4R + 6 + 2/R$. For $\rho = 2$ the two coefficients are 3 and 64. $\square$

The high-mass argument in fact permits replacing $8\eta^{-3}$ by $1/g(1)$. Indeed $g(1) \leq 1 - e^{-\eta}$, since their difference is $\eta - \eta^2/2 \geq 0$; thus the same coefficient controls both the constant and the J_I term in C_L . This gives the distinct sharper finite expression $2(1 + 1/\rho) + \max\{4(1 + \rho), 1/g(1)\}$.

A tempting all-high simplification $T \leq 2D_I$ is false. On $[1, 2]^3$ with failure probabilities $(1/2, 49/100, 1/100)$, divide values by 2^3 . Exact values are

$$C = 501/800, \quad V = 1/2, \quad P = 90147/160000, \quad T = 20200/160000 > 2D_I = 20106/160000.$$

The boundary mean $1/2$ is allowed in this all-high calculation. It does not invalidate the mixed-law inequality.

E.2 The asymmetric split

Proposition E.2 (An earlier leading-order bound). *For $\rho \geq 64$, positive products on boxes of aspect ratios at most ρ satisfy*

$$T \leq \left(2 + \frac{\rho + 1}{1 - 3/\sqrt{\rho}}\right) H.$$

Proof. Put $\delta = \rho^{-1/2} \leq 1/8$ and $b = (\rho + 1)/(1 - 3\delta)$. Now classify normalized means as low if $p_i \leq 1 - \delta$, and high otherwise. High deviations are $q_i = 1 - p_i < \delta$. Normalize by ρ^h again. Reset the counts to $L(s) = \#\{i \text{ low} : p_i \geq s\}$ and $H(s) = \#\{i \text{ high} : q_i \geq s\}$, with $Q_L = \sum_{\text{low}} p_i$ and $Q_H = \sum_{\text{high}} q_i$. The separate groups' common-threshold upper values and independent values are now

$$\begin{aligned} C_L &= 1 + \int_0^{1-\delta} (\rho^{L(s)} - 1) ds, & P_L &= \prod_{\text{low}} (1 + tp_i), \\ C_H &= 1 + \int_0^\delta (\alpha^{H(s)} - 1) ds, & P_H &= \prod_{\text{high}} (1 - \eta q_i). \end{aligned}$$

Low successes and high failures under common-threshold rounding are disjoint, so (E.1), the definitions $A = C_L - 1 - tQ_L$, $B = C_H - 1 + \eta Q_H$, and (E.4) remain valid. The dispersion bounds (E.2) hold with maxima a_L, a_H of low means and high failures; they did not require low means at most $1/2$. A different estimate now replaces $D_{OL} \geq A/2$:

$$D_{IL} = C_L - P_L \geq \delta A.$$

Expand the low physical product only for this calculation. Every subset of degree at least two has independent product at most $(1 - \delta)$ times its minimum mean, and the sum of its common upper values is exactly A . Constant and affine terms cancel.

For $s \in [0, \delta]$ and $U = s$ or $1 - s$, conditional low expectations are $1 + (t/2)\mathbf{1}\{s \leq p_i\}$, while high normalized expectations are $1 - (\eta/2)\mathbf{1}\{s \leq q_i\}$. Outside these two end strips the conditional high

product is one. Conditional independence of the orientation coins gives $D_O = D_{OL} + D_{OH} + J_O \geq J_O$, where

$$J_O = 2 \int_0^\delta \left(\prod_{\text{low}} [1 + (t/2) \mathbf{1}\{s \leq p_i\}] - 1 \right) \left(1 - \prod_{\text{high}} [1 - (\eta/2) \mathbf{1}\{s \leq q_i\}] \right) ds \geq (t\eta/2) \min(a_L, a_H).$$

Combining this with dispersion and the target inequality yields $T \leq (1 + 1/\rho)A + (1 + \rho)B + 2J_O$. If $Q_H \leq 4\delta$, the same Bonferroni argument with $q = a_H \leq \delta$ and $R' = Q_H - q \leq 4\delta$ gives $D_{IH} \geq (1 - q - R'/2)B \geq (1 - 3\delta)B$. Since $(1 + 1/\rho)/\delta \leq b$, it follows that $T \leq bD_I + 2D_O$.

If $Q_H \geq 4\delta$, the common high failure set has measure at most δ , so $C_H \geq 1 - \delta$. Also $P_H \leq e^{-4\eta\delta} \leq 1 - 2\delta$, by $e^{-z} \leq 1 - z + z^2/2$ and $4\eta - 8\eta^2\delta \geq 3 - 1 = 2$. Hence $D_{IH} \geq \delta$ and $J_I \geq \delta t Q_L$, giving $D_I \geq \delta(A + 1 + tQ_L) = \delta C_L$. As $T \leq C_L$, again $T \leq bD_I + 2D_O$. Mix the two global laws with weights proportional to $b, 2$, sum over products, and use positive affine transfer for unequal aspects. $\square$

Its expansion is $\rho + 3\sqrt{\rho} + O(1)$; the coefficient theorem improves the additive error to two.

E.3 Optimality within a fixed two-law mixture

Proposition E.3 (A limitation of fixed independent/fair-orientation mixtures). *Fix $\rho > 1$ and an orientation weight $w \in [0, 1]$. For mixtures of I and independently fair O , the best uniform per-physical-monomial captured fraction over all dimensions is at most $\min\{w/2, (1-w)/\rho\}$. Optimizing the weight gives exactly $1/(\rho + 2)$, attained at $w = 2/(\rho + 2)$.*

Proof. For a bilinear product at normalized means $(\epsilon, 1 - \epsilon)$, $0 < \epsilon < 1/2$, deleting affine parts gives $D_I/T = \epsilon$ and $D_O/T = 1/2$. Letting $\epsilon \downarrow 0$ bounds the mixture fraction by $w/2$. For the other obstruction use means ϵ and k copies of $1 - \epsilon/k$, where $k \geq 1$. Divide physical values by ρ^{k+1} , and write $\alpha = 1/\rho$, $\eta = 1 - \alpha$, $\gamma = (1 + \alpha)/2$. The sum of success means is k , giving $V = \alpha$. Common-threshold and fair-orientation integration, split at ϵ/k and ϵ , and direct independent multiplication give

$$\begin{aligned} C &= \alpha + \epsilon[\eta - \alpha(1 - \alpha^k)/k], \\ P &= \alpha[1 + (\rho - 1)\epsilon][1 - \eta\epsilon/k]^k, \\ O &= \alpha + \epsilon[\eta - 2\gamma(1 - \gamma^k)/k]. \end{aligned}$$

For clarity, in the orientation law the two end intervals of length ϵ/k have normalized expected product γ^{k+1} , the two next intervals of total length $2(\epsilon - \epsilon/k)$ have value γ , and the middle interval of length $1 - 2\epsilon$ has value α ; these yield the stated O . For each fixed k , divide deficiencies by ϵ and let $\epsilon \downarrow 0$. Then

$$\begin{aligned} T/\epsilon &= \eta - \alpha(1 - \alpha^k)/k, \\ D_I/\epsilon &\longrightarrow \alpha\eta - \alpha(1 - \alpha^k)/k, \\ D_O/\epsilon &= [2\gamma(1 - \gamma^k) - \alpha(1 - \alpha^k)]/k. \end{aligned}$$

The denominator coefficient is at least $\eta^2 > 0$ because $1 - \alpha^k \leq k\eta$. Now let $k \rightarrow \infty$ to obtain $D_I/T \rightarrow 1/\rho$ and $D_O/T \rightarrow 0$, bounding the mixture by $(1 - w)/\rho$. The maximum of the minimum of these two bounds is $1/(\rho + 2)$ at their intersection. Lemma 10.1 with fair orientations supplies the matching guarantee. $\square$

This is only an optimality statement for a fixed two-law mixture’s per-monomial guarantee over unrestricted dimensions. Other laws, coefficient-dependent mixtures, and the true optimum joint law for a sum of monomials are outside it. In particular it does not prove $C_{\text{box}}(\rho) = \rho + 2$, nor conflict with the finite-ambient balanced-orientation refinement.

References

- [1] Warren P. Adams, Akshay Gupte, and Yibo Xu. Error bounds for monomial convexification in polynomial optimization. *Mathematical Programming*, 175(1–2):355–393, 2019. doi: 10.1007/s10107-018-1246-8. Proposition numbers refer to the author manuscript, arXiv:1704.00424.
- [2] Jason M. Altschuler and Enric Boix-Adserà. Polynomial-time algorithms for multimarginal optimal transport problems with structure. *Mathematical Programming*, 199:1107–1178, 2023. doi: 10.1007/s10107-022-01868-7.
- [3] Michael D. Barrus. On fractional realizations of graph degree sequences. *Electronic Journal of Combinatorics*, 21(2):P2.18, 2014. URL <https://www.combinatorics.org/ojs/index.php/eljc/article/download/v21i2p18/pdf/>.
- [4] Natashia Boland, Santanu S. Dey, Thomas Kalinowski, Marco Molinaro, and Fabian Rigterink. Bounding the gap between the McCormick relaxation and the convex hull for bilinear functions. *Mathematical Programming*, 162(1–2):523–535, 2017. doi: 10.1007/s10107-016-1031-5.
- [5] Stefano Coniglio. Solving the 2-norm k-hyperplane clustering problem via multi-norm formulations. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=VJAqqtVXfD>. The comparison uses Assumption 1, Propositions 2–4 and Appendix C of the anonymous review version, <https://openreview.net/pdf/369974754b073802afab412ee5f715561c39adbc.pdf>; identity with the published version has not been verified.
- [6] Gérard Cornuéjols. Combinatorial optimization: Packing and covering, 2000. URL <https://www.andrew.cmu.edu/user/gc0v/webpub/notes.pdf>. July 2000 author manuscript; Theorems 6.5 and 6.13.
- [7] Kenneth R. Davidson and Allan P. Donsig. Norms of Schur multipliers. *Illinois Journal of Mathematics*, 51(3), 2007. doi: 10.1215/ijm/1258131101.
- [8] Alberto Del Pia and Aida Khajavirad. The multilinear polytope for acyclic hypergraphs. *SIAM Journal on Optimization*, 28(2):1049–1076, 2018. doi: 10.1137/16M1095998.
- [9] Gabriel Deza and Shmuel Onn. Optimization over degree sequences of graphs. *Discrete Applied Mathematics*, 296:2–8, 2021. doi: 10.1016/j.dam.2019.12.016. Theorem 1.2 and Section 3; author manuscript arXiv:1908.09278.
- [10] Jack Edmonds. Maximum matching and a polyhedron with 0,1-vertices. *Journal of Research of the National Bureau of Standards, Section B*, 69B(1–2):125–130, 1965. doi: 10.6028/jres.069B.013.
- [11] Dima Grigoriev. Complexity of Positivstellensatz proofs for the knapsack. *Computational Complexity*, 10(2):139–154, 2001. doi: 10.1007/s00037-001-8192-0.

- [12] Martin Grötschel, László Lovász, and Alexander Schrijver. *Geometric Algorithms and Combinatorial Optimization*, volume 2 of *Algorithms and Combinatorics*. Springer, 1988. doi: 10.1007/978-3-642-97881-4.
- [13] Refael Hassin and Arie Tamir. Efficient algorithms for optimization and selection on series-parallel graphs. *SIAM Journal on Algebraic and Discrete Methods*, 7(3):379–389, 1986. doi: 10.1137/0607043. URL <https://www.math.tau.ac.il/~hassin/sp.pdf>.
- [14] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. URL <https://www.cs.rpi.edu/academics/courses/spring06/random/hoeffding.pdf>.
- [15] Florian Jarre. Best case exponential running time of a branch-and-bound algorithm using an optimal semidefinite relaxation, 2018. URL <https://optimization-online.org/wp-content/uploads/2018/07/6729.pdf>. Author preprint, August 4, 2018.
- [16] Richard M. Karp. Reducibility among combinatorial problems. In Raymond E. Miller and James W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, 1972. doi: 10.1007/978-1-4684-2001-2_9. URL <https://www.cs.umd.edu/~gasarch/BLOGPAPERS/Karp.pdf>.
- [17] James Luedtke, Mahdi Namazifar, and Jeff Linderoth. Some results on the strength of relaxations of multilinear functions. *Mathematical Programming*, 136(2):325–351, 2012. doi: 10.1007/s10107-012-0606-z. Theorem numbers refer to the 23-page author manuscript: <https://jlinderoth.github.io/papers/Luedtke-Namazifar-Linderoth-12-TR.pdf>.
- [18] Garth P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I—convex underestimating problems. *Mathematical Programming*, 10(1):147–175, 1976. doi: 10.1007/BF01580665.
- [19] Manfred Padberg. The boolean quadric polytope: Some characteristics, facets and relatives. *Mathematical Programming*, 45:139–172, 1989. doi: 10.1007/BF01589101.
- [20] Aaron Potechin. Sum of squares lower bounds from symmetry and a good story. In *10th Innovations in Theoretical Computer Science Conference (ITCS 2019)*, volume 124 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 61:1–61:20. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2019. doi: 10.4230/LIPIcs.ITCS.2019.61.
- [21] Hanif D. Sherali. Convex envelopes of multilinear functions over a unit hypercube and over special discrete sets. *Acta Mathematica Vietnamica*, 22(1):245–270, 1997. URL <https://math.ac.vn/uploads/files/9701245.pdf>.
- [22] Stanisław J. Szarek. On the best constants in the Khinchin inequality. *Studia Mathematica*, 58(2):197–208, 1976. doi: 10.4064/sm-58-2-197-208.